Differential Equations Workbook Solutions

Fourier series

krista king MATH

FOURIER SERIES REPRESENTATIONS

■ 1. Find the Fourier series representation of $f(x) = (2x + 1)^2$ on $-L \leq x \leq L$.

Solution:

For A_0, we get

$$A_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\ dx$$

$$A_0 = \frac{1}{2L}\int_{-L}^{L} (2x + 1)^2\ dx$$

$$A_0 = \frac{1}{6L}(2x + 1)^3 \cdot \frac{1}{2}\Bigg|_{-L}^{L}$$

$$A_0 = \frac{1}{12L}((1 + 2L)^3 - (1 - 2L)^3)$$

$$A_0 = \frac{1}{12L}(12L + 16L^3)$$

$$A_0 = 1 + \frac{4}{3}L^2$$

For A_n, we get

$$A_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)\ dx$$

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$$A_n = \frac{1}{L}\int_{-L}^{L} (2x+1)^2 \cos\left(\frac{n\pi x}{L}\right)\, dx$$

Use integration by parts with $u = (2x+1)^2$, $du = 4(2x+1)\, dx$, $dv = \cos(n\pi x/L)\, dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

$$A_n = \frac{1}{L}\left[(2x+1)^2 \frac{L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_{-L}^{L} - \frac{4L}{n\pi}\int_{-L}^{L}(2x+1)\sin\left(\frac{n\pi x}{L}\right)\, dx\right]$$

$$A_n = -\frac{4}{n\pi}\int_{-L}^{L}(2x+1)\sin\left(\frac{n\pi x}{L}\right)\, dx$$

Use integration by parts with $u = 2x+1$, $du = 2\, dx$, $dv = \sin(n\pi x/L)\, dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$A_n = -\frac{4}{n\pi}\left[-(2x+1)\frac{L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_{-L}^{L} + \frac{2L}{n\pi}\int_{-L}^{L}\cos\left(\frac{n\pi x}{L}\right)\, dx\right]$$

$$A_n = -\frac{4}{n\pi}\left[-(2L+1)\frac{L}{n\pi}(-1)^n + (1-2L)\frac{L}{n\pi}(-1)^n + \frac{2L^2}{n^2\pi^2}\sin\left(\frac{n\pi x}{L}\right)\Big|_{-L}^{L}\right]$$

$$A_n = \frac{16L^2}{n^2\pi^2}(-1)^n$$

For B_n, we get

$$B_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)\, dx$$

$$B_n = \frac{1}{L}\int_{-L}^{L} (2x+1)^2\sin\left(\frac{n\pi x}{L}\right)\, dx$$

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Use integration by parts with $u = (2x + 1)^2$, $du = 4(2x + 1)\ dx$, $dv = \sin(n\pi x/L)\ dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$B_n = \frac{1}{L}\left[-(2x+1)^2\frac{L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_{-L}^{L} + \frac{4L}{n\pi}\int_{-L}^{L}(2x+1)\cos\left(\frac{n\pi x}{L}\right)\ dx\right]$$

Use integration by parts with $u = 2x + 1$, $du = 2\ dx$, $dv = \cos(n\pi x/L)\ dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

$$\int_{-L}^{L}(2x+1)\cos\left(\frac{n\pi x}{L}\right)\ dx$$

$$= (2x+1)\frac{L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_{-L}^{L} - \frac{2L}{n\pi}\int_{-L}^{L}\sin\left(\frac{n\pi x}{L}\right)\ dx$$

$$\int_{-L}^{L}(2x+1)\cos\left(\frac{n\pi x}{L}\right)\ dx = \frac{2L^2}{(n\pi)^2}\cos\left(\frac{n\pi x}{L}\right)\Big|_{-L}^{L} = 0$$

Therefore,

$$B_n = \frac{1}{L}\left[-(2x+1)^2\frac{L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_{-L}^{L} + 0\right]$$

$$B_n = \frac{1}{L}\left(-(2L+1)^2\frac{L}{n\pi}(-1)^n + (1-2L)^2\frac{L}{n\pi}(-1)^n\right)$$

$$B_n = \frac{(-1)^n}{n\pi}(-8L)$$

$$B_n = -\frac{8L(-1)^n}{n\pi}$$

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Then the Fourier series representation of $f(x) = (2x+1)^2$ on $-L \leq x \leq L$ is

$$f(x) = 1 + \frac{4}{3}L^2 + \sum_{n=1}^{\infty} \frac{16L^2}{n^2\pi^2}(-1)^n \cos\left(\frac{n\pi x}{L}\right) - \sum_{n=1}^{\infty} \frac{8L}{n\pi}(-1)^n \sin\left(\frac{n\pi x}{L}\right)$$

■ 2. Find the Fourier series representation of $f(x) = e^x$ on $-\pi \leq x \leq \pi$.

Solution:

For A_0, we get

$$A_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^x \, dx$$

$$A_0 = \frac{1}{2\pi} e^x \Big|_{-\pi}^{\pi}$$

$$A_0 = \frac{1}{2\pi}(e^{\pi} - e^{-\pi})$$

For A_n, we get

$$A_n = \frac{1}{\pi}\int_{-\pi}^{\pi} e^x \cos(nx) \, dx$$

Use integration by parts with $u = e^x$, $du = e^x \, dx$, $dv = \cos(nx) \, dx$, and $v = (1/n)\sin(nx)$.

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$$A_n = \frac{1}{\pi}\left[e^x \frac{1}{n}\sin(nx)\Big|_{-\pi}^{\pi} - \frac{1}{n}\int_{-\pi}^{\pi} e^x \sin(nx)\ dx\right]$$

$$A_n = -\frac{1}{n\pi}\int_{-\pi}^{\pi} e^x \sin(nx)\ dx$$

Use integration by parts with $u = e^x$, $du = e^x\ dx$, $dv = \sin(nx)\ dx$, and $v = -(1/n)\cos(nx)$.

$$A_n = -\frac{1}{n\pi}\left[-\frac{1}{n}e^x \cos(nx)\Big|_{-\pi}^{\pi} + \frac{1}{n}\int_{-\pi}^{\pi} e^x \cos(nx)\ dx\right]$$

$$A_n = -\frac{1}{n\pi}\left(-\frac{1}{n}(e^{\pi} - e^{-\pi})(-1)^n + \frac{\pi}{n}A_n\right)$$

$$A_n\left(1 + \frac{1}{n^2}\right) = \frac{(e^{\pi} - e^{-\pi})(-1)^n}{n^2\pi}$$

$$A_n = \frac{(e^{\pi} - e^{-\pi})(-1)^n}{\pi(n^2 + 1)}$$

For B_n, we get

$$B_n = \frac{1}{\pi}\int_{-\pi}^{\pi} e^x \sin(nx)\ dx$$

Use integration by parts with $u = e^x$, $du = e^x\ dx$, $dv = \sin(nx)\ dx$, and $v = -(1/n)\cos(nx)$.

$$B_n = \frac{1}{\pi}\left[-\frac{1}{n}e^x \cos(nx)\Big|_{-\pi}^{\pi} + \frac{1}{n}\int_{-\pi}^{\pi} e^x \cos(nx)\ dx\right]$$

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$$B_n = \frac{1}{\pi}\left(-\frac{1}{n}(-1)^n(e^{\pi} - e^{-\pi}) + \frac{\pi}{n}A_n\right)$$

$$B_n = -\frac{n}{\pi(n^2+1)}(e^{\pi} - e^{-\pi})(-1)^n$$

Then the Fourier series representation of $f(x) = e^x$ on $-\pi \leq x \leq \pi$ is

$$f(x) = \frac{1}{2\pi}(e^{\pi} - e^{-\pi}) + \sum_{n=1}^{\infty} \frac{(e^{\pi} - e^{-\pi})(-1)^n}{\pi(n^2+1)}\cos(nx) + \sum_{n=1}^{\infty} -\frac{n(-1)^n(e^{\pi} - e^{-\pi})}{\pi(n^2+1)}\sin(nx)$$

3. Find the Fourier series representation of the function on $-\pi \leq x \leq \pi$.

$$f(x) = \sin\left(\frac{x}{2}\right)$$

Solution:

Since $f(x)$ is odd, we know $A_0 = A_n = 0$. For B_n, we get

$$B_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \sin\left(\frac{x}{2}\right)\sin\left(\frac{n\pi x}{\pi}\right)\,dx$$

$$B_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \sin\left(\frac{x}{2}\right)\sin(nx)\,dx$$

$$B_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \frac{1}{2}\cos\left(\left(n-\frac{1}{2}\right)x\right) - \cos\left(\left(n+\frac{1}{2}\right)x\right)\,dx$$

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$$B_n = \frac{1}{2\pi}\left(\frac{1}{n-\frac{1}{2}} \sin\left(n-\frac{1}{2}\right)x - \frac{1}{n+\frac{1}{2}} \sin\left(n+\frac{1}{2}\right)x \right)\Bigg|_{-\pi}^{\pi}$$

$$B_n = \frac{1}{2\pi}\left(\frac{1}{n-\frac{1}{2}} \sin\left(n-\frac{1}{2}\right)\pi - \frac{1}{n+\frac{1}{2}} \sin\left(n+\frac{1}{2}\right)\pi \right.$$

$$\left. + \frac{1}{n-\frac{1}{2}} \sin\left(n-\frac{1}{2}\right)\pi - \frac{1}{n+\frac{1}{2}} \sin\left(n+\frac{1}{2}\right)\pi \right)$$

$$B_n = \frac{1}{2\pi}\left(\frac{2}{n-\frac{1}{2}} \sin\left(n\pi-\frac{\pi}{2}\right) - \frac{2}{n+\frac{1}{2}} \sin\left(n\pi+\frac{\pi}{2}\right) \right)$$

$$B_n = \frac{1}{2\pi}\left(\frac{2}{n-\frac{1}{2}} \left(\sin(n\pi)\cos\left(\frac{\pi}{2}\right) - \cos(n\pi)\sin\left(\frac{\pi}{2}\right) \right) \right.$$

$$\left. - \frac{2}{n+\frac{1}{2}} \left(\sin(n\pi)\cos\left(\frac{\pi}{2}\right) + \cos(n\pi)\sin\left(\frac{\pi}{2}\right) \right) \right)$$

$$B_n = \frac{1}{2\pi}\left(\frac{2}{n-\frac{1}{2}}(0-\cos(n\pi)) - \frac{2}{n+\frac{1}{2}}(0+\cos(n\pi)) \right)$$

$$B_n = \frac{1}{2\pi}\left(-\frac{4}{2n-1}\cos(n\pi) - \frac{4}{2n+1}\cos(n\pi) \right)$$

$$B_n = -\frac{8n(-1)^n}{\pi(4n^2-1)}$$

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Then the Fourier series representation of $f(x)$ on $-\pi \leq x \leq \pi$ is

$$f(x) = \sum_{n=1}^{\infty} -\frac{8n(-1)^n}{\pi(4n^2-1)} \sin(nx)$$

■ 4. Find the Fourier series representation of $f(x) = x \sin x$ on $-\pi \leq x \leq \pi$.

Solution:

Since $f(x)$ is even, we know $B_n = 0$. For A_0, we get

$$A_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} x \sin x \ dx$$

$$A_0 = \frac{1}{\pi} \int_{0}^{\pi} x \sin x \ dx$$

Use integration by parts with $u = x$, $du = dx$, $dv = \sin x \ dx$, and $v = -\cos x$.

$$A_0 = \frac{1}{\pi} \left(-x \cos x \Big|_0^{\pi} + \int_0^{\pi} \cos x \ dx \right)$$

$$A_0 = \frac{1}{\pi} \left(-x \cos x \Big|_0^{\pi} + \int_0^{\pi} \cos x \ dx \right)$$

$$A_0 = \frac{1}{\pi} \left(\pi + \sin x \Big|_0^{\pi} \right) = 1$$

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For A_n, we get

$$A_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\ dx$$

$$A_n = \frac{2}{\pi}\int_{0}^{\pi} f(x)\ \cos(nx)\ dx$$

$$A_n = \frac{2}{\pi}\int_{0}^{\pi} x\sin x\cos(nx)\ dx$$

$$A_n = \frac{2}{\pi}\int_{0}^{\pi} \frac{x}{2}(\sin(1+n)x + \sin(1-n)x)\ dx$$

$$A_n = \frac{1}{\pi}\int_{0}^{\pi} x(\sin(1+n)x + \sin(1-n)x)\ dx$$

Use integration by parts with $u = x$, $du = dx$, $dv = \sin(1+n)x + \sin(1-n)x\ dx$, and

$$v = -\frac{1}{n+1}\cos(n+1)x + \frac{1}{n-1}\cos(n-1)x \text{ with } n \neq 1$$

Then we get

$$A_n = \frac{1}{\pi}\left(x\left(-\frac{1}{n+1}\cos(n+1)x + \frac{1}{n-1}\cos(n-1)x\right)\Big|_0^{\pi} \right.$$

$$\left. + \int_0^{\pi} \frac{1}{n+1}\cos(n+1)x\ dx - \int_0^{\pi} \frac{1}{n-1}\cos(n-1)x\ dx \right)$$

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$$A_n = \frac{1}{\pi}\left(\pi \left(-\frac{1}{n+1}(-1)^{n+1} + \frac{1}{n-1}(-1)^{n+1} \right) \right.$$

$$\left. + \frac{1}{(n+1)^2}\sin(n+1)x\Big|_0^{\pi} - \frac{1}{(n-1)^2}\sin(n-1)x\Big|_0^{\pi} \right)$$

$$A_n = \frac{2}{n^2-1}(-1)^{n+1} \text{ with } n \neq 1$$

Because the value we found for A_n wasn't valid for $n = 1$, we'll also find A_1.

$$A_1 = \frac{2}{\pi}\int_0^{\pi} x \sin x \cos x \; dx$$

$$A_1 = \frac{1}{\pi}\int_0^{\pi} x \sin(2x) \; dx$$

Use integration by parts with $u = x$, $du = dx$, $dv = \sin(2x)\, dx$, and $v = -(1/2)\cos(2x)$.

$$A_1 = \frac{1}{\pi}\left(-\frac{1}{2}x\cos(2x)\Big|_0^{\pi} + \frac{1}{2}\int_0^{\pi} \cos(2x) \; dx \right)$$

$$A_1 = \frac{1}{\pi}\left(-\frac{\pi}{2} + \frac{1}{4}\sin(2x)\Big|_0^{\pi} \right)$$

$$A_1 = -\frac{1}{2}$$

Then the Fourier series representation of $f(x) = x \sin x$ on $-\pi \leq x \leq \pi$ is

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$$f(x) = 1 - \frac{1}{2}\cos x + \sum_{n=2}^{\infty} \frac{2(-1)^{n+1}}{n^2 - 1}\cos(nx)$$

■ 5. Find the Fourier series representation of $f(x) = x^2 + \sin x$ on $-L \leq x \leq L$.

Solution:

The function x^2 is even, so $B_n = 0$ in its Fourier representation. For A_0, we get

$$A_0 = \frac{1}{L}\int_0^L x^2 \, dx$$

$$A_0 = \frac{1}{L} \cdot \frac{x^3}{3}\bigg|_0^L = \frac{L^2}{3}$$

For A_n, we get

$$A_n = \frac{2}{L}\int_0^L x^2 \cos\left(\frac{n\pi x}{L}\right) dx$$

Use integration by parts with $u = x^2$, $du = 2x \, dx$, $dv = \cos(n\pi x/L) \, dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

$$A_n = \frac{2}{L}\left(x^2 \frac{L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\bigg|_0^L - \frac{2L}{n\pi}\int_0^L x \sin\left(\frac{n\pi x}{L}\right) dx\right)$$

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$$A_n = \frac{2}{L}\left(0 - \frac{2L}{n\pi}\int_0^L x \sin\left(\frac{n\pi x}{L}\right)\, dx\right)$$

$$A_n = -\frac{4}{n\pi}\int_0^L x \sin\left(\frac{n\pi x}{L}\right) dx$$

Use integration by parts with $u = x$, $du = dx$, $dv = \sin(n\pi x/L)\, dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$A_n = -\frac{4}{n\pi}\left(-x\frac{L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Bigg|_0^L + \int_0^L \frac{L}{n\pi}\cos\left(\frac{n\pi x}{L}\right) dx\right)$$

$$A_n = -\frac{4}{n\pi}\left(-\frac{L^2}{n\pi}\cos(n\pi) + \int_0^L \frac{L}{n\pi}\cos\left(\frac{n\pi x}{L}\right) dx\right)$$

$$A_n = \frac{4L^2}{n^2\pi^2}(-1)^n + \frac{4}{n\pi}\cdot\frac{L^2}{n^2\pi^2}\sin\left(\frac{n\pi x}{L}\right)\Bigg|_0^L$$

$$A_n = \frac{4L^2}{n^2\pi^2}(-1)^n$$

The function $\sin(x)$ is odd, so $A_0 = A_n = 0$ in its Fourier representation. For B_n, we get

$$B_n = \frac{2}{L}\int_0^L \sin x \sin\left(\frac{n\pi x}{L}\right) dx$$

$$B_n = \frac{1}{L}\int_0^L \left(\cos\left(\frac{n\pi}{L} - 1\right)x - \cos\left(\frac{n\pi}{L} + 1\right)x\right) dx$$

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$$B_n = \frac{1}{L}\left(\frac{L}{n\pi - L}\sin\left(\frac{n\pi}{L} - 1\right)x - \frac{L}{n\pi + L}\sin\left(\frac{n\pi}{L} + 1\right)x\right)\Bigg|_0^L$$

$$B_n = \frac{1}{L}\Bigg(\frac{L}{n\pi - L}\left(\sin\left(\frac{n\pi x}{L}\right)\cos x - \cos\left(\frac{n\pi x}{L}\right)\sin x\right)$$

$$-\frac{L}{n\pi + L}\left(\sin\left(\frac{n\pi x}{L}\right)\cos x + \cos\left(\frac{n\pi x}{L}\right)\sin x\right)\Bigg)\Bigg|_0^L$$

$$B_n = \frac{1}{L}\left(\frac{L}{n\pi - L}(-\cos(n\pi)\sin L) - \frac{L}{n\pi + L}(\cos(n\pi)\sin L)\right)$$

$$B_n = \frac{-\sin L\cos(n\pi)}{n\pi - L} - \frac{\sin L\cos(n\pi)}{n\pi + L}$$

$$B_n = -\frac{2n\pi(-1)^n\sin L}{n^2\pi^2 - L^2}$$

Therefore, the Fourier series representation of $f(x) = x^2 + \sin x$ on $-L \le x \le L$ is

$$f(x) = \frac{L^2}{3} + \sum_{n=1}^{\infty}\frac{4L^2}{n^2\pi^2}(-1)^n\cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} -\frac{2n\pi\sin L(-1)^n}{n^2\pi^2 - L^2}\sin\left(\frac{n\pi x}{L}\right)$$

■ 6. Find the Fourier series representation of $f(x) = \cosh x$ on $-\pi \le x \le \pi$.

Solution:

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Since $f(x)$ is even, we know $B_n = 0$. For A_0, we get

$$A_0 = \frac{1}{\pi}\int_0^{\pi} \cosh x \; dx$$

$$A_0 = \frac{1}{\pi}\sinh x\Big|_0^{\pi} = \frac{\sinh \pi}{\pi}$$

For A_n, we get

$$A_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos(nx) \; dx$$

$$A_n = \frac{1}{\pi}\int_0^{\pi} (e^x + e^{-x})\cos(nx) \; dx$$

Use integration by parts with $u = e^x + e^{-x}$, $du = (e^x - e^{-x})\; dx$, $dv = \cos(nx)\; dx$, and $v = (1/n)\sin(nx)$.

$$A_n = \frac{1}{\pi}\left((e^x + e^{-x})\frac{1}{n}\sin(nx)\Big|_0^{\pi} - \frac{1}{n}\int_0^{\pi} (e^x - e^{-x})\sin(nx) \; dx \right)$$

$$A_n = \frac{1}{\pi}\left(0 - \frac{1}{n}\int_0^{\pi} (e^x - e^{-x})\sin(nx) \; dx \right)$$

$$A_n = -\frac{1}{n\pi}\int_0^{\pi} (e^x - e^{-x})\sin(nx) \; dx$$

Use integration by parts with $u = e^x - e^{-x}$, $du = (e^x + e^{-x})\; dx$, $dv = \sin(nx)\; dx$, and $v = -(1/n)\cos(nx)$.

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$$A_n = -\frac{1}{n\pi}\left(-\frac{1}{n}(e^x - e^{-x})\cos(nx)\,dx\Big|_0^{\pi} + \frac{1}{n}\int_0^{\pi}(e^x + e^{-x})\cos(nx)\,dx\right)$$

$$A_n = -\frac{1}{n\pi}\left(-\frac{2}{n}(-1)^n \sinh\pi + \frac{\pi}{n}A_n\right)$$

$$A_n = \frac{2}{n^2\pi}(-1)^n \sinh\pi - \frac{1}{n^2}A_n$$

$$A_n\left(1 + \frac{1}{n^2}\right) = \frac{2(-1)^n \sinh\pi}{n^2\pi}$$

$$A_n = \frac{2(-1)^n \sinh\pi}{\pi(n^2+1)}$$

Then the Fourier series representation of $f(x) = \cosh x$ on $-\pi \le x \le \pi$ is

$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos(nx)$$

$$f(x) = \frac{\sinh\pi}{\pi} + \sum_{n=1}^{\infty} \frac{2(-1)^n \sinh\pi}{\pi(n^2+1)}\cos(nx)$$

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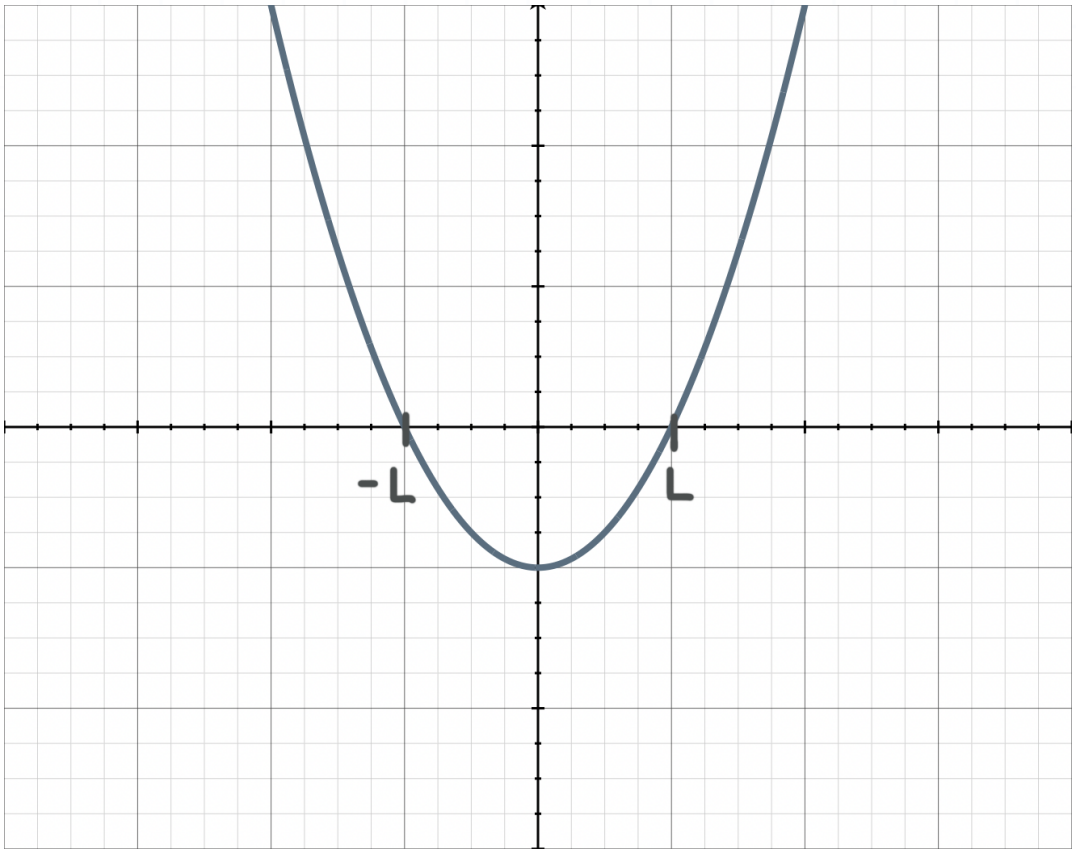
PERIODIC FUNCTIONS AND PERIODIC EXTENSIONS

■ 1. Sketch the function's periodic extension on the interval $-L \leq x \leq L$, given some positive value of L.

$$f(x) = \frac{1}{L}x^2 - L$$

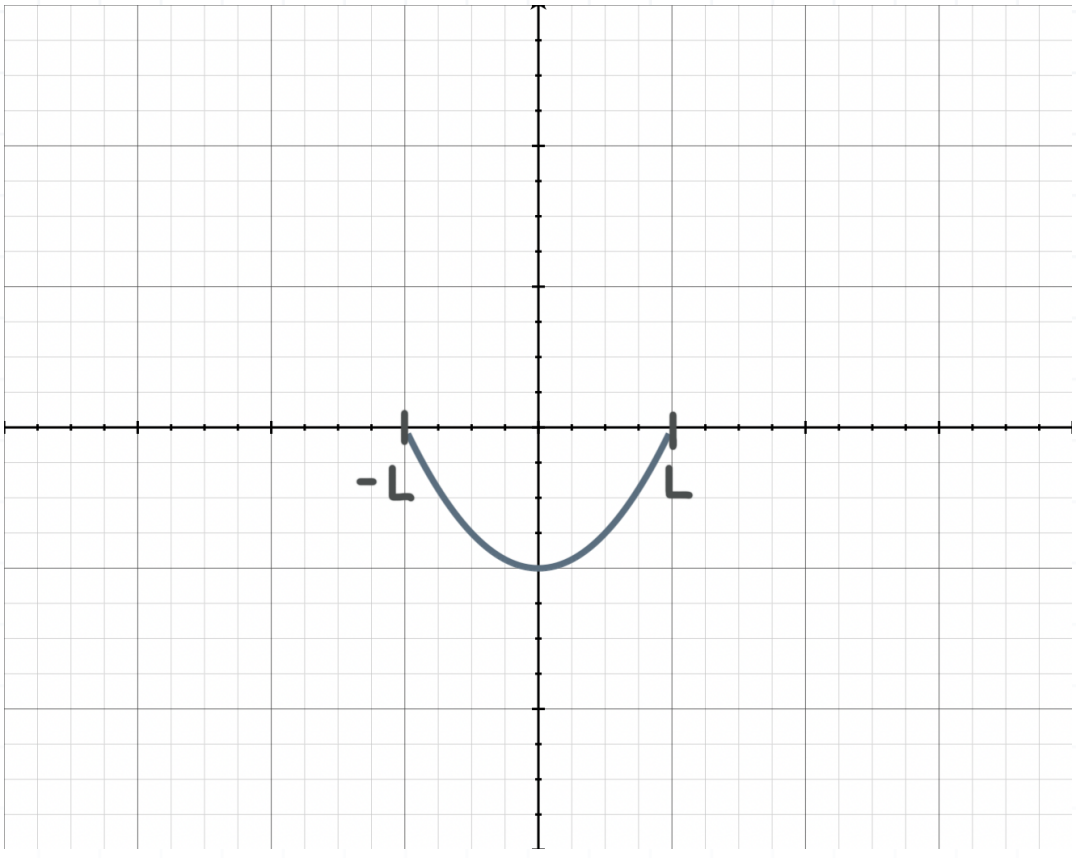
Solution:

The sketch of $f(x)$ is

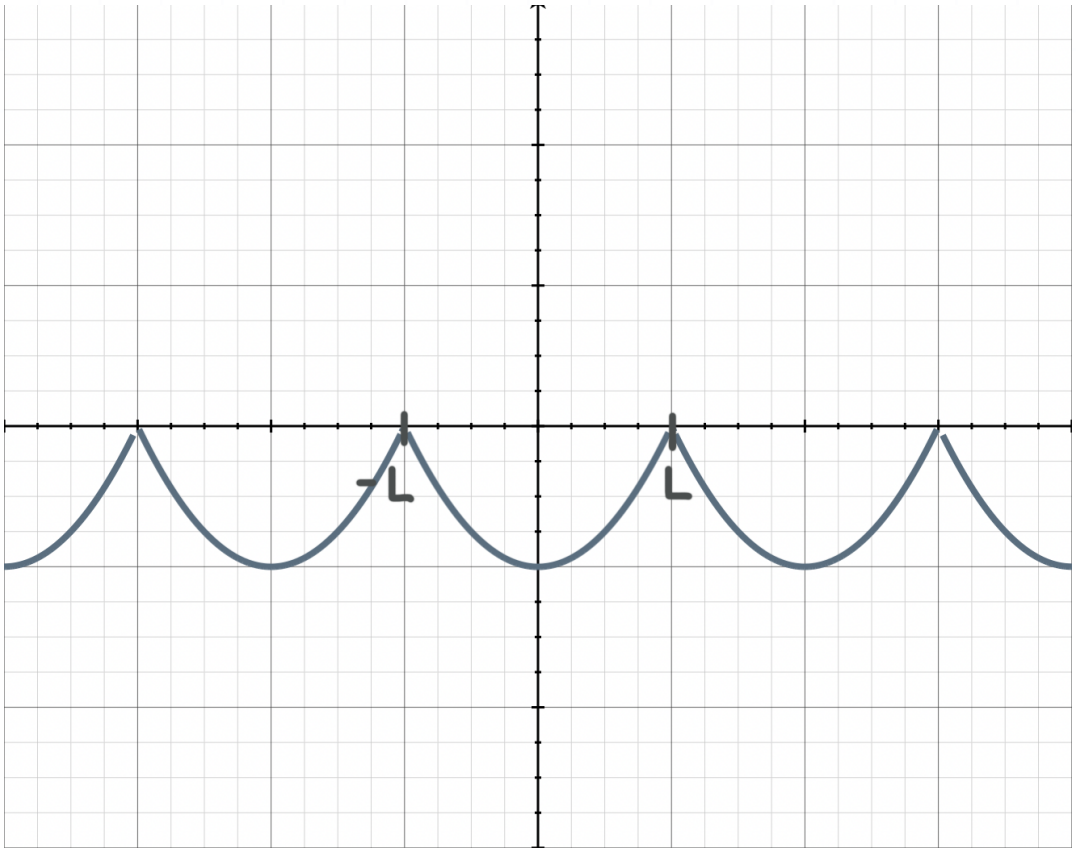


If we limit the graph to the interval $-L \leq x \leq L$, then the section of the graph on $-L \leq x \leq L$ is

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If we repeat this section over and over again on both sides, we get a sketch of the periodic extension of $f(x)$.



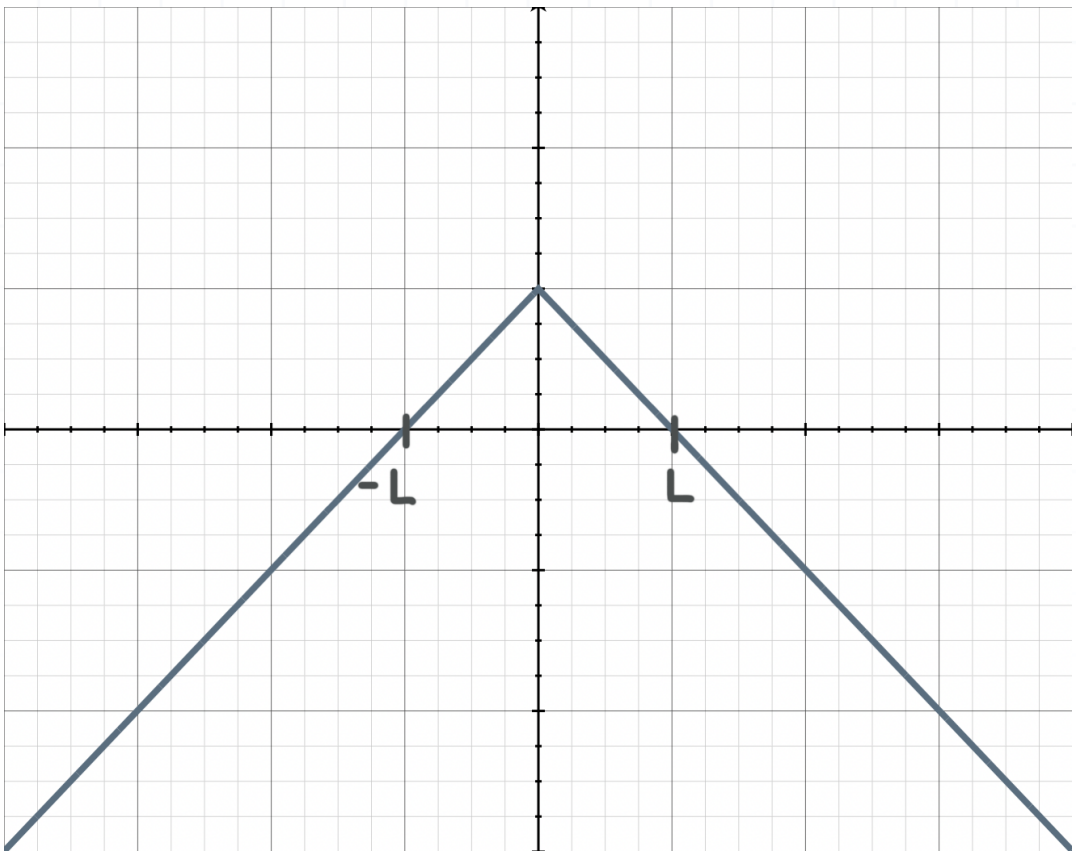
■ 2. Sketch the function's periodic extension on the interval $-L \leq x \leq L$.

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$$f(x) = \begin{cases} L + x & x < 0 \\ L - x & x \geq 0 \end{cases}$$

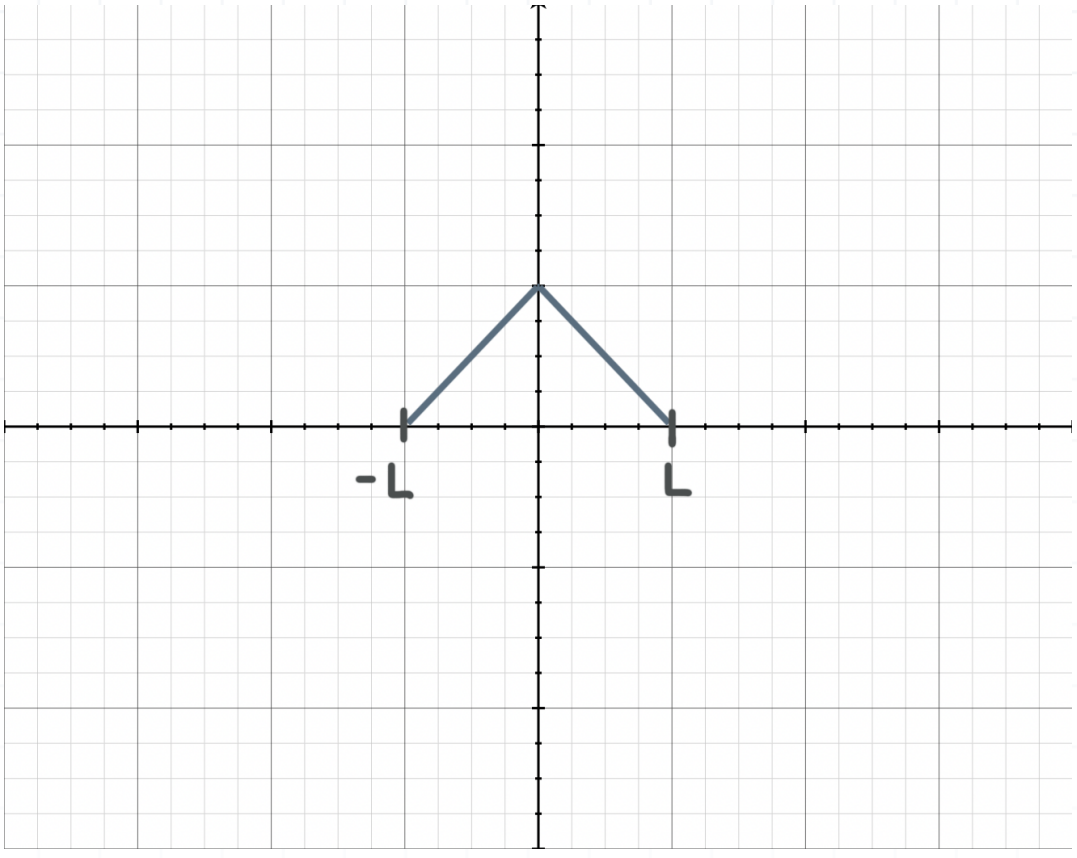
Solution:

The sketch of $f(x)$ is

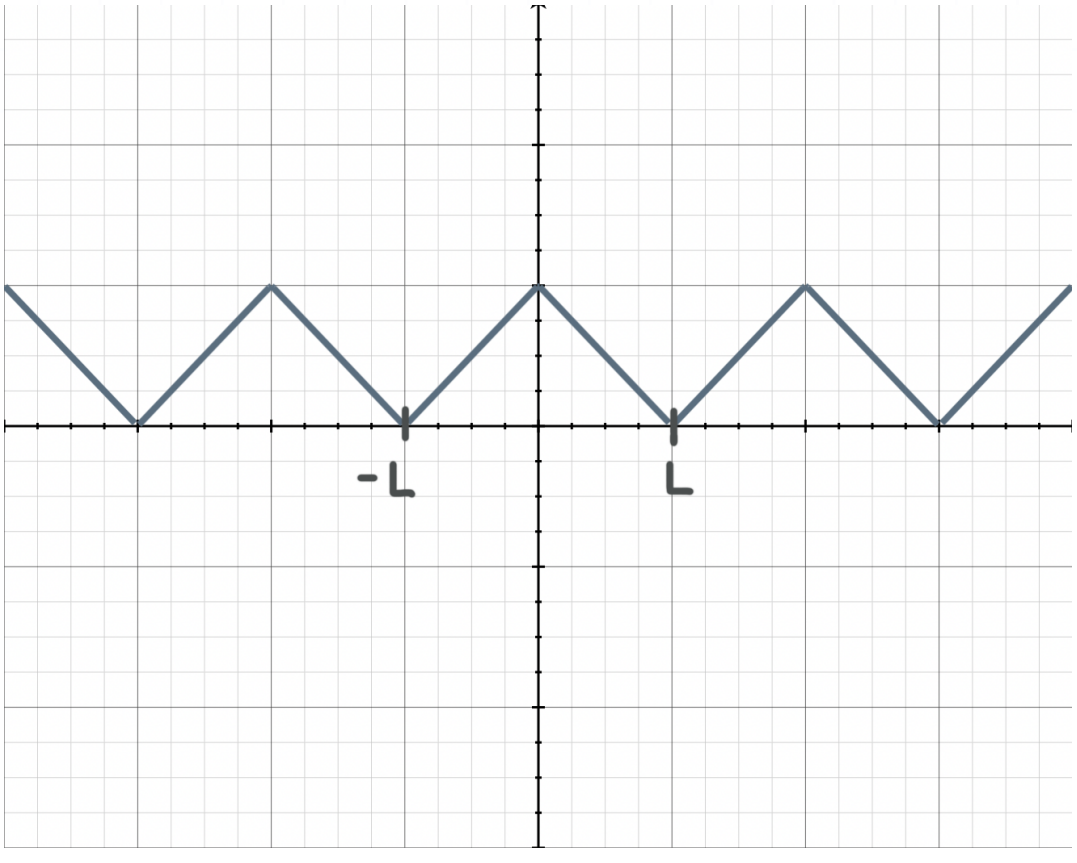


If we limit the graph to the interval $-L \leq x \leq L$, then the section of the graph on $-L \leq x \leq L$ is

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If we repeat this section over and over again on both sides, we get a sketch of the periodic extension of $f(x)$.



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■ 3. Find the even and odd extensions of the function, given some positive value of L. Then sketch both extensions.

$$f(x) = \frac{1}{L^2}x^3 - \frac{1}{L}x^2$$

Solution:

The even extension of $f(x)$ is

$$g(x) = \begin{cases} f(x) & 0 \le x \le L \\ f(-x) & -L \le x < 0 \end{cases}$$

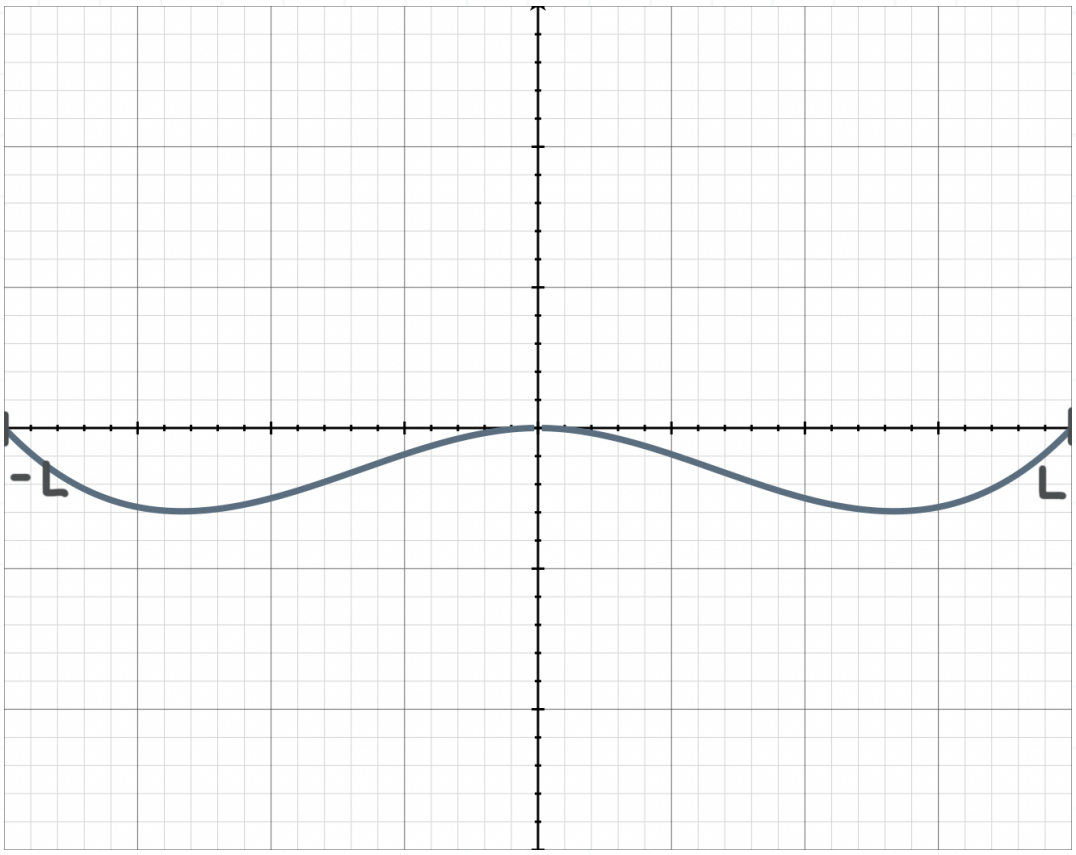
$$g(x) = \begin{cases} \frac{1}{L^2}x^3 - \frac{1}{L}x^2 & 0 \le x \le L \\ \frac{1}{L^2}(-x)^3 - \frac{1}{L}(-x)^2 & -L \le x < 0 \end{cases}$$

$$g(x) = \begin{cases} \frac{1}{L^2}x^3 - \frac{1}{L}x^2 & 0 \le x \le L \\ -\frac{1}{L^2}x^3 - \frac{1}{L}x^2 & -L \le x < 0 \end{cases}$$

The sketch of the even extension is

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The odd extension of $f(x)$ is

$$g(x) = \begin{cases} f(x) & 0 \le x \le L \\ -f(-x) & -L \le x < 0 \end{cases}$$

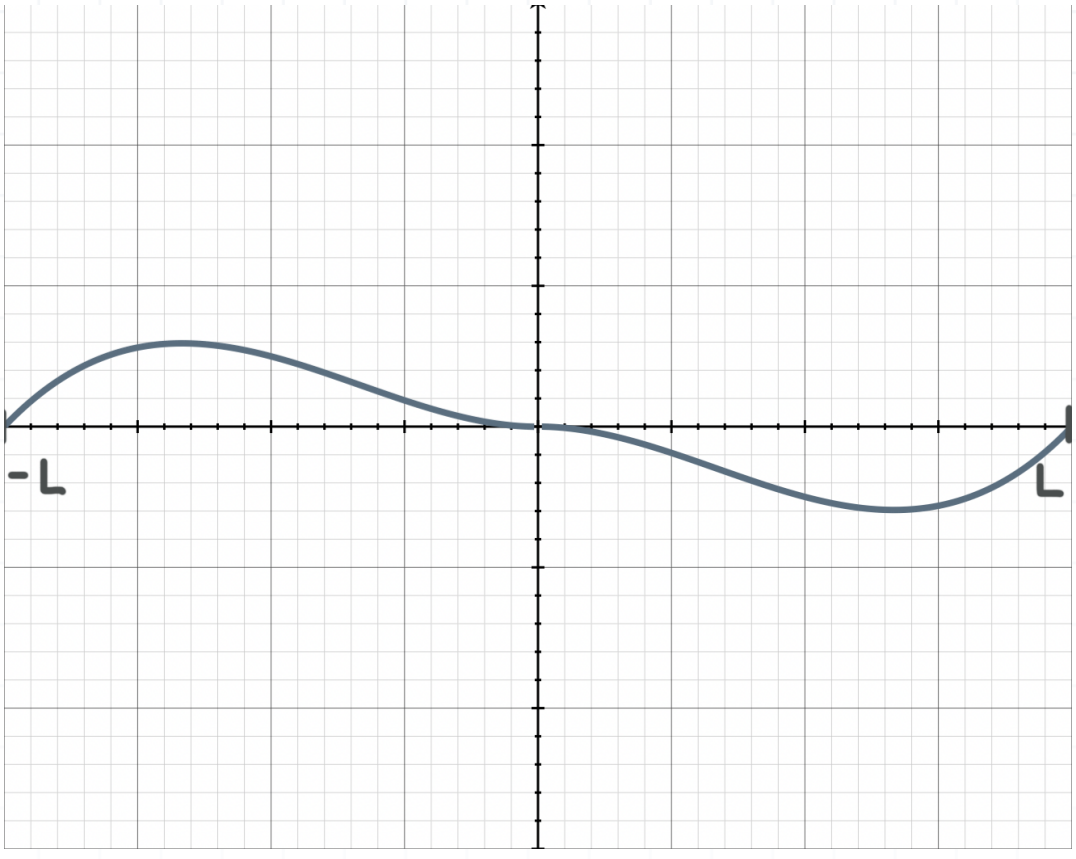
$$g(x) = \begin{cases} \frac{1}{L^2}x^3 - \frac{1}{L}x^2 & 0 \le x \le L \\ -\left(\frac{1}{L^2}(-x)^3 - \frac{1}{L}(-x)^2\right) & -L \le x < 0 \end{cases}$$

$$g(x) = \begin{cases} \frac{1}{L^2}x^3 - \frac{1}{L}x^2 & 0 \le x \le L \\ -\left(-\frac{1}{L^2}x^3 - \frac{1}{L}x^2\right) & -L \le x < 0 \end{cases}$$

$$g(x) = \begin{cases} \frac{1}{L^2}x^3 - \frac{1}{L}x^2 & 0 \le x \le L \\ \frac{1}{L^2}x^3 + \frac{1}{L}x^2 & -L \le x < 0 \end{cases}$$

The sketch of the odd extension is

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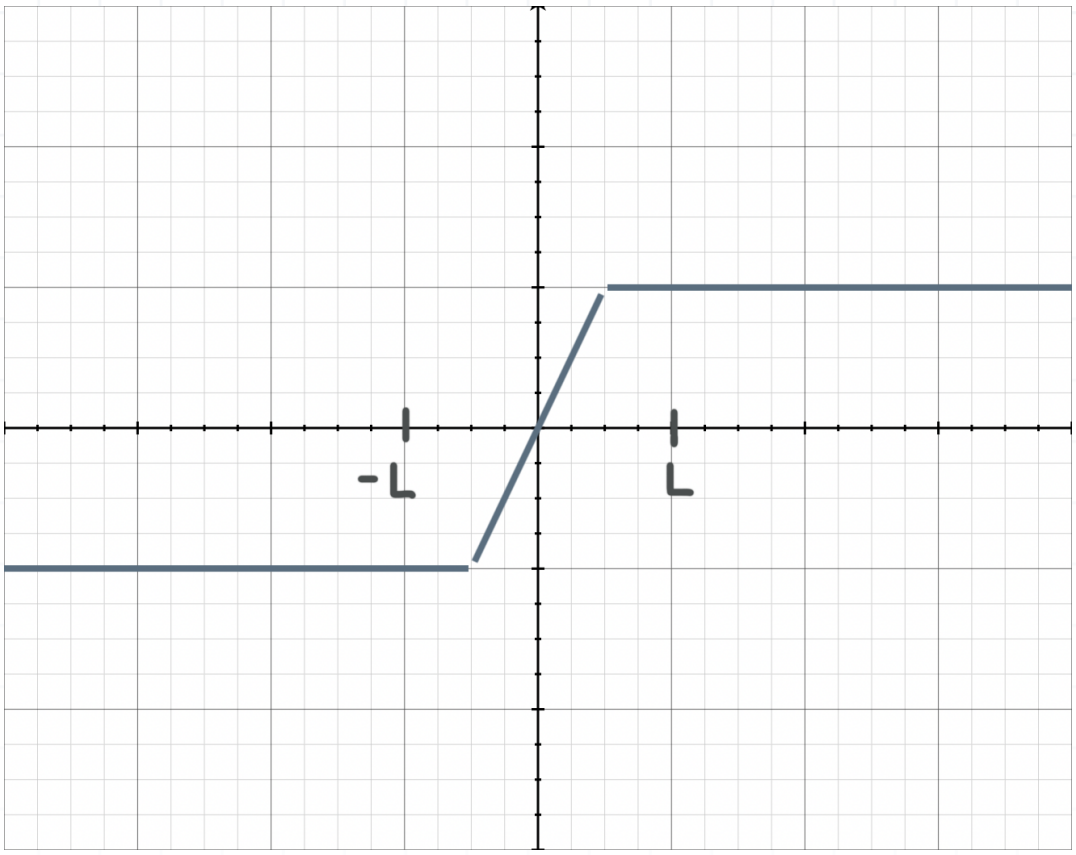
4. Find the even and odd extensions of the function, given some positive value of L. Compare both sketches.

$$f(x) = \begin{cases} L^2 & \frac{L}{2} < x \\ 2Lx & -\frac{L}{2} \leq x \leq \frac{L}{2} \\ -L^2 & x < -\frac{L}{2} \end{cases}$$

Solution:

The sketch of $f(x)$ is

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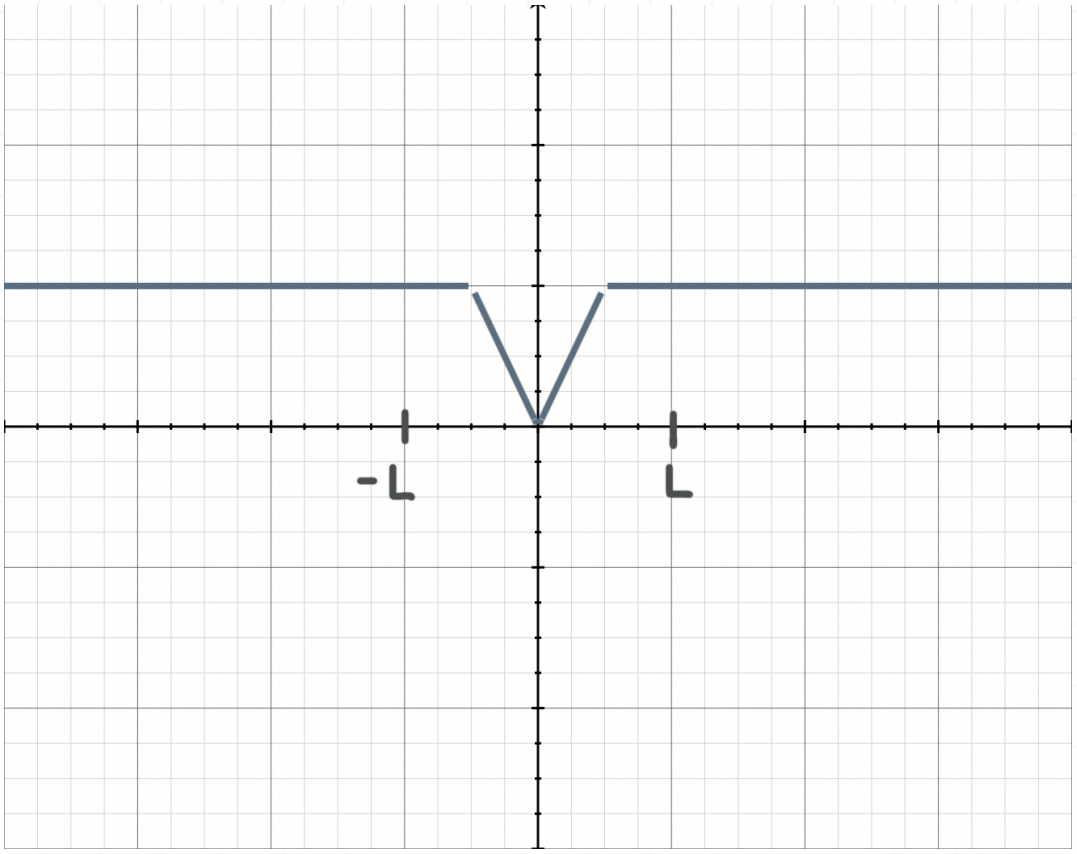
The even extension of $f(x)$ is

$$g(x) = \begin{cases} f(x) & 0 \le x \le L \\ f(-x) & -L \le x < 0 \end{cases}$$

$$g(x) = \begin{cases} L^2 & \frac{L}{2} < x \le L \\ 2Lx & 0 \le x \le \frac{L}{2} \\ 2L(-x) & -\frac{L}{2} \le x < 0 \\ -L^2 & -L \le x < -\frac{L}{2} \end{cases} = \begin{cases} L^2 & \frac{L}{2} < x \le L \\ 2Lx & 0 \le x \le \frac{L}{2} \\ -2Lx & -\frac{L}{2} \le x < 0 \\ -L^2 & -L \le x < -\frac{L}{2} \end{cases}$$

The sketch of the even extension is

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The odd extension of $f(x)$ is

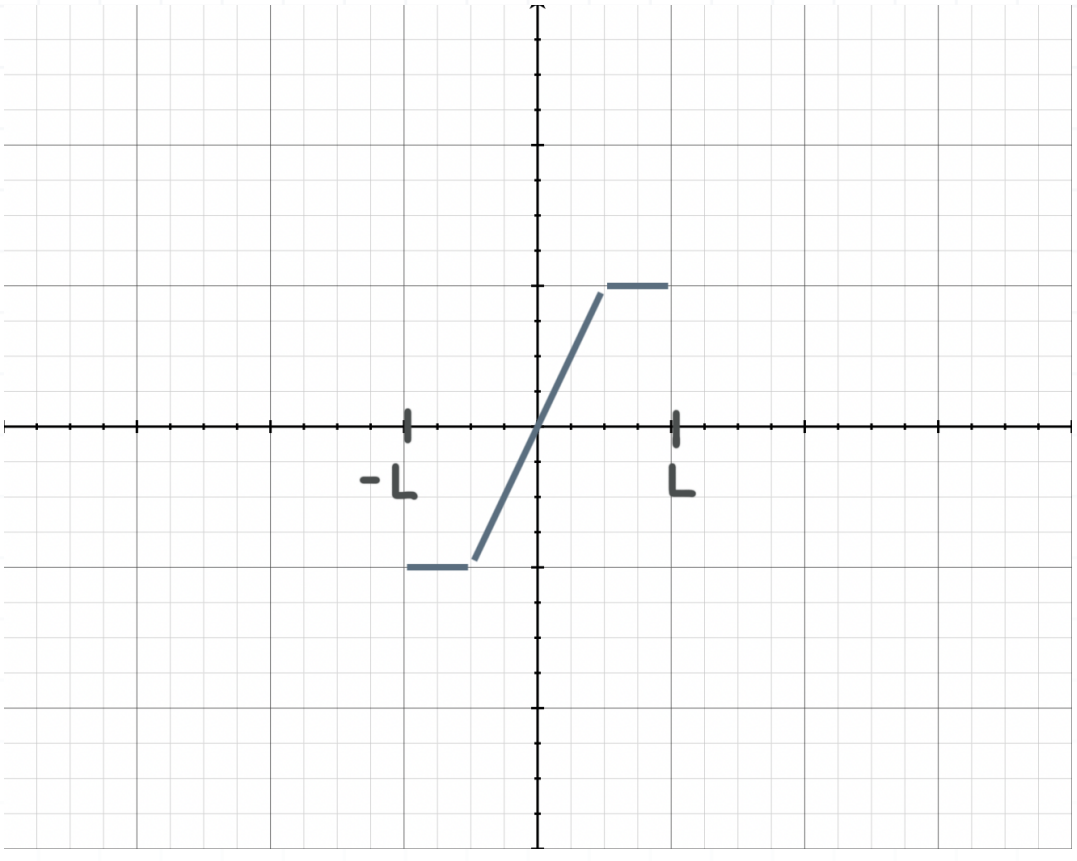
$$g(x) = \begin{cases} f(x) & 0 \le x \le L \\ -f(-x) & -L \le x < 0 \end{cases}$$

$$g(x) = \begin{cases} L^2 & \frac{L}{2} < x \le L \\ 2Lx & 0 \le x \le \frac{L}{2} \\ -(2L(-x)) & -\frac{L}{2} \le x < 0 \\ -L^2 & -L \le x < -\frac{L}{2} \end{cases}$$

$$g(x) = \begin{cases} L^2 & \frac{L}{2} < x \le L \\ 2Lx & 0 \le x \le \frac{L}{2} \\ 2Lx & -\frac{L}{2} \le x < 0 \\ -L^2 & -L \le x < -\frac{L}{2} \end{cases} = \begin{cases} L^2 & \frac{L}{2} < x \le L \\ 2Lx & -\frac{L}{2} \le x \le \frac{L}{2} \\ -L^2 & -L \le x < -\frac{L}{2} \end{cases}$$

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The sketch of the odd extension is



Since the sketches of the function and its odd extension are equivalent on $-L \leq x \leq L$, the function is odd.

■ 5. Find the even and odd extensions of the function, given some positive value of L.

$$f(x) = e^{L-2x}$$

Solution:

The even extension of $f(x)$ is

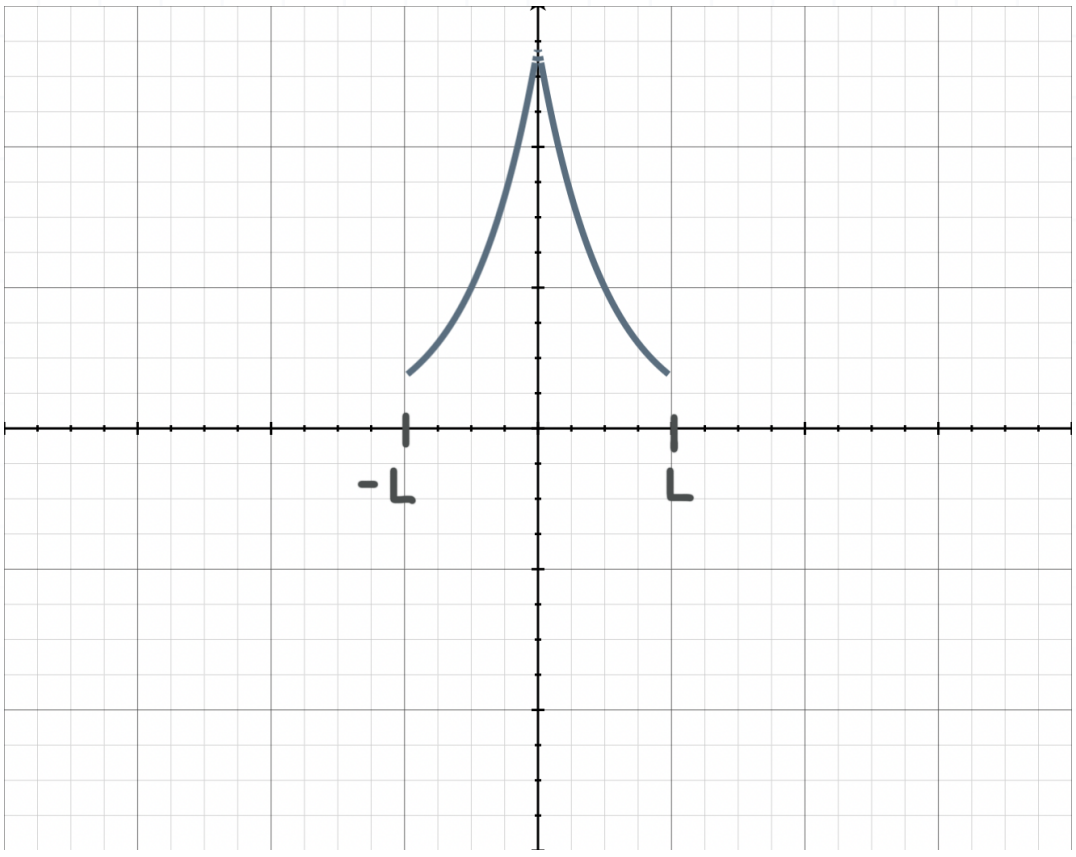
$$g(x) = \begin{cases} f(x) & 0 \leq x \leq L \\ f(-x) & -L \leq x < 0 \end{cases}$$

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$$g(x) = \begin{cases} e^{L-2x} & 0 \le x \le L \\ e^{L-2(-x)} & -L \le x < 0 \end{cases}$$

$$g(x) = \begin{cases} e^{L-2x} & 0 \le x \le L \\ e^{L+2x} & -L \le x < 0 \end{cases}$$

The sketch of the even extension is



The odd extension of $f(x)$ is

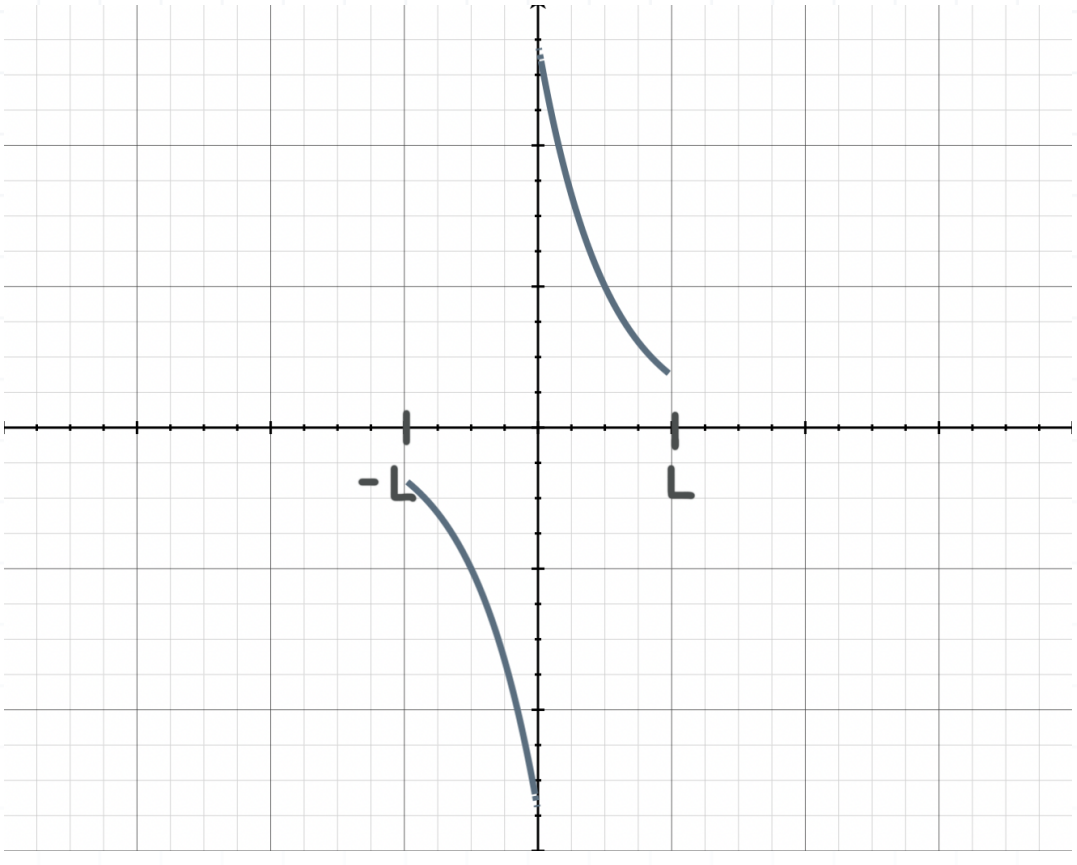
$$g(x) = \begin{cases} f(x) & 0 \le x \le L \\ -f(-x) & -L \le x < 0 \end{cases}$$

$$g(x) = \begin{cases} e^{L-2x} & 0 \le x \le L \\ -e^{L-2(-x)} & -L \le x < 0 \end{cases}$$

$$g(x) = \begin{cases} e^{L-2x} & 0 \le x \le L \\ -e^{L+2x} & -L \le x < 0 \end{cases}$$

The sketch of the odd extension is

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■ 6. Sketch the periodic extensions of the even and odd extensions of the function $f(x) = x$ on $-L \leq x \leq L$.

Solution:

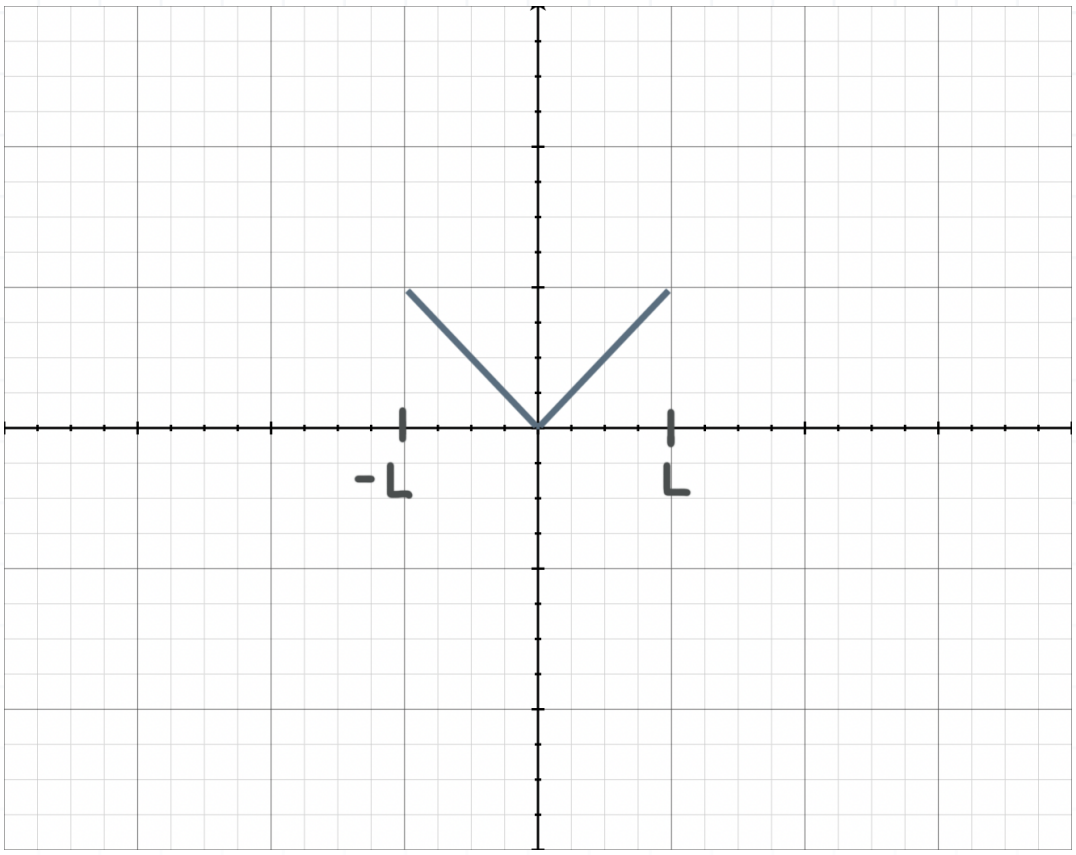
The even extension of $f(x)$ is

$$g(x) = \begin{cases} f(x) & 0 \leq x \leq L \\ f(-x) & -L \leq x < 0 \end{cases}$$

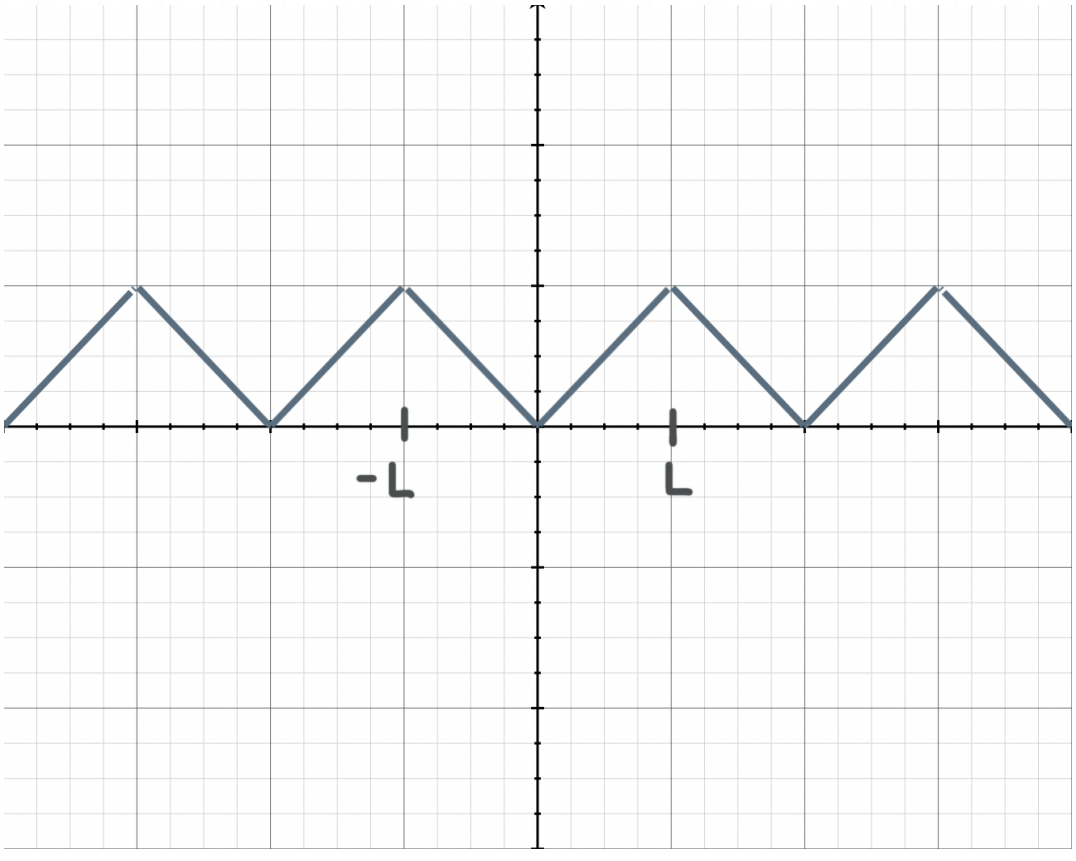
$$g(x) = \begin{cases} x & 0 \leq x \leq L \\ -x & -L \leq x < 0 \end{cases}$$

The sketch of the even extension is

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Then the sketch of the periodic extension of the even extension is



The odd extension of $f(x)$ is

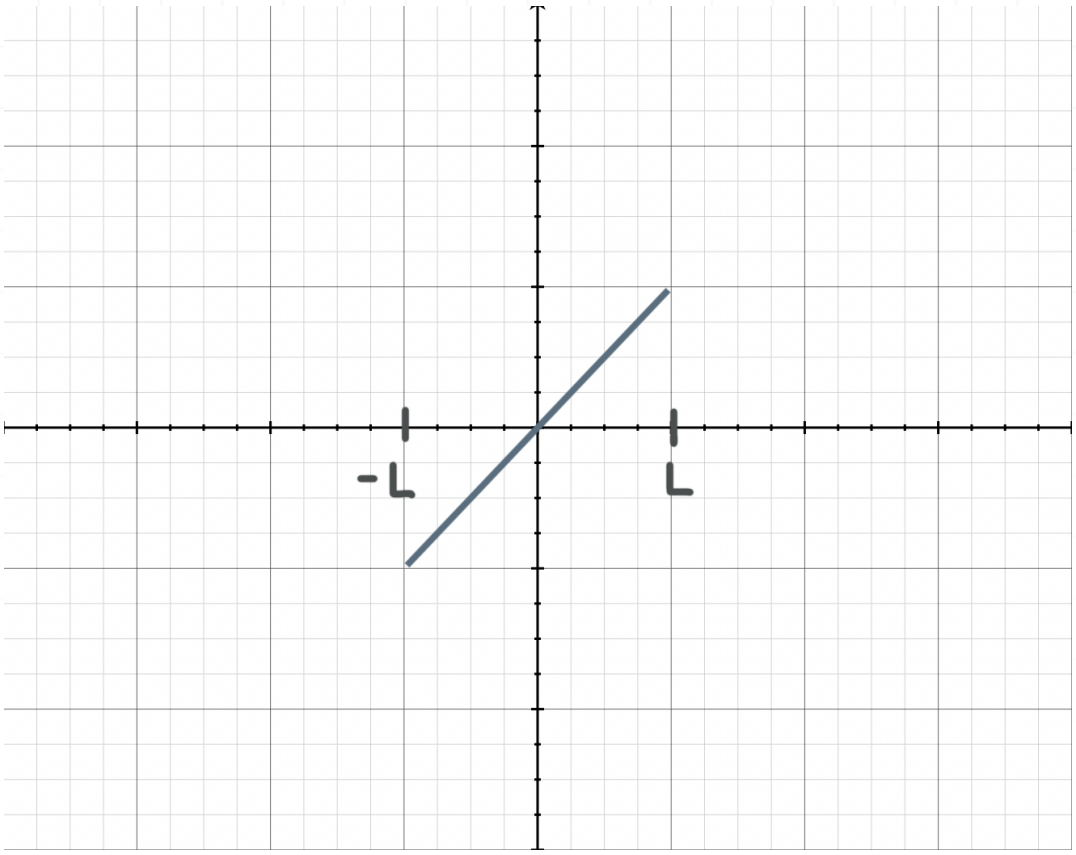
$$g(x) = \begin{cases} f(x) & 0 \le x \le L \\ -f(-x) & -L \le x < 0 \end{cases}$$

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$$g(x) = \begin{cases} x & 0 \le x \le L \\ -(-x) & -L \le x < 0 \end{cases}$$

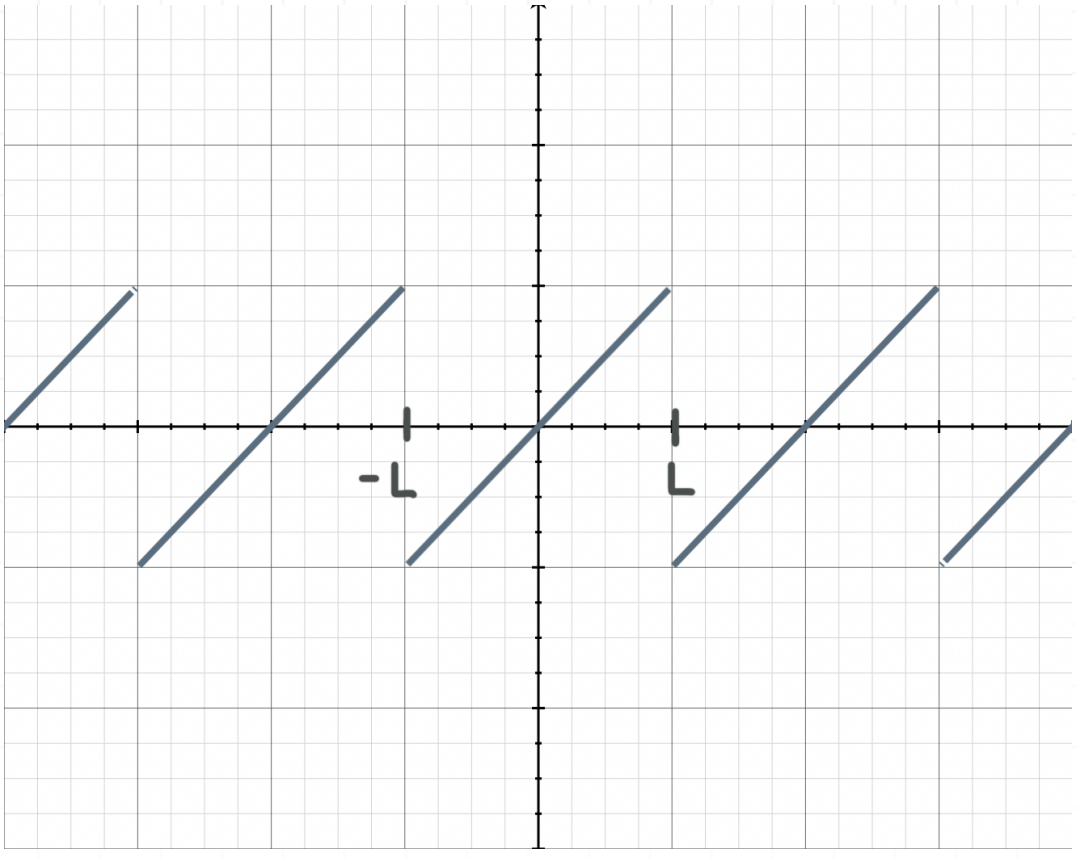
$g(x) = x$ on $-L \le x \le L$

The sketch of the odd extension is



Then the sketch of the periodic extension of the odd extension is

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REPRESENTING PIECEWISE FUNCTIONS

■ 1. Find the Fourier series representation of the piecewise function on $-L \leq x \leq L$.

$$f(x) = \begin{cases} -2x & -L \leq x < 0 \\ 3x & 0 \leq x \leq L \end{cases}$$

Solution:

For A_0 we get

$$A_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\, dx$$

$$A_0 = \frac{1}{2L}\left(\int_{-L}^{0} -2x\, dx + \int_{0}^{L} 3x\, dx\right)$$

$$A_0 = \frac{1}{2L}\left(-x^2\Big|_{-L}^{0} + \frac{3}{2}x^2\Big|_{0}^{L}\right)$$

$$A_0 = \frac{5}{4}L$$

For A_n we get

$$A_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right) dx$$

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$$A_n = \frac{1}{L}\left(\int_{-L}^{0} -2x\cos\left(\frac{n\pi x}{L}\right)\,dx + \int_{0}^{L} 3x\cos\left(\frac{n\pi x}{L}\right)\,dx \right)$$

$$A_n = -\frac{2}{L}\int_{-L}^{0} x\cos\left(\frac{n\pi x}{L}\right)\,dx + \frac{3}{L}\int_{0}^{L} x\cos\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = x$, $du = dx$, $dv = \cos(n\pi x/L)\,dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

$$A_n = -\frac{2}{L}\left(\frac{xL}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_{-L}^{0} - \frac{L}{n\pi}\int_{-L}^{0} \sin\left(\frac{n\pi x}{L}\right)\,dx \right)$$

$$+\frac{3}{L}\left(\frac{xL}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_{0}^{L} - \frac{L}{n\pi}\int_{0}^{L} \sin\left(\frac{n\pi x}{L}\right)\,dx \right)$$

$$A_n = -\frac{2}{L}\left(\left(\frac{L}{n\pi}\right)^2 \cos\left(\frac{n\pi x}{L}\right)\Big|_{-L}^{0} \right) + \frac{3}{L}\left(\left(\frac{L}{n\pi}\right)^2 \cos\left(\frac{n\pi x}{L}\right)\Big|_{0}^{L} \right)$$

$$A_n = -\frac{2}{L}\left(\left(\frac{L}{n\pi}\right)^2 - \left(\frac{L}{n\pi}\right)^2 (-1)^n \right) + \frac{3}{L}\left(\left(\frac{L}{n\pi}\right)^2 (-1)^n - \left(\frac{L}{n\pi}\right)^2 \right)$$

$$A_n = \frac{5L}{n^2\pi^2}((-1)^n - 1)$$

For B_n we get

$$B_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

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$$B_n = \frac{1}{L}\left(\int_{-L}^{0} -2x \sin\left(\frac{n\pi x}{L}\right)\, dx + \int_{0}^{L} 3x \sin\left(\frac{n\pi x}{L}\right)\, dx \right)$$

$$B_n = -\frac{2}{L}\left(\int_{-L}^{0} x \sin\left(\frac{n\pi x}{L}\right)\, dx \right) + \frac{3}{L}\left(\int_{0}^{L} x \sin\left(\frac{n\pi x}{L}\right)\, dx \right)$$

Use integration by parts with $u = x$, $du = dx$, $dv = \sin(n\pi x/L)\, dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$B_n = -\frac{2}{L}\left(-\frac{xL}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_0^L + \int_0^L \frac{L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\, dx \right)$$

$$+\frac{3}{L}\left(-\frac{xL}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_0^L + \int_0^L \frac{L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\, dx \right)$$

$$B_n = -\frac{2}{L}\left(-\frac{L^2}{n\pi}(-1)^n + \left(\frac{L}{n\pi}\right)^2 \sin\left(\frac{n\pi x}{L}\right)\Big|_0^L \right)$$

$$+\frac{3}{L}\left(-\frac{L^2}{n\pi}(-1)^n + \left(\frac{L}{n\pi}\right)^2 \sin\left(\frac{n\pi x}{L}\right)\Big|_0^L \right)$$

$$B_n = -\frac{2}{L}\left(-\frac{L^2}{n\pi}(-1)^n \right) + \frac{3}{L}\left(-\frac{L^2}{n\pi}(-1)^n \right)$$

$$B_n = -\frac{L}{n\pi}(-1)^n$$

Then the Fourier series representation of the piecewise function on $-L \le x \le L$ is

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$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{5}{4}L + \sum_{n=1}^{\infty} \frac{5L}{n^2\pi^2}((-1)^n - 1)\cos\left(\frac{n\pi x}{L}\right) - \sum_{n=1}^{\infty} \frac{L}{n\pi}(-1)^n \sin\left(\frac{n\pi x}{L}\right)$$

■ 2. Find the Fourier series representation of the piecewise function on $-L \leq x \leq L$.

$$f(x) = \begin{cases} 4 - 2x & -L \leq x < 0 \\ 2x + 4 & 0 \leq x \leq L \end{cases}$$

Solution:

Since $f(x)$ is even,

$$f(-x) = \begin{cases} 2x + 4 & 0 \leq x \leq L \\ 4 - 2x & -L \leq x < 0 \end{cases}$$

$$f(-x) = \begin{cases} 4 - 2x & -L \leq x < 0 \\ 2x + 4 & 0 \leq x \leq L \end{cases}$$

$$f(-x) = f(x)$$

we know $B_n = 0$. For A_0 we get

$$A_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\, dx$$

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$$A_0 = \frac{1}{2L}\left(\int_{-L}^{0} -2x+4\ dx + \int_{0}^{L} 4+2x\ dx\right)$$

$$A_0 = \frac{1}{2L}\left((-x^2+4x)\Big|_{-L}^{0} + (4x+x^2)\Big|_{0}^{L}\right)$$

$$A_0 = \frac{1}{2L}(L^2+4L+4L+L^2)$$

$$A_0 = \frac{1}{2L}(8L+2L^2)$$

$$A_0 = 4+L$$

For A_n, we get

$$A_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)\ dx$$

$$A_n = \frac{1}{L}\int_{-L}^{0} (4-2x)\cos\left(\frac{n\pi x}{L}\right)\ dx + \frac{1}{L}\int_{0}^{L} (4+2x)\cos\left(\frac{n\pi x}{L}\right)\ dx$$

Use integration by parts with $u = 4-2x$, $du = -2\ dx$, $dv = \cos(n\pi x/L)\ dx$, and $v = (L/n\pi)\sin(n\pi x/L)$ in the first integral and with $u = 4+2x$, $du = 2\ dx$, $dv = \cos(n\pi x/L)\ dx$, and $v = (L/n\pi)\sin(n\pi x/L)$ in the second integral.

$$A_n = \frac{1}{L}\left((4-2x)\frac{L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_{-L}^{0} + \frac{2L}{n\pi}\int_{-L}^{0}\sin\left(\frac{n\pi x}{L}\right)\ dx\right.$$

$$\left.+(4+2x)\frac{L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_{0}^{L} - \frac{2L}{n\pi}\int_{0}^{L}\sin\left(\frac{n\pi x}{L}\right)\ dx\right)$$

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$$A_n = \frac{1}{L}\left(\frac{2L}{n\pi}\left(-\frac{L}{n\pi}\right)\cos\left(\frac{n\pi x}{L}\right)\Big|_{-L}^{0} - \frac{2L}{n\pi}\left(-\frac{L}{n\pi}\right)\cos\left(\frac{n\pi x}{L}\right)\Big|_{0}^{L}\right)$$

$$A_n = \frac{1}{L}\left(-\frac{2L^2}{(n\pi)^2}(1-(-1)^n) + \frac{2L^2}{(n\pi)^2}((-1)^n - 1)\right)$$

$$A_n = \frac{4L}{n^2\pi^2}((-1)^n - 1)$$

Then the Fourier series representation of the piecewise function on $-L \leq x \leq L$ is

$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = 4 + L + \sum_{n=1}^{\infty} \frac{4L}{n^2\pi^2}((-1)^n - 1)\cos\left(\frac{n\pi x}{L}\right)$$

■ 3. Find the Fourier series representation of the piecewise function on $-L \leq x \leq L$.

$$f(x) = \begin{cases} 1 & -L \leq x < 0 \\ 3\left(x + \frac{L}{2}\right) + 1 & 0 \leq x \leq L \end{cases}$$

Solution:

For A_0, we get

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$$A_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\ dx$$

$$A_0 = \frac{1}{2L}\left(\int_{-L}^{0} 1\ dx + \int_{0}^{L}\left(3\left(x+\frac{L}{2}\right)+1\right)\ dx\right)$$

$$A_0 = \frac{1}{2L}\left(x\Big|_{-L}^{0} + \left(\frac{3}{2}x^2+\frac{3L}{2}x+x\right)\Big|_{0}^{L}\right)$$

$$A_0 = \frac{1}{2L}\left(L+\left(\frac{3L^2}{2}+\frac{3L^2}{2}+L\right)\right)$$

$$A_0 = \frac{1}{2L}(2L+3L^2)$$

$$A_0 = 1+\frac{3L}{2}$$

For A_n, we get

$$A_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)\ dx$$

$$A_n = \frac{1}{L}\left(\int_{-L}^{0}\cos\left(\frac{n\pi x}{L}\right)\ dx + \int_{0}^{L}\left(3\left(x+\frac{L}{2}\right)+1\right)\cos\left(\frac{n\pi x}{L}\right)\ dx\right)$$

$$A_n = \frac{1}{L}\left(\frac{L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_{-L}^{0} + \int_{0}^{L}\left(3\left(x+\frac{L}{2}\right)+1\right)\cos\left(\frac{n\pi x}{L}\right)\ dx\right)$$

Use integration by parts with $u = 3(x+L/2)+1$, $du = 3\ dx$, $dv = \cos(n\pi x/L)\ dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

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$$A_n = \frac{1}{L}\left(\frac{L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_{-L}^{0} + \left(3\left(x+\frac{L}{2}\right)+1\right)\frac{L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_{0}^{L} \right.$$

$$\left. -\frac{3L}{n\pi}\int_0^L \sin\left(\frac{n\pi x}{L}\right)\,dx \right)$$

$$A_n = \frac{1}{L}\left(\frac{L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_{-L}^{0} + \left(3\left(x+\frac{L}{2}\right)+1\right)\frac{L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_{0}^{L} \right.$$

$$\left. +\frac{3L^2}{(n\pi)^2}\cos\left(\frac{n\pi x}{L}\right)\Big|_{0}^{L} \right)$$

$$A_n = \frac{1}{L}\left(-\frac{L}{n\pi}\sin(-n\pi) + \left(3\left(L+\frac{L}{2}\right)+1\right)\frac{L}{n\pi}\sin(n\pi) \right.$$

$$\left. +\frac{3L^2}{(n\pi)^2}\cos(n\pi) - \frac{3L^2}{(n\pi)^2} \right)$$

$$A_n = \frac{1}{L}\left(\frac{L}{n\pi}\sin(n\pi) + \frac{3L^2}{(n\pi)^2}(-1)^n - \frac{3L^2}{(n\pi)^2} \right)$$

$$A_n = \frac{3L}{(n\pi)^2}((-1)^n - 1)$$

For B_n, we get

$$B_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

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$$B_n = \frac{1}{L}\left(\int_{-L}^{0} \sin\left(\frac{n\pi x}{L}\right)\, dx + \int_{0}^{L} \left(3\left(x+\frac{L}{2}\right)+1\right) \sin\left(\frac{n\pi x}{L}\right)\, dx \right)$$

Use integration by parts with $u = 3(x + L/2) + 1$, $du = 3\ dx$, $dv = \sin(n\pi x/L)\ dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$B_n = \frac{1}{L}\left(-\frac{L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_{-L}^{0} - \left(3\left(x+\frac{L}{2}\right)+1\right)\left(\frac{L}{n\pi}\right)\cos\left(\frac{n\pi x}{L}\right)\Big|_{0}^{L} \right.$$

$$\left. + \frac{3L}{n\pi}\int_{0}^{L} \cos\left(\frac{n\pi x}{L}\right)\, dx \right)$$

$$B_n = \frac{1}{L}\left(-\frac{L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_{-L}^{0} - \left(3\left(x+\frac{L}{2}\right)+1\right)\left(\frac{L}{n\pi}\right)\cos\left(\frac{n\pi x}{L}\right)\Big|_{0}^{L} \right.$$

$$\left. + \frac{3L^2}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\Big|_{0}^{L} \right)$$

$$B_n = \frac{1}{L}\left(-\frac{L}{n\pi} + \frac{L}{n\pi}\cos(-n\pi) - \left(3\left(L+\frac{L}{2}\right)+1\right)\left(\frac{L}{n\pi}\right)\cos(n\pi) \right.$$

$$\left. + \left(3\left(\frac{L}{2}\right)+1\right)\left(\frac{L}{n\pi}\right) + \frac{3L^2}{(n\pi)^2}\sin(n\pi) \right)$$

$$B_n = \frac{1}{L}\left(-\frac{L}{n\pi} + \frac{L}{n\pi}(-1)^n - \left(\frac{9L}{2}+1\right)\left(\frac{L}{n\pi}\right)(-1)^n + \left(\frac{3L}{2}+1\right)\left(\frac{L}{n\pi}\right) \right)$$

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$$B_n = \left(-\frac{1}{n\pi} + \frac{1}{n\pi}(-1)^n - \frac{1}{n\pi}\left(\frac{9L}{2} + 1\right)(-1)^n + \frac{1}{n\pi}\left(\frac{3L}{2} + 1\right)\right)$$

$$B_n = \frac{1}{n\pi}\left((-1)^n - \left(\frac{9L}{2} + 1\right)(-1)^n + \frac{3L}{2}\right)$$

$$B_n = \frac{3L}{2n\pi}(1 - 3(-1)^n)$$

Then the Fourier series representation of the piecewise function on $-L \leq x \leq L$ is

$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = 1 + \frac{3L}{2} + \sum_{n=1}^{\infty} \frac{3L}{(n\pi)^2}((-1)^n - 1)\cos\left(\frac{n\pi x}{L}\right)$$

$$+ \sum_{n=1}^{\infty} \frac{3L}{2n\pi}(1 - 3(-1)^n)\sin\left(\frac{n\pi x}{L}\right)$$

■ 4. Find the Fourier series representation of the piecewise function on $-\pi \leq x \leq \pi$.

$$f(x) = \begin{cases} (x+1)^2 & -\pi \leq x < 0 \\ 2 - (x-1)^2 & 0 \leq x \leq \pi \end{cases}$$

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Solution:

For A_0, we get

$$A_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\ dx$$

$$A_0 = \frac{1}{2\pi}\left(\int_{-\pi}^{0} (x+1)^2\ dx + \int_{0}^{\pi} (2-(x-1)^2)\ dx\right)$$

$$A_0 = \frac{1}{2\pi}\left(\left(\frac{x^3}{3}+x^2+x\right)\Big|_{-\pi}^{0} + \left(2x-\left(\frac{x^3}{3}-x^2+x\right)\right)\Big|_{0}^{\pi}\right)$$

$$A_0 = \frac{1}{2\pi}\left(\left(\frac{x^3}{3}+x^2+x\right)\Big|_{-\pi}^{0} + \left(-\frac{x^3}{3}+x^2+x\right)\Big|_{0}^{\pi}\right)$$

$$A_0 = \frac{1}{2\pi}\left(\frac{\pi^3}{3}-\pi^2+\pi-\frac{\pi^3}{3}+\pi^2+\pi\right)$$

$$A_0 = 1$$

For A_n, we get

$$A_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\ dx$$

$$A_n = \frac{1}{\pi}\left(\int_{-\pi}^{0} (x+1)^2\cos(nx)\ dx + \int_{0}^{\pi} (2-(x-1)^2)\cos(nx)\ dx\right)$$

Since

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$$\int_{-\pi}^{0} (x+1)^2 \cos(nx)\ dx = \int_{0}^{\pi} (1-x)^2 \cos(nx)\ dx$$

we get

$$A_n = \frac{1}{\pi}\left(\int_{0}^{\pi} (1-x)^2 \cos(nx)\ dx \right.$$

$$\left. + \int_{0}^{\pi} 2\cos(nx)\ dx \ - \int_{0}^{\pi} (x-1)^2 \cos(nx)\ dx \right)$$

$$A_n = \frac{1}{\pi} \int_{0}^{\pi} 2\cos(nx)\ dx$$

$$A_n = \frac{1}{\pi} \cdot \frac{2}{n} \sin(nx) \Big|_{0}^{\pi} = 0$$

For B_n, we get

$$B_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx)\ dx$$

$$B_n = \frac{1}{\pi} \left(\int_{-\pi}^{0} (x+1)^2 \sin(nx)\ dx + \int_{0}^{\pi} (2-(x-1)^2) \sin(nx)\ dx \right)$$

Since

$$\int_{-\pi}^{0} (x+1)^2 \sin(nx)\ dx = \int_{0}^{\pi} -(x-1)^2 \sin(nx)\ dx$$

we get

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$$B_n = \frac{1}{\pi}\left(\int_0^{\pi} -(x-1)^2 \sin(nx)\ dx + \int_0^{\pi} (2-(x-1)^2)\sin(nx)\ dx \right)$$

$$B_n = \frac{1}{\pi}\int_0^{\pi} (2-2(x-1)^2)\sin(nx)\ dx$$

$$B_n = -\frac{1}{\pi}\int_0^{\pi} (2x^2-4x)\sin(nx)\ dx$$

Use integration by parts with $u = 2x^2 - 4x$, $du = (4x-4)\ dx$, $dv = \sin(nx)\ dx$, and $v = -(1/n)\cos(nx)$.

$$B_n = -\frac{1}{\pi}\left(-\frac{1}{n}(2x^2-4x)\cos(nx)\Big|_0^{\pi} + \frac{4}{n}\int_0^{\pi} (x-1)\cos(nx)\ dx \right)$$

$$B_n = -\frac{1}{\pi}\left(-\frac{1}{n}(2\pi^2-4\pi)(-1)^n + \frac{4}{n}\int_0^{\pi} (x-1)\cos(nx)\ dx \right)$$

Use integration by parts with $u = x - 1$, $du =\ dx$, $dv = \cos(nx)\ dx$, and $v = (1/n)\sin(nx)$.

$$B_n = -\frac{1}{\pi}\left(-\frac{2\pi}{n}(\pi-2)(-1)^n + \frac{4}{n}\left(\frac{(x-1)}{n}\sin(nx)\Big|_0^{\pi} - \frac{1}{n}\int_0^{\pi} \sin(nx)\ dx \right) \right)$$

$$B_n = -\frac{1}{\pi}\left(-\frac{2\pi}{n}(\pi-2)(-1)^n + \frac{4}{n}\left(0 + \frac{1}{n^2}\cos(nx)\Big|_0^{\pi} \right) \right)$$

$$B_n = \frac{2(\pi-2)}{n}(-1^n) - \frac{4}{\pi n^3}\cos(nx)\Big|_0^{\pi}$$

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$$B_n = \frac{2(\pi - 2)}{n}(-1^n) - \frac{4}{\pi n^3}((-1)^n - 1)$$

Then the Fourier series representation of the piecewise function on $-\pi \leq x \leq \pi$ is

$$f(x) = 1 + \sum_{n=1}^{\infty} \left(\left(\frac{2(\pi - 2)}{n} - \frac{4}{\pi n^3} \right)(-1)^n + \frac{4}{\pi n^3} \right) \sin(nx)$$

■ 5. Find the Fourier series representation of the piecewise function.

$$f(x) = \begin{cases} 3x + 6 & -L \leq x < -1 \\ -3x & -1 \leq x \leq 1 \\ 3x - 6 & 1 < x \leq L \end{cases}$$

Solution:

The function $f(x)$ is odd,

$$f(-x) = \begin{cases} 3(-x) + 6 & -L \leq -x < -1 \\ -3(-x) & -1 \leq -x \leq 1 \\ 3(-x) - 6 & 1 < -x \leq L \end{cases}$$

$$f(-x) = \begin{cases} -3x - 6 & -L \leq x < -1 \\ 3x & -1 \leq x \leq 1 \\ -3x + 6 & 1 < x \leq L \end{cases}$$

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$$f(-x) = \begin{cases} -(3x+6) & -L \le x < -1 \\ -(-3x) & -1 \le x \le 1 \\ -(3x-6) & 1 < x \le L \end{cases}$$

$$f(-x) = -f(x)$$

so we know $A_0 = A_n = 0$. For B_n, we get

$$B_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{1}{L}\left(\int_{-L}^{-1}(3x+6)\sin\left(\frac{n\pi x}{L}\right)\,dx + \int_{-1}^{1} -3x\sin\left(\frac{n\pi x}{L}\right)\,dx\right.$$

$$\left. + \int_{1}^{L}(3x-6)\sin\left(\frac{n\pi x}{L}\right)\,dx\right)$$

Use integration by parts with $u = 3x+6$, $du = 3\,dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$ for the first integral.

$$B_n = \frac{1}{L}\left((3x+6)\left(-\frac{L}{n\pi}\right)\cos\left(\frac{n\pi x}{L}\right)\Big|_{-L}^{-1} + \frac{3L}{n\pi}\int_{-L}^{1}\cos\left(\frac{n\pi x}{L}\right)\,dx\right.$$

$$\left. + \int_{-1}^{1} -3x\sin\left(\frac{n\pi x}{L}\right)\,dx + \int_{1}^{L}(3x-6)\sin\left(\frac{n\pi x}{L}\right)\,dx\right)$$

$$B_n = \frac{1}{L}\left(-\frac{3L}{n\pi}\cos\left(\frac{n\pi}{L}\right)\frac{L(-3L+6)}{n\pi}(-1)^n + \frac{3L^2}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\Big|_{-L}^{-1}\right.$$

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$$+\int_{-1}^{1} -3x\sin\left(\frac{n\pi x}{L}\right)\,dx+\int_{1}^{L}(3x-6)\sin\left(\frac{n\pi x}{L}\right)\,dx\Bigg)$$

$$B_n=\frac{1}{L}\Bigg(-\frac{3L}{n\pi}\cos\left(\frac{n\pi}{L}\right)+\frac{L(-3L+6)}{n\pi}(-1)^n-\frac{3L^2}{(n\pi)^2}\sin\left(\frac{n\pi}{L}\right)$$

$$+\int_{-1}^{1} -3x\sin\left(\frac{n\pi x}{L}\right)\,dx+\int_{1}^{L}(3x-6)\sin\left(\frac{n\pi x}{L}\right)\,dx\Bigg)$$

Use integration by parts with $u=-3x$, $du=-3\,dx$, $dv=\sin(n\pi x/L)\,dx$, and $v=-(L/n\pi)\cos(n\pi x/L)$ for the second integral.

$$B_n=\frac{1}{L}\Bigg(-\frac{3L}{n\pi}\cos\left(\frac{n\pi}{L}\right)+\frac{L(-3L+6)}{n\pi}(-1)^n-\frac{3L^2}{(n\pi)^2}\sin\left(\frac{n\pi}{L}\right)$$

$$+\frac{3xL}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Bigg|_{-1}^{1}-\frac{3L}{n\pi}\int_{-1}^{1}\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$+\int_{1}^{L}(3x-6)\sin\left(\frac{n\pi x}{L}\right)\,dx\Bigg)$$

$$B_n=\frac{1}{L}\Bigg(-\frac{3L}{n\pi}\cos\left(\frac{n\pi}{L}\right)+\frac{L(-3L+6)}{n\pi}(-1)^n-\frac{3L^2}{(n\pi)^2}\sin\left(\frac{n\pi}{L}\right)$$

$$+\frac{3L}{n\pi}\cos\left(\frac{n\pi}{L}\right)+\frac{3L}{n\pi}\cos\left(\frac{n\pi}{L}\right)-\frac{3L}{n\pi}\int_{-1}^{1}\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$+\int_{1}^{L}(3x-6)\sin\left(\frac{n\pi x}{L}\right)\,dx\Bigg)$$

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$$B_n = \frac{1}{L}\left(\frac{L(-3L+6)}{n\pi}(-1)^n - \frac{3L^2}{(n\pi)^2}\sin\left(\frac{n\pi}{L}\right) \right.$$

$$\left. + \frac{3L}{n\pi}\cos\left(\frac{n\pi}{L}\right) - \frac{3L^2}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\Big|_{-1}^{1} + \int_1^L (3x-6)\sin\left(\frac{n\pi x}{L}\right)\,dx \right)$$

$$B_n = \frac{1}{L}\left(\frac{L(-3L+6)}{n\pi}(-1)^n - \frac{3L^2}{(n\pi)^2}\sin\left(\frac{n\pi}{L}\right) \right.$$

$$\left. + \frac{3L}{n\pi}\cos\left(\frac{n\pi}{L}\right) - \frac{6L^2}{(n\pi)^2}\sin\left(\frac{n\pi}{L}\right) + \int_1^L (3x-6)\sin\left(\frac{n\pi x}{L}\right)\,dx \right)$$

Use integration by parts with $u = 3x - 6$, $du = 3\,dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$ for the second integral.

$$B_n = \frac{1}{L}\left(\frac{L(-3L+6)}{n\pi}(-1)^n - \frac{9L^2}{(n\pi)^2}\sin\left(\frac{n\pi}{L}\right) + \frac{3L}{n\pi}\cos\left(\frac{n\pi}{L}\right) \right.$$

$$\left. + (3x-6)\left(-\frac{L}{n\pi}\right)\cos\left(\frac{n\pi x}{L}\right)\Big|_1^L + \frac{3L}{n\pi}\int_1^L \cos\left(\frac{n\pi x}{L}\right)\,dx \right)$$

$$B_n = \frac{1}{L}\left(\frac{L(-3L+6)}{n\pi}(-1)^n - \frac{9L^2}{(n\pi)^2}\sin\left(\frac{n\pi}{L}\right) + \frac{3L}{n\pi}\cos\left(\frac{n\pi}{L}\right) \right.$$

$$\left. - (3L-6)\frac{L}{n\pi}(-1)^n - \frac{3L}{n\pi}\cos\left(\frac{n\pi}{L}\right) + \frac{3L^2}{n^2\pi^2}\sin\left(\frac{n\pi x}{L}\right)\Big|_1^L \right)$$

$$B_n = -\frac{12L}{n^2\pi^2}\sin\left(\frac{n\pi}{L}\right) - \frac{6(L-2)}{n\pi}(-1)^n$$

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Then the Fourier series representation is

$$f(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \sum_{n=1}^{\infty} \left(-\frac{12L}{n^2\pi^2}\sin\left(\frac{n\pi}{L}\right) - \frac{6(L-2)}{n\pi}(-1)^n\right)\sin\left(\frac{n\pi x}{L}\right)$$

■ 6. Find the Fourier series representation of the piecewise function.

$$f(x) = \begin{cases} 2-2x & -L \le x < 0 \\ 4x+3 & 0 \le x \le L \end{cases}$$

Solution:

For A_0, we get

$$A_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\,dx$$

$$A_0 = \frac{1}{2L}\left(\int_{-L}^{0}(2-2x)\,dx + \int_{0}^{L}(4x+3)\,dx\right)$$

$$A_0 = \frac{1}{2L}\left(2x - x^2\Big|_{-L}^{0} + (2x^2+3x)\Big|_{0}^{L}\right)$$

$$A_0 = \frac{1}{2L}(2L + L^2 + 2L^2 + 3L)$$

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$$A_0 = \frac{1}{2}(3L + 5)$$

For A_n, we get

$$A_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = \frac{1}{L}\left(\int_{-L}^{0}(2 - 2x)\cos\left(\frac{n\pi x}{L}\right)\,dx + \int_{0}^{L}(4x + 3)\cos\left(\frac{n\pi x}{L}\right)\,dx\right)$$

Since

$$\int_{-L}^{0}(2 - 2x)\cos\left(\frac{n\pi x}{L}\right)\,dx = \int_{0}^{L}(2 + 2x)\cos\left(\frac{n\pi x}{L}\right)\,dx$$

we get

$$A_n = \frac{1}{L}\int_{0}^{L}(2 + 2x + 4x + 3)\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = \frac{1}{L}\int_{0}^{L}(6x + 5)\cos\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = 6x + 5$, $du = 6\,dx$, $dv = \cos(n\pi x/L)\,dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

$$A_n = \frac{1}{L}\left((6x + 5)\frac{L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_{0}^{L} - \frac{6L}{n\pi}\int_{0}^{L}\sin\left(\frac{n\pi x}{L}\right)\,dx\right)$$

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$$A_n = \frac{1}{L}\left(0 - \frac{6L}{n\pi}\int_0^L \sin\left(\frac{n\pi x}{L}\right)\,dx\right)$$

$$A_n = \frac{1}{L}\left(\frac{6L}{n\pi}\cdot\frac{L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_0^L\right)$$

$$A_n = \frac{6L}{n^2\pi^2}((-1)^n - 1)$$

For B_n, we get

$$B_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{1}{L}\left(\int_{-L}^{0}(2-2x)\sin\left(\frac{n\pi x}{L}\right)\,dx + \int_0^L (4x+3)\sin\left(\frac{n\pi x}{L}\right)\,dx\right)$$

Since

$$\int_{-L}^{0}(2-2x)\sin\left(\frac{n\pi x}{L}\right)\,dx = \int_0^L(-2-2x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

we get

$$B_n = \frac{1}{L}\int_{-L}^{0}(-2-2x+4x+3)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{1}{L}\int_{-L}^{0}(2x+1)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

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Use integration by parts with $u = 2x + 1$, $du = 2\ dx$, $dv = \sin(n\pi x/L)\ dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$B_n = \frac{1}{L}\left(-(2x+1)\frac{L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_0^L + \frac{2L}{n\pi}\int_0^L \cos\left(\frac{n\pi x}{L}\right)\ dx\right)$$

$$B_n = \frac{1}{L}\left(-\frac{L}{n\pi}((2L+1)(-1)^n - 1) + \frac{2L^2}{n^2\pi^2}\ \sin\left(\frac{n\pi x}{L}\right)\Big|_0^L\right)$$

$$B_n = \frac{1}{L}\left(-\frac{L}{n\pi}((2L+1)(-1)^n - 1) + 0\right)$$

$$B_n = \frac{1-(2L+1)(-1)^n}{n\pi}$$

Then the Fourier series representation of the piecewise function is

$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{1}{2}(3L+5) + \sum_{n=1}^{\infty}\frac{6L}{n^2\pi^2}((-1)^n - 1)\cos\left(\frac{n\pi x}{L}\right)$$

$$+ \sum_{n=1}^{\infty}\frac{1-(2L+1)(-1)^n}{n\pi}\sin\left(\frac{n\pi x}{L}\right)$$

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CONVERGENCE OF A FOURIER SERIES

■ 1. Given the function $f(x)$ and its Fourier series representation, say whether or not the Fourier series converges to $f(x)$.

$$f(x) = \begin{cases} -1 & -L \leq x < 0 \\ x^2 & 0 \leq x \leq L \end{cases}$$

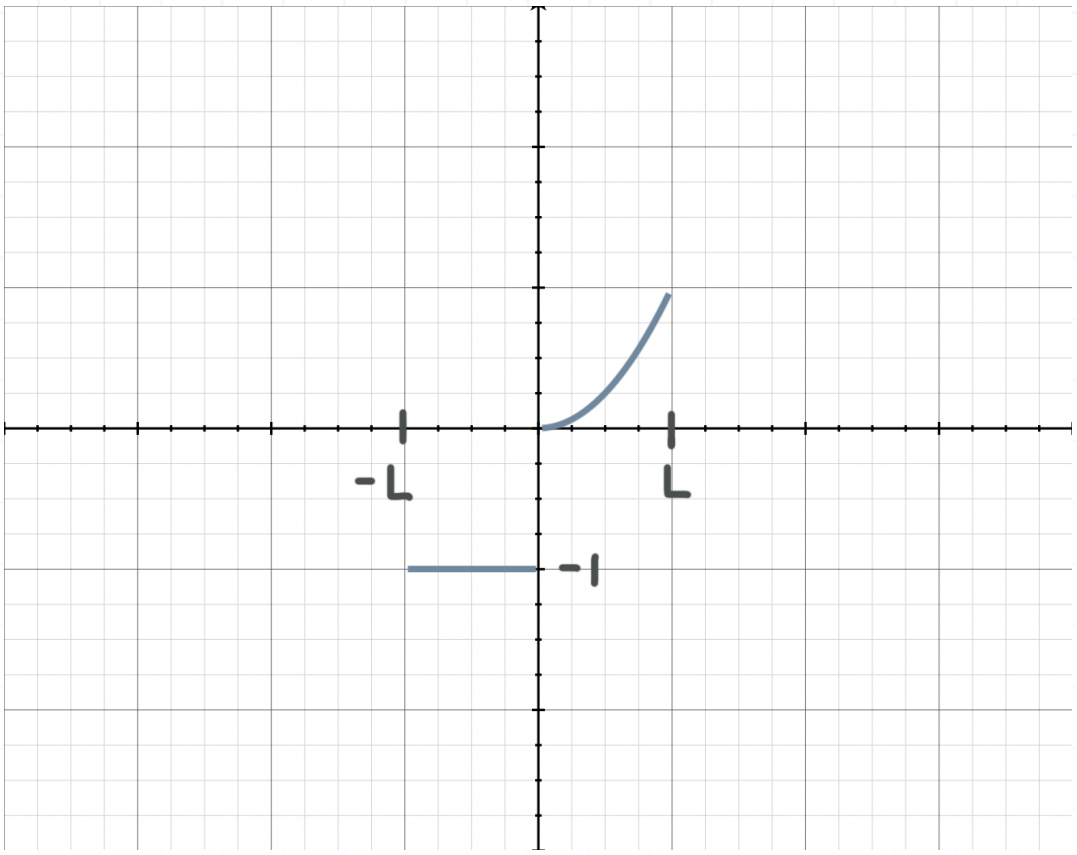
$$f(x) = -\frac{1}{2} + \frac{L^2}{6} + \frac{2L^2}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos\left(\frac{n\pi x}{L}\right)$$

$$+ \sum_{n=1}^{\infty} \frac{1}{n^3\pi^3} \left[(2L^2 - L^2n^2\pi^2 - n^2\pi^2)(-1)^n + n^2\pi^2 - 2L^2\right] \sin\left(\frac{n\pi x}{L}\right)$$

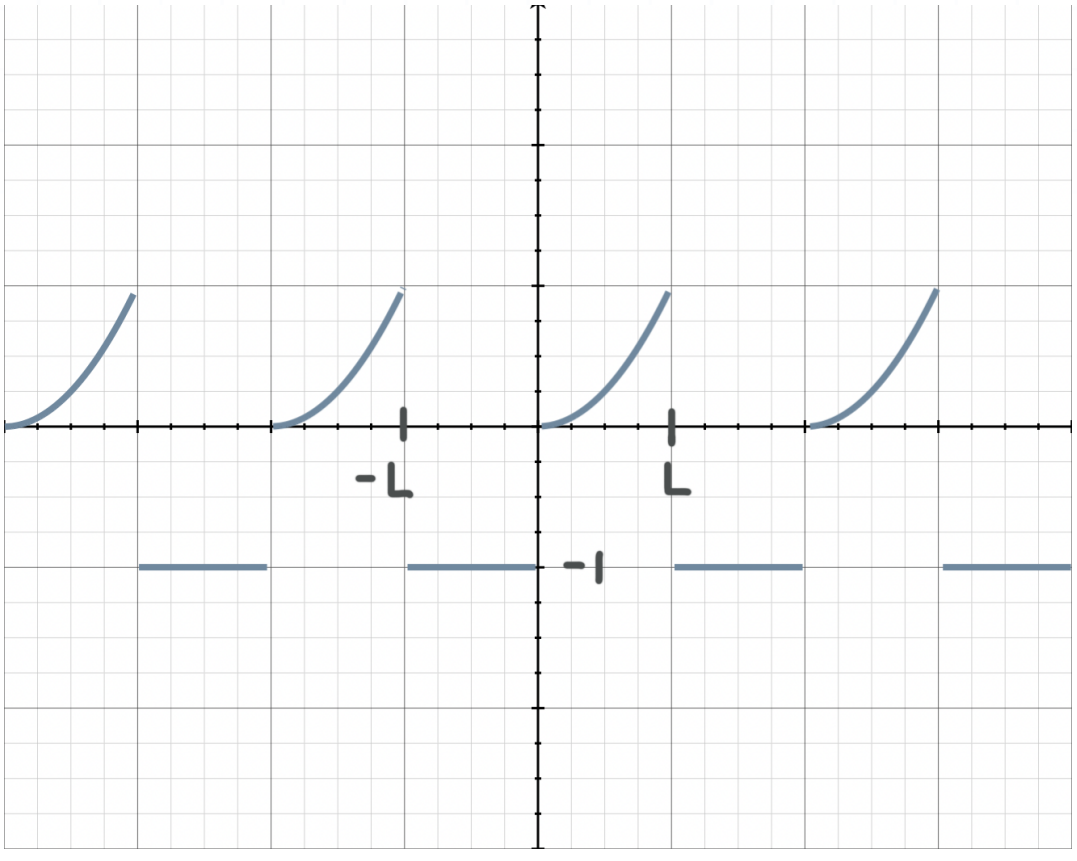
Solution:

A sketch of the function $f(x)$ is

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The sketch of its periodic extension, $g(x)$, is



From the graph of the periodic extension, we can see that $g(x)$ has a jump discontinuity at $x = 0$, within the interval $-L \leq x \leq L$, as well as jump discontinuities at $-L$ and L.

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Because the function is piecewise smooth on the individual intervals $-L \leq x \leq 0$ and $0 \leq x \leq L$, the function $f(x)$ and the periodic extension $g(x)$ are both continuous there, so the Fourier series representation will converge to the periodic extension $g(x)$, and therefore also to the original function $f(x)$. Because of the jump discontinuity at $x = 0$, the Fourier series representation at that point will converge to

$$\frac{\lim_{x \to 0^-} g(x) + \lim_{x \to 0^+} g(x)}{2}$$

$$\frac{-1 + 0}{2}$$

$$-\frac{1}{2}$$

At $x = -L$, the Fourier series representation will converge to

$$\frac{\lim_{x \to -L^-} g(x) + \lim_{x \to -L^+} g(x)}{2}$$

$$\frac{L^2 + (-1)}{2}$$

$$\frac{L^2 - 1}{2}$$

and at $x = L$ the Fourier series representation will converge to

$$\frac{\lim_{x \to L^-} g(x) + \lim_{x \to L^+} g(x)}{2}$$

$$\frac{L^2 - 1}{2}$$

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The Fourier series representation converges to $f(x)$, since $f(x)$ is piecewise smooth and has a finite number of jump discontinuities on any interval with finite endpoints.

■ 2. Given the function $f(x)$ and its Fourier series representation, say whether or not the Fourier series converges to $f(x)$.

$$f(x) = \begin{cases} -x^2 + 1 & -L \leq x < 0 \\ L - x & 0 \leq x \leq L \end{cases}$$

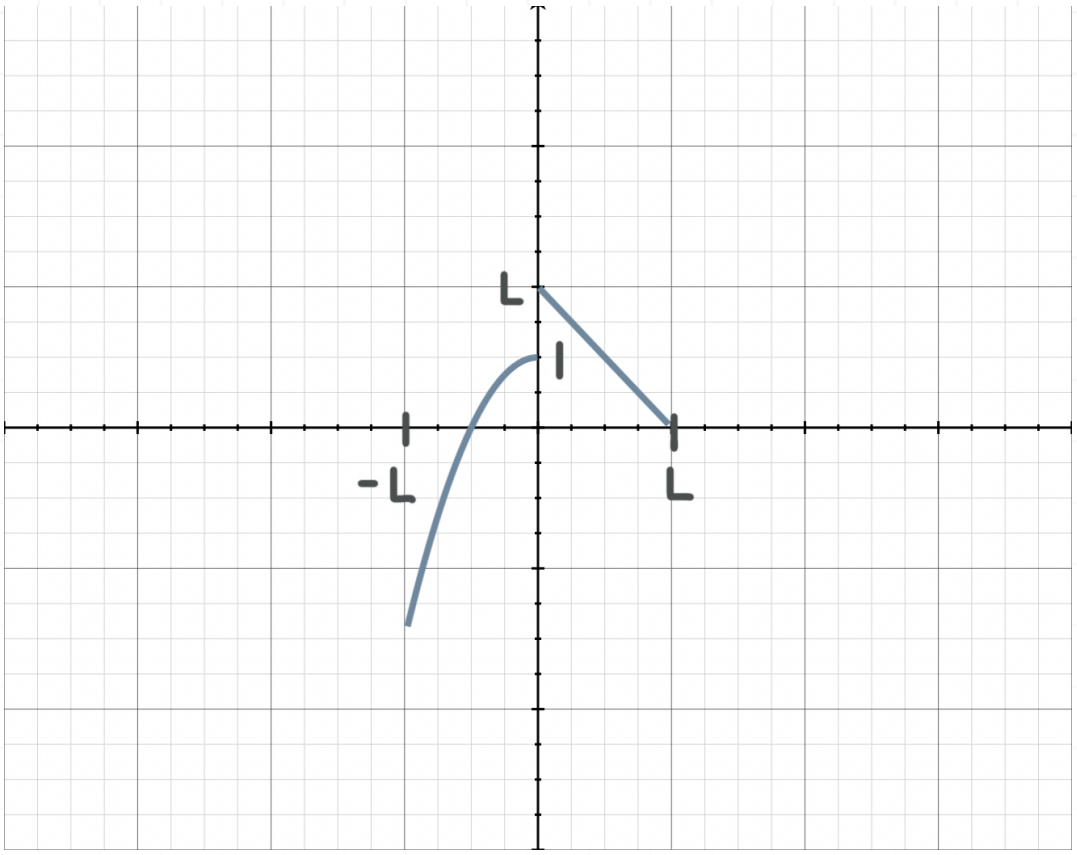
$$f(x) = \frac{6 + 3L - 2L^2}{12} + \frac{L}{\pi^2} \sum_{n=1}^{\infty} \frac{1 - (2L+1)(-1)^n}{n^2} \cos\left(\frac{n\pi x}{L}\right)$$

$$+ \frac{1}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{n^3} \left[(n^2\pi^2(1 - L^2) + 2L^2)(-1)^n + n^2\pi^2(L - 1) - 2L^2\right] \sin\left(\frac{n\pi x}{L}\right)$$

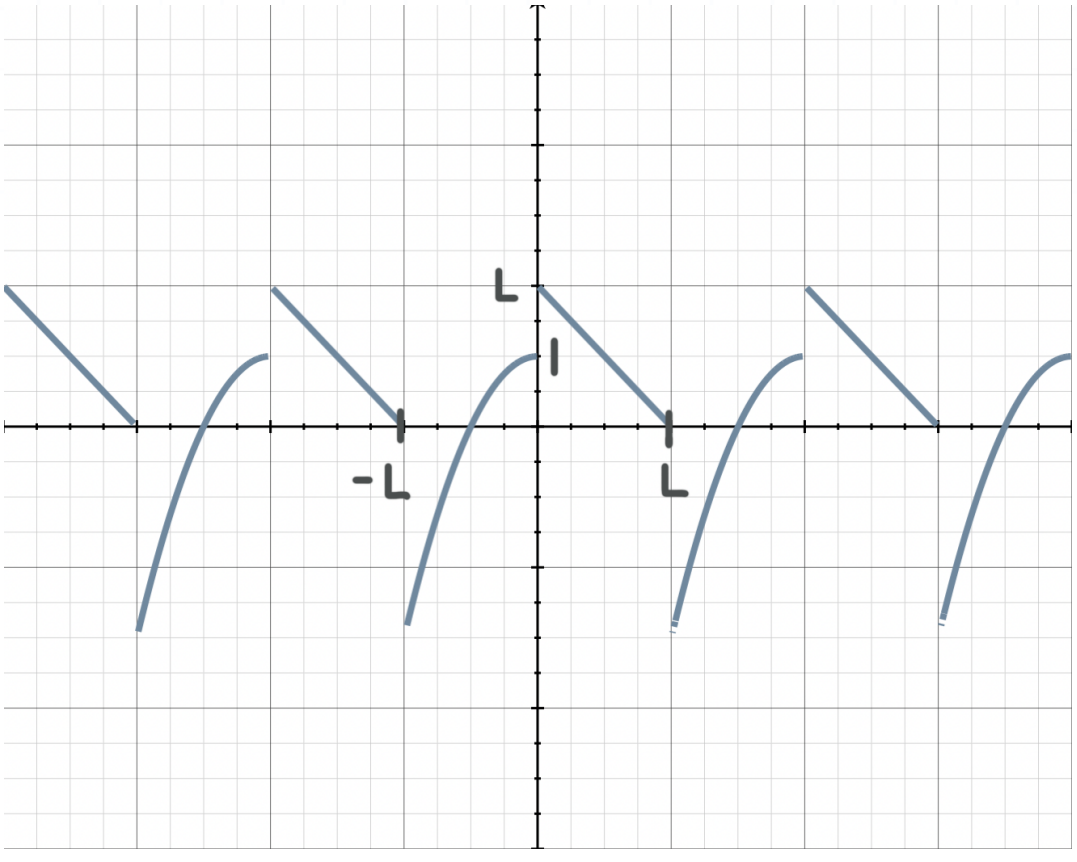
Solution:

A sketch of the function $f(x)$ is

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The sketch of its periodic extension, $g(x)$, is



From the graph of the periodic extension, we can see that $g(x)$ has a jump discontinuity at $x = 0$, within the interval $-L \leq x \leq L$, as well as jump discontinuities at $-L$ and L, assuming $L \neq 1$.

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Because the function is piecewise smooth on the individual intervals $-L \leq x \leq 0$ and $0 \leq x \leq L$, the function $f(x)$ and the periodic extension $g(x)$ are both continuous there, so the Fourier series representation will converge to the periodic extension $g(x)$, and therefore also to the original function $f(x)$. Because of the jump discontinuity at $x = 0$, the Fourier series representation at that point will converge to

$$\frac{\lim_{x \to 0^-} g(x) + \lim_{x \to 0^+} g(x)}{2}$$

$$\frac{1 + L}{2}$$

At $x = -L$, the Fourier series representation will converge to

$$\frac{\lim_{x \to -L^-} g(x) + \lim_{x \to -L^+} g(x)}{2}$$

$$\frac{0 + (-L^2 + 1)}{2}$$

$$\frac{1 - L^2}{2}$$

and at $x = L$ the Fourier series representation will converge to

$$\frac{\lim_{x \to L^-} g(x) + \lim_{x \to L^+} g(x)}{2}$$

$$\frac{0 + (-L^2 + 1)}{2}$$

$$\frac{1 - L^2}{2}$$

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The Fourier series representation converges to $f(x)$, since $f(x)$ is piecewise smooth and has a finite number of jump discontinuities on any interval with finite endpoints.

■ 3. Given the function $f(x)$ and its Fourier series representation, say whether or not the Fourier series converges to $f(x)$, assuming $L > 1$.

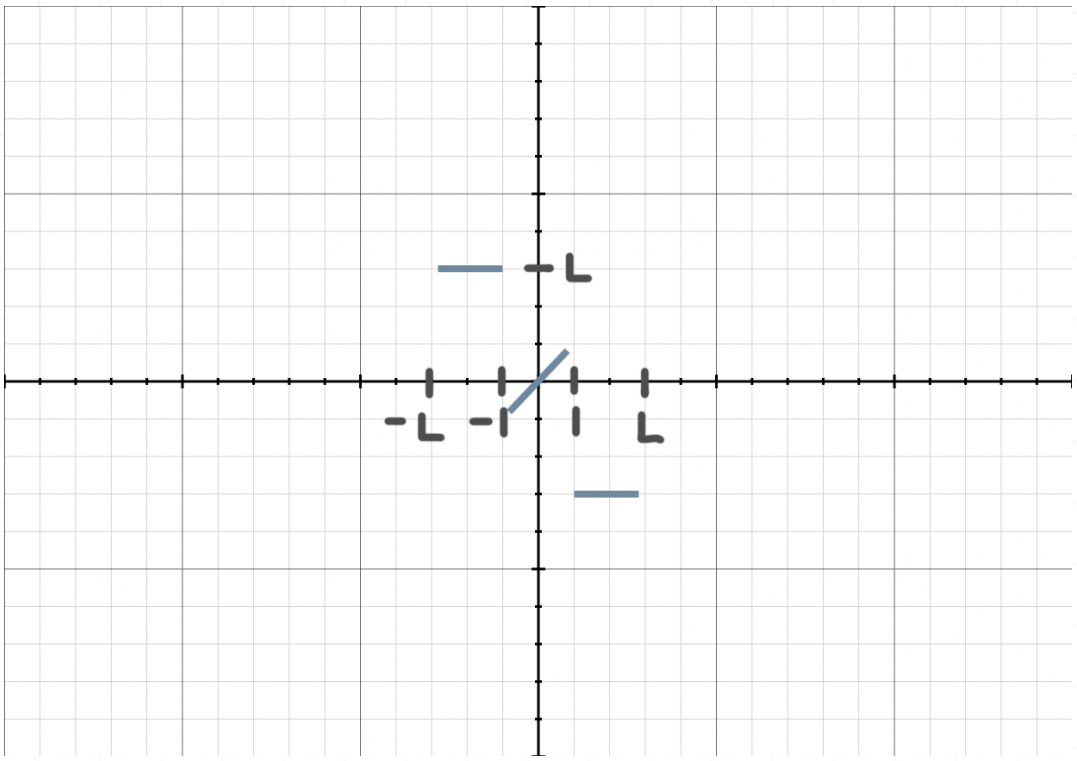
$$f(x) = \begin{cases} L & -L \leq x < -1 \\ x & -1 \leq x \leq 1 \\ -L & 1 < x \leq L \end{cases}$$

$$f(x) = \sum_{n=1}^{\infty} \left[\frac{2L}{n\pi}(-1)^n - \frac{2(L+1)}{n\pi} \cos\left(\frac{n\pi}{L}\right) + \frac{2L}{n^2\pi^2} \sin\left(\frac{n\pi}{L}\right) \right] \sin\left(\frac{n\pi x}{L}\right)$$

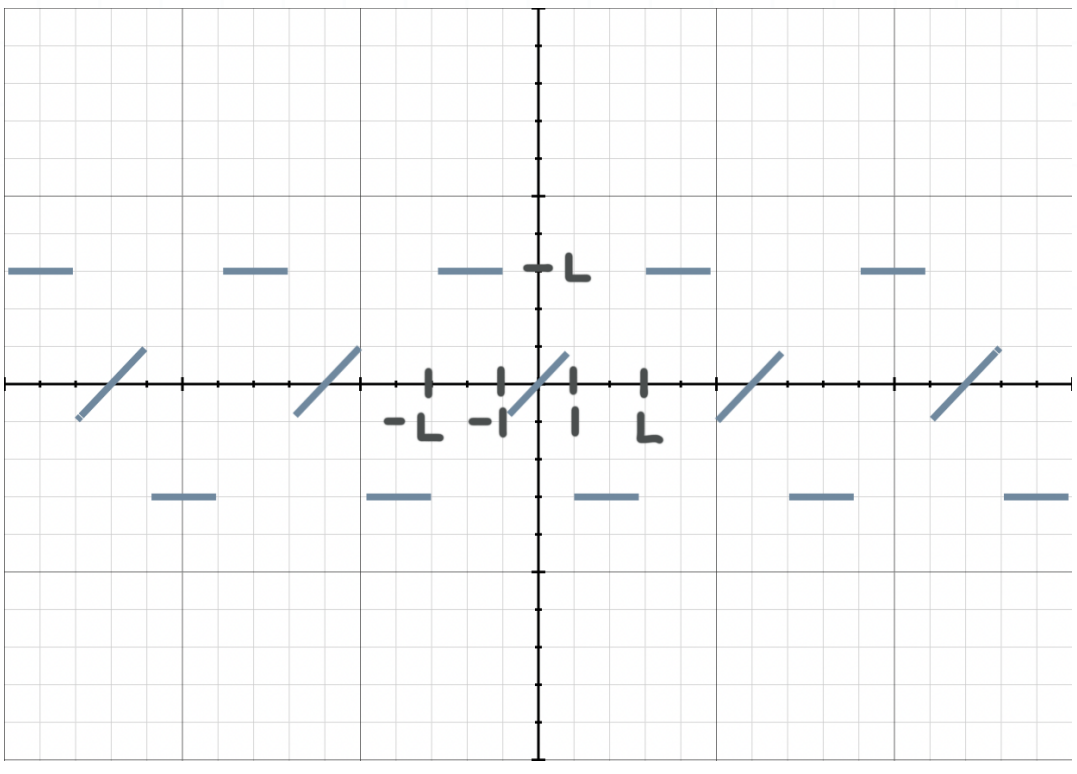
Solution:

A sketch of the function $f(x)$ is

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The sketch of its periodic extension, $g(x)$, is



From the graph of the periodic extension, we can see that $g(x)$ has a jump discontinuity at $x = \pm 1$, within the interval $-L \leq x \leq L$, as well as jump discontinuities at $-L$ and L.

Because the function is piecewise smooth on the individual intervals $-L \leq x \leq -1$, $-1 \leq x \leq 1$, and $1 \leq x \leq L$, the function $f(x)$ and the periodic

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extension $g(x)$ are both continuous there, so the Fourier series representation will converge to the periodic extension $g(x)$, and therefore also to the original function $f(x)$. Because of the jump discontinuities at $x = \pm 1$, the Fourier series representation at $x = 1$ will converge to

$$\frac{\lim_{x\to 1^-} g(x) + \lim_{x\to 1^+} g(x)}{2}$$

$$\frac{1 + (-L)}{2}$$

$$\frac{1 - L}{2}$$

and the Fourier series representation at $x = -1$ will converge to

$$\frac{\lim_{x\to -1^-} g(x) + \lim_{x\to -1^+} g(x)}{2}$$

$$\frac{L + (-1)}{2}$$

$$\frac{L - 1}{2}$$

At $x = -L$, the Fourier series representation will converge to

$$\frac{\lim_{x\to -L^-} g(x) + \lim_{x\to -L^+} g(x)}{2}$$

$$\frac{-L + L}{2} = 0$$

and at $x = L$ the Fourier series representation will converge to

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$$\frac{\lim_{x\to L^-} g(x) + \lim_{x\to L^+} g(x)}{2}$$

$$\frac{-L + L}{2} = 0$$

The Fourier series representation converges to $f(x)$, since $f(x)$ is piecewise smooth and has a finite number of jump discontinuities on any interval with finite endpoints.

■ 4. Given the function $f(x)$ and its Fourier series representation, say whether or not the Fourier series converges to $f(x)$.

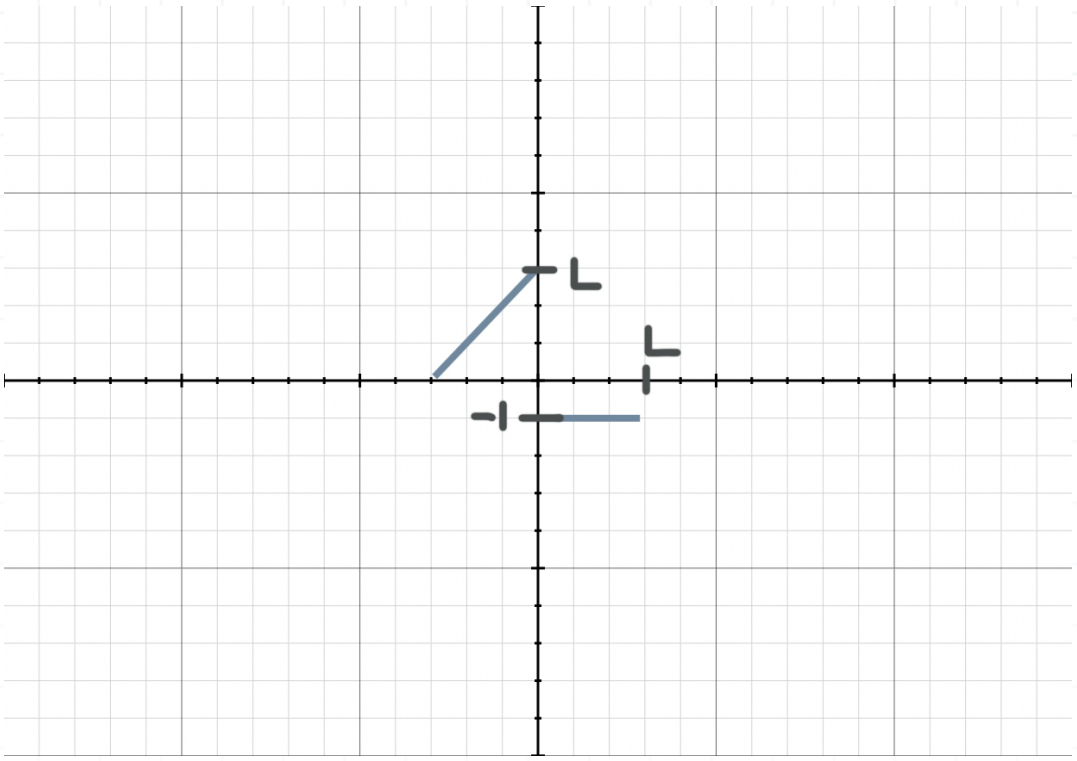
$$f(x) = \begin{cases} x + L & -L \le x < 0 \\ -1 & 0 \le x \le L \end{cases}$$

$$f(x) = \frac{L-2}{4} + \frac{L}{\pi^2}\sum_{n=1}^{\infty} \frac{1-(-1)^n}{n^2} \cos\left(\frac{n\pi x}{L}\right) + \frac{1}{\pi}\sum_{n=1}^{\infty} \frac{(-1)^n - 1 - L}{n} \sin\left(\frac{n\pi x}{L}\right)$$

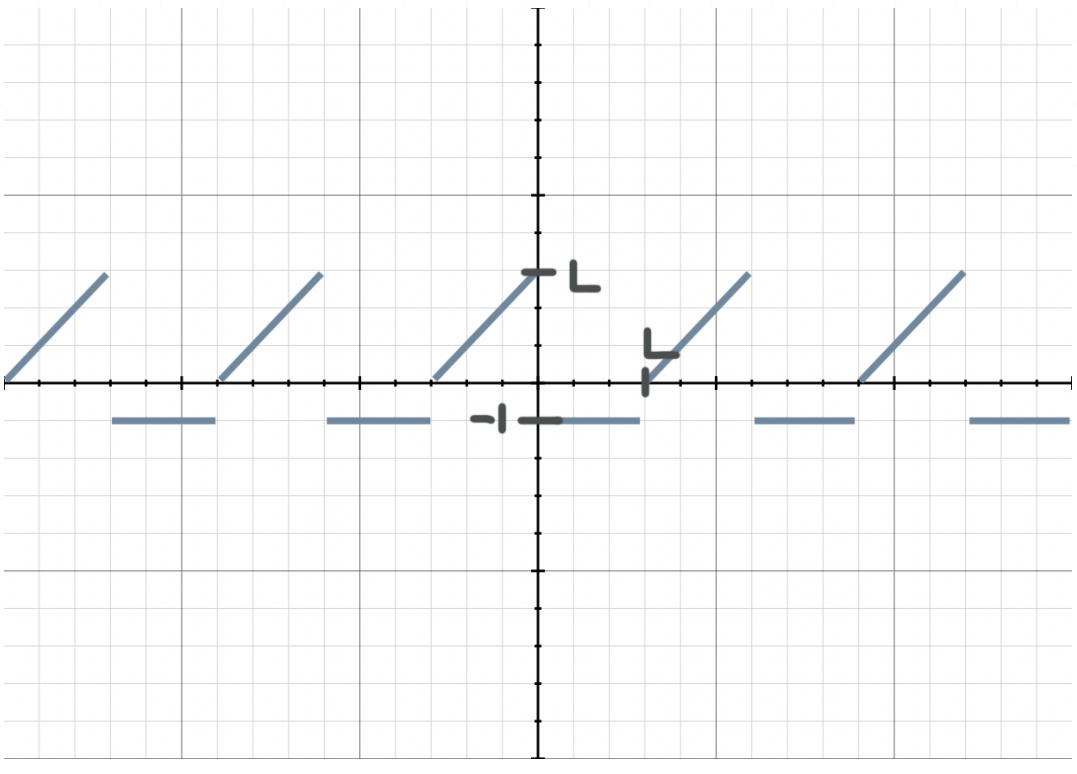
Solution:

A sketch of the function $f(x)$ is

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The sketch of its periodic extension, $g(x)$, is



From the graph of the periodic extension, we can see that $g(x)$ has a jump discontinuity at $x = 0$, within the interval $-L \leq x \leq L$, as well as jump discontinuities at $-L$ and L.

Because the function is piecewise smooth on the individual intervals $-L \leq x \leq 0$ and $0 \leq x \leq L$, the function $f(x)$ and the periodic extension $g(x)$

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are both continuous there, so the Fourier series representation will converge to the periodic extension $g(x)$, and therefore also to the original function $f(x)$. Because of the jump discontinuity at $x = 0$, the Fourier series representation at that point will converge to

$$\frac{\lim_{x\to 0^-} g(x) + \lim_{x\to 0^+} g(x)}{2}$$

$$\frac{L + (-1)}{2}$$

$$\frac{L - 1}{2}$$

At $x = -L$, the Fourier series representation will converge to

$$\frac{\lim_{x\to -L^-} g(x) + \lim_{x\to -L^+} g(x)}{2}$$

$$\frac{-1 + 0}{2}$$

$$-\frac{1}{2}$$

and at $x = L$ the Fourier series representation will converge to

$$\frac{\lim_{x\to L^-} g(x) + \lim_{x\to L^+} g(x)}{2}$$

$$\frac{-1 + 0}{2}$$

$$-\frac{1}{2}$$

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The Fourier series representation converges to $f(x)$, since $f(x)$ is piecewise smooth and has a finite number of jump discontinuities on any interval with finite endpoints.

■ 5. Given the function $f(x)$ and its Fourier series representation, say whether or not the Fourier series converges to $f(x)$, assuming $L > 1$.

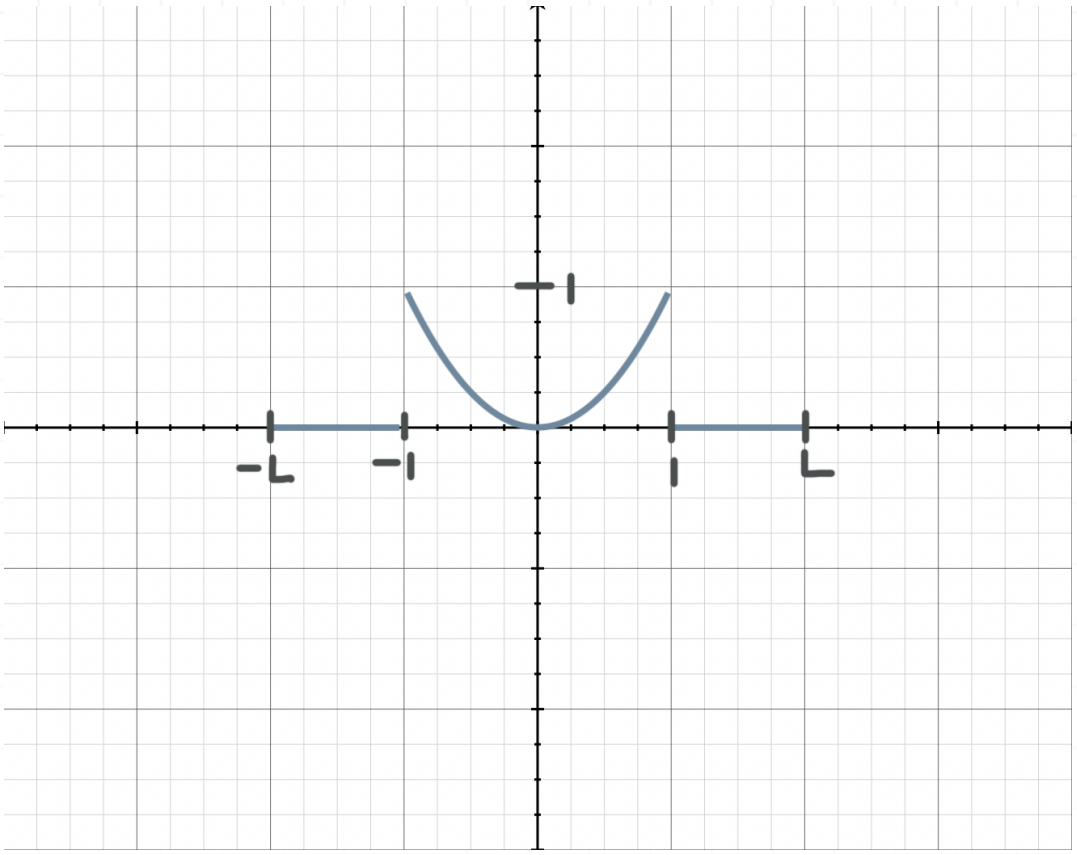
$$f(x) = \begin{cases} x^2 & |x| < 1 \\ 0 & 1 \leq |x| \leq L \end{cases}$$

$$f(x) = \frac{1}{3L} + \frac{2}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{n^3} \left[(n^2\pi^2 - 2L^2) \sin\left(\frac{n\pi}{L}\right) + n\pi L \cos\left(\frac{n\pi}{L}\right) \right] \cos\left(\frac{n\pi x}{L}\right)$$

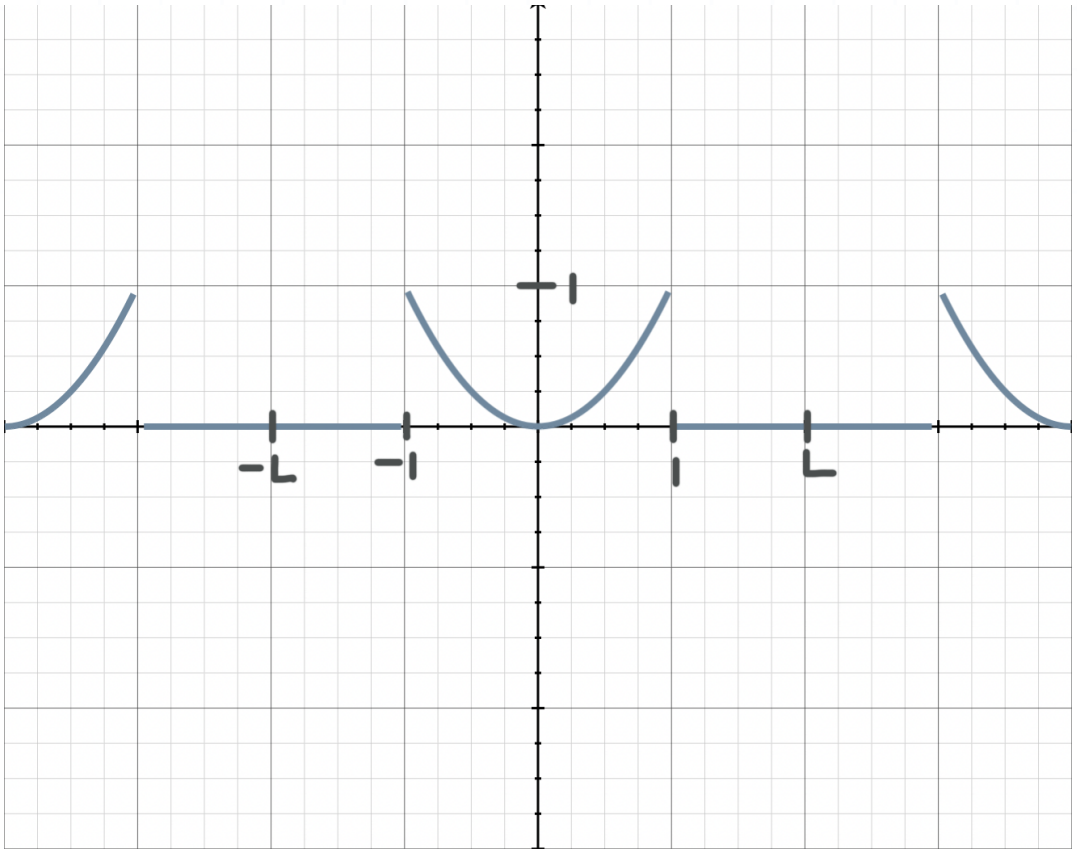
Solution:

A sketch of the function $f(x)$ is

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The sketch of its periodic extension, $g(x)$, is



From the graph of the periodic extension, we can see that $g(x)$ has a jump discontinuity at $x = \pm 1$, within the interval $-L \leq x \leq L$.

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Because the function is piecewise smooth on the individual intervals $-L \leq x \leq -1$, $-1 \leq x \leq 1$, and $1 \leq x \leq L$, the function $f(x)$ and the periodic extension $g(x)$ are both continuous there, so the Fourier series representation will converge to the periodic extension $g(x)$, and therefore also to the original function $f(x)$. Because of the jump discontinuities at $x = \pm 1$, the Fourier series representation at $x = 1$ will converge to

$$\frac{\lim_{x \to 1^-} g(x) + \lim_{x \to 1^+} g(x)}{2}$$

$$\frac{1 + 0}{2}$$

$$\frac{1}{2}$$

and the Fourier series representation at $x = -1$ will converge to

$$\frac{\lim_{x \to -1^-} g(x) + \lim_{x \to -1^+} g(x)}{2}$$

$$\frac{0 + 1}{2}$$

$$\frac{1}{2}$$

The Fourier series representation converges to $f(x)$, since $f(x)$ is piecewise smooth and has a finite number of jump discontinuities on any interval with finite endpoints.

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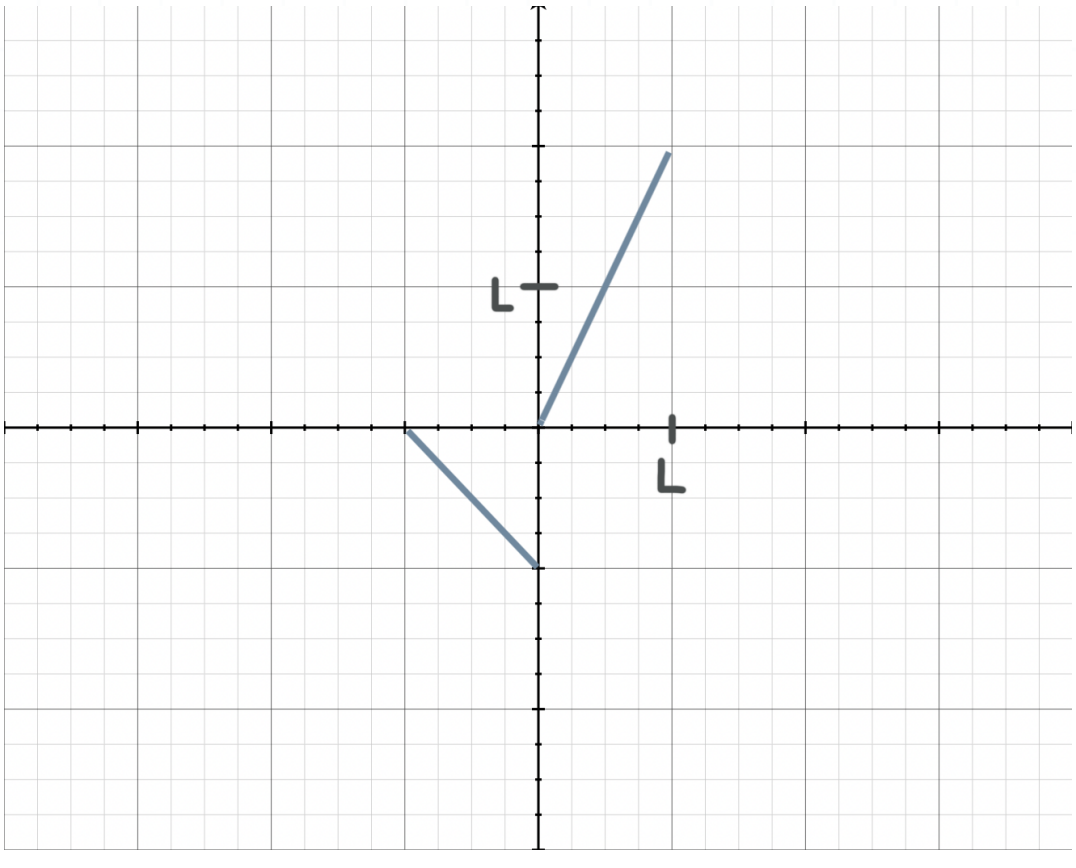
■ 6. Given the function $f(x)$ and its Fourier series representation, say whether or not the Fourier series converges to $f(x)$.

$$f(x) = \begin{cases} Lx & 0 \le x \le L \\ -x - L & -L \le x < 0 \end{cases}$$

$$f(x) = \frac{L^2 - L}{4} + \frac{L}{\pi} \sum_{n=1}^{\infty} \frac{1 - L(-1)^n}{n} \sin\left(\frac{n\pi x}{L}\right) + \frac{L(L+1)}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{n^2} \cos\left(\frac{n\pi x}{L}\right)$$

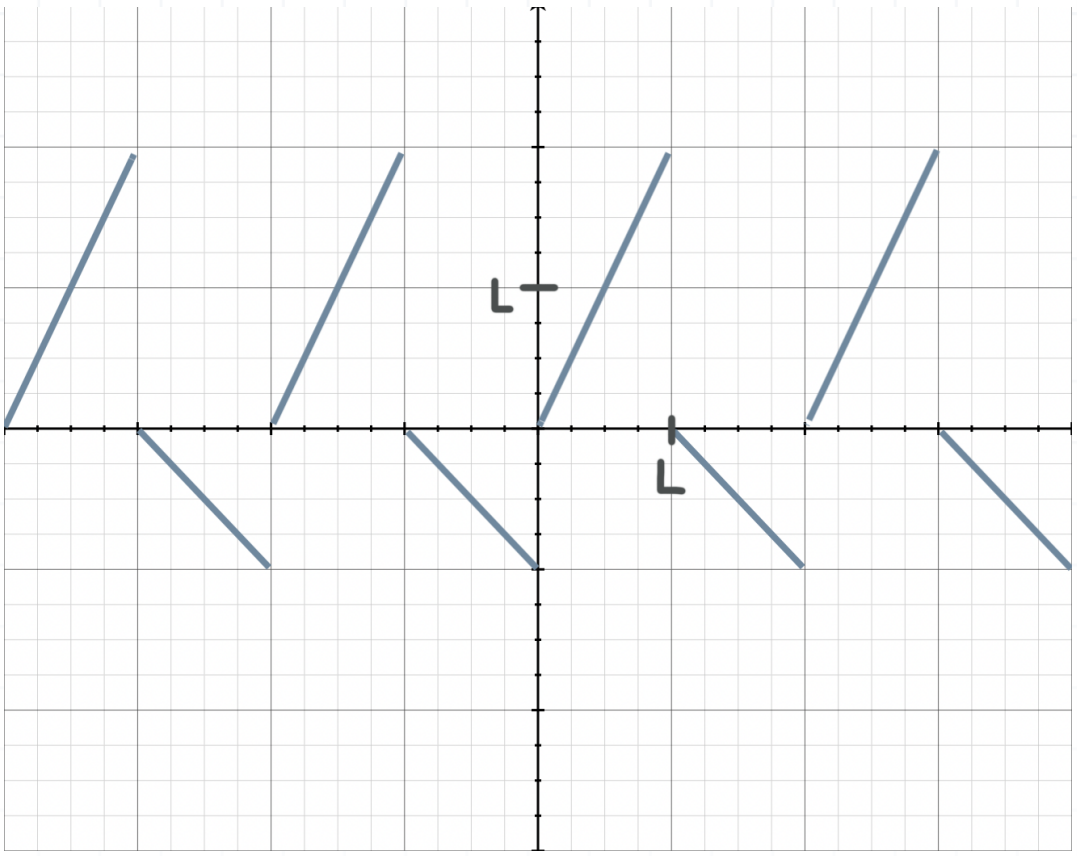
Solution:

A sketch of the function $f(x)$ is



The sketch of its periodic extension, $g(x)$, is

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From the graph of the periodic extension, we can see that $g(x)$ has a jump discontinuity at $x = 0$, within the interval $-L \leq x \leq L$, as well as jump discontinuities at $-L$ and L.

Because the function is piecewise smooth on the individual intervals $-L \leq x \leq 0$ and $0 \leq x \leq L$, the function $f(x)$ and the periodic extension $g(x)$ are both continuous there, so the Fourier series representation will converge to the periodic extension $g(x)$, and therefore also to the original function $f(x)$. Because of the jump discontinuity at $x = 0$, the Fourier series representation at that point will converge to

$$\frac{\lim_{x\to 0^-} g(x) + \lim_{x\to 0^+} g(x)}{2}$$

$$\frac{-L + 0}{2}$$

$$-\frac{L}{2}$$

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At $x = -L$, the Fourier series representation will converge to

$$\frac{\lim_{x\to -L^-} g(x) + \lim_{x\to -L^+} g(x)}{2}$$

$$\frac{L^2 + 0}{2}$$

$$\frac{L^2}{2}$$

and at $x = L$ the Fourier series representation will converge to

$$\frac{\lim_{x\to L^-} g(x) + \lim_{x\to L^+} g(x)}{2}$$

$$\frac{L^2 + 0}{2}$$

$$\frac{L^2}{2}$$

The Fourier series representation converges to $f(x)$, since $f(x)$ is piecewise smooth and has a finite number of jump discontinuities on any interval with finite endpoints.

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FOURIER COSINE SERIES

■ 1. Find the Fourier cosine series representation of the function $f(x) = 3x + 3$ on $0 \leq x \leq L$.

Solution:

For A_0, we get

$$A_0 = \frac{1}{L}\int_0^L f(x)\ dx$$

$$A_0 = \frac{1}{L}\int_0^L 3x + 3\ dx$$

$$A_0 = \frac{1}{L}\left(\frac{3}{2}x^2 + 3x\Big|_0^L\right)$$

$$A_0 = \frac{3}{2}L + 3$$

For A_n, we get

$$A_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)\ dx$$

$$A_n = \frac{2}{L}\int_0^L (3x + 3)\cos\left(\frac{n\pi x}{L}\right)\ dx$$

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Use integration by parts with $u = 3x + 3$, $du = 3\ dx$, $dv = \cos(n\pi x/L)\ dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

$$A_n = \frac{2}{L}\left((3x+3)\frac{L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_0^L - \frac{3L}{n\pi}\int_0^L \sin\left(\frac{n\pi x}{L}\right)\ dx\right)$$

$$A_n = \frac{2}{L}\left(0 + \frac{3L^2}{(n\pi)^2}\cos\left(\frac{n\pi x}{L}\right)\Big|_0^L\right)$$

$$A_n = \frac{6L}{n^2\pi^2}\cos\left(\frac{n\pi x}{L}\right)\Big|_0^L$$

$$A_n = \frac{6L}{n^2\pi^2}((-1)^n - 1)$$

Then the Fourier cosine series representation is

$$f(x) = \frac{3}{2}L + 3 + \sum_{n=1}^{\infty}\frac{6L}{n^2\pi^2}((-1)^n - 1)\cos\left(\frac{n\pi x}{L}\right)$$

■ 2. Find the Fourier cosine series representation of the function $f(x) = 3x^2 + 2x$ on $0 \le x \le L$.

Solution:

For A_0, we get

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$$A_0 = \frac{1}{L}\int_0^L f(x)\ dx$$

$$A_0 = \frac{1}{L}\int_0^L 3x^2 + 2x\ dx$$

$$A_0 = \frac{1}{L}\left(x^3 + x^2\Big|_0^L\right)$$

$$A_0 = L^2 + L$$

For A_n, we get

$$A_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)\ dx$$

$$A_n = \frac{2}{L}\int_0^L (3x^2 + 2x)\cos\left(\frac{n\pi x}{L}\right)\ dx$$

Use integration by parts with $u = 3x^2 + 2x$, $du = 6x + 2\ dx$, $dv = \cos(n\pi x/L)\ dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

$$A_n = \frac{2}{L}\left((3x^2 + 2x)\frac{L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_0^L - \frac{L}{n\pi}\int_0^L (6x + 2)\sin\left(\frac{n\pi x}{L}\right)\ dx\right)$$

Use integration by parts with $u = 6x + 2$, $du = 6\ dx$, $dv = \sin(n\pi x/L)\ dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$A_n = \frac{2}{L}\left(0 - \frac{L}{n\pi}\left(-(6x + 2)\frac{L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_0^L + \frac{6L}{n\pi}\int_0^L \cos\left(\frac{n\pi x}{L}\right)\ dx\right)\right)$$

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$$A_n = -\frac{2}{n\pi}\left(-(6x+2)\frac{L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_0^L + \frac{12}{n\pi}\int_0^L \cos\left(\frac{n\pi x}{L}\right)\,dx\right)$$

$$A_n = -\frac{2}{n\pi}\left(-\frac{6L^2+2L}{n\pi}(-1)^n + \frac{2L}{n\pi} + \frac{12L}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\Big|_0^L\right)$$

$$A_n = -\frac{2}{n\pi}\left(-\frac{6L^2+2L}{n\pi}(-1)^n + \frac{2L}{n\pi} + 0\right)$$

$$A_n = \frac{12L^2+4L}{(n\pi)^2}(-1)^n - \frac{4L}{(n\pi)^2}$$

Then the Fourier cosine series representation is

$$f(x) = L^2 + L + \sum_{n=1}^{\infty}\left(\frac{12L^2+4L}{(n\pi)^2}(-1)^n - \frac{4L}{(n\pi)^2}\right)\cos\left(\frac{n\pi x}{L}\right)$$

■ 3. Find the Fourier cosine series representation of the function $f(x) = x^3$ function on $0 \leq x \leq \pi$.

Solution:

For A_0, we get

$$A_0 = \frac{1}{L}\int_0^L f(x)\,dx$$

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$$A_0 = \frac{1}{\pi}\int_0^{\pi} f(x)\ dx$$

$$A_0 = \frac{1}{\pi}\int_0^{\pi} x^3\ dx$$

$$A_0 = \frac{1}{\pi}\cdot\frac{x^4}{4}\Big|_0^{\pi}$$

$$A_0 = \frac{\pi^3}{4}$$

For A_n, we get

$$A_n = \frac{2}{L}\int_0^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)\ dx$$

$$A_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos(nx)\ dx$$

$$A_n = \frac{2}{\pi}\int_0^{\pi} x^3\cos(nx)\ dx$$

Use integration by parts with $u = x^3$, $du = 3x^2\ dx$, $dv = \cos(nx)\ dx$, and $v = (1/n)\sin(nx)$.

$$A_n = \frac{2}{\pi}\left(\frac{x^3}{n}\sin(nx)\Big|_0^{\pi} - \frac{3}{n}\int_0^{\pi} x^2\sin(nx)\ dx\right)$$

$$A_n = \frac{2}{\pi}\left(0 - \frac{3}{n}\int_0^{\pi} x^2\sin(nx)\ dx\right)$$

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Use integration by parts with $u = x^2$, $du = 2x\ dx$, $dv = \sin(nx)\ dx$, and $v = -(1/n)\cos(nx)$.

$$A_n = -\frac{6}{n\pi}\left(-\frac{x^2}{n}\cos(nx)\Big|_0^\pi + \frac{2}{n}\int_0^\pi x\cos(nx)\ dx\right)$$

$$A_n = -\frac{6}{n\pi}\left(-\frac{\pi^2}{n}(-1)^n + \frac{2}{n}\int_0^\pi x\cos(nx)\ dx\right)$$

Use integration by parts with $u = x$, $du = dx$, $dv = \cos(nx)\ dx$, and $v = (1/n)\sin(nx)$.

$$A_n = \frac{6\pi^2(-1)^n}{n^2\pi} - \frac{12}{n^2\pi}\left(\frac{x}{n}\sin(nx)\Big|_0^\pi - \frac{1}{n}\int_0^\pi \sin(nx)\ dx\right)$$

$$A_n = \frac{6\pi(-1)^n}{n^2} - \frac{12}{n^2\pi}\left(0 - \frac{1}{n}\int_0^\pi \sin(nx)\ dx\right)$$

$$A_n = \frac{6\pi(-1)^n}{n^2} + \frac{12}{n^3\pi}\left(-\frac{1}{n}\cos(nx)\right)\Big|_0^\pi$$

$$A_n = \frac{6\pi}{n^2}(-1)^n - \frac{12}{n^4\pi}((-1)^n - 1)$$

Then the Fourier cosine series representation is

$$f(x) = \frac{\pi^3}{4} + \sum_{n=1}^{\infty}\left(\frac{6\pi}{n^2}(-1)^n + \frac{12}{n^4\pi}(1-(-1)^n)\right)\cos(nx)$$

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■ 4. Find the Fourier cosine series representation of the function $f(x) = e^x$ on $0 \leq x \leq L$.

Solution:

For A_0, we get

$$A_0 = \frac{1}{L}\int_0^L f(x)\ dx$$

$$A_0 = \frac{1}{L}\int_0^L e^x\ dx$$

$$A_0 = \frac{1}{L}e^x\Big|_0^L$$

$$A_0 = \frac{1}{L}(e^L - 1)$$

For A_n, we get

$$A_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)\ dx$$

$$A_n = \frac{2}{L}\int_0^L e^x\cos\left(\frac{n\pi x}{L}\right)\ dx$$

Use integration by parts with $u = e^x$, $du = e^x\ dx$, $dv = \cos(n\pi x/L)\ dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

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$$A_n = \frac{2}{L}\left(\frac{e^x L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_0^L - \frac{L}{n\pi}\int_0^L e^x \sin\left(\frac{n\pi x}{L}\right)\,dx\right)$$

$$A_n = \frac{2}{L}\left(0 - \frac{L}{n\pi}\int_0^L e^x \sin\left(\frac{n\pi x}{L}\right)\,dx\right)$$

Use integration by parts with $u = e^x$, $du = e^x\,dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$A_n = -\frac{2}{n\pi}\left(-\frac{e^x L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_0^L + \frac{L}{n\pi}\int_0^L e^x \cos\left(\frac{n\pi x}{L}\right)\,dx\right)$$

$$A_n = -\frac{2}{n\pi}\left(-\frac{e^L L}{n\pi}(-1)^n + \frac{L}{n\pi} + \frac{L}{n\pi}\cdot\frac{LA_n}{2}\right)$$

$$A_n = \frac{2Le^L(-1)^n}{n^2\pi^2} - \frac{2L}{n^2\pi^2} - \frac{L^2 A_n}{n^2\pi^2}$$

$$A_n\left(1 + \frac{L^2}{n^2\pi^2}\right) = \frac{2L}{n^2\pi^2}(e^L(-1)^n - 1)$$

$$A_n = \frac{2L(e^L(-1)^n - 1)}{n^2\pi^2 + L^2}$$

Then the Fourier cosine series representation is

$$f(x) = \frac{1}{L}(e^L - 1) + \sum_{n=1}^{\infty}\frac{2L(e^L(-1)^n - 1)}{n^2\pi^2 + L^2}\cos\left(\frac{n\pi x}{L}\right)$$

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■ 5. Find the Fourier cosine series representation of the function $f(x) = x\cos x$ on $0 \le x \le 2\pi$.

Solution:

For A_0, we get

$$A_0 = \frac{1}{L}\int_0^L f(x)\,dx$$

$$A_0 = \frac{1}{2\pi}\int_0^{2\pi} f(x)\,dx$$

$$A_0 = \frac{1}{2\pi}\int_0^{2\pi} x\cos x\;dx$$

Use integration by parts with $u = x$, $du = dx$, $dv = \cos x\;dx$, and $v = \sin x$.

$$A_0 = \frac{1}{2\pi}\left(x\sin x\Big|_0^{2\pi} - \int_0^{2\pi}\sin x\;dx\right)$$

$$A_0 = \frac{1}{2\pi}\left(0 - \int_0^{2\pi}\sin x\;dx\right)$$

$$A_0 = \frac{1}{2\pi}\cos x\Big|_0^{2\pi} = 0$$

For A_n, we get

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$$A_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = \frac{2}{2\pi}\int_0^{2\pi} x\cos x\cos\left(\frac{nx}{2}\right)\,dx$$

Use the trig identity

$$\cos A\cos B = \frac{1}{2}[\cos(A-B)+\cos(A+B)]$$

to get

$$A_n = \frac{1}{2\pi}\int_0^{2\pi} x\left(\cos\left(1+\frac{n}{2}\right)x+\cos\left(1-\frac{n}{2}\right)x\right)\,dx$$

Use integration by parts with $u = x$, $du = dx$, and

$$dv = \left(\cos\left(1+\frac{n}{2}\right)x+\cos\left(1-\frac{n}{2}\right)x\right)\,dx$$

$$v = \frac{2}{2+n}\sin\left(1+\frac{n}{2}\right)x+\frac{2}{2-n}\sin\left(1-\frac{n}{2}\right)x$$

Then we get

$$A_n = \frac{1}{2\pi}\left[x\left(\frac{2}{2+n}\sin\left(1+\frac{n}{2}\right)x+\frac{2}{2-n}\sin\left(1-\frac{n}{2}\right)x\right)\Bigg|_0^{2\pi}\right.$$

$$\left.-\int_0^{2\pi}\frac{2}{2+n}\sin\left(1+\frac{n}{2}\right)x+\frac{2}{2-n}\sin\left(1-\frac{n}{2}\right)x\,dx\right]$$

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$$A_n = \frac{1}{2\pi}\left[0 + \left(\left(\frac{2}{2+n}\right)^2 \cos\left(1+\frac{n}{2}\right)x + \left(\frac{2}{2-n}\right)^2 \cos\left(1-\frac{n}{2}\right)x\right)\Big|_0^{2\pi}\right]$$

$$A_n = \frac{4}{2\pi}\left(\frac{1}{(2+n)^2}(\cos(2\pi+\pi n)-1) + \frac{1}{(2-n)^2}(\cos(2\pi-\pi n)-1)\right)$$

$$A_n = \frac{2}{\pi}\left(\frac{1}{(2+n)^2}(\cos \pi n - 1) + \frac{1}{(2-n)^2}(\cos \pi n - 1)\right)$$

$$A_n = \frac{2}{\pi} \cdot \frac{2n^2+8}{(2+n)^2(2-n)^2}((-1)^n - 1)$$

$$A_n = \frac{4(n^2+4)((-1)^n-1)}{\pi(n^2-4)^2} \text{ with } n \neq 2$$

We need to find A_2, so

$$A_2 = \frac{1}{\pi}\int_0^{2\pi} x\cos^2 x \; dx$$

Use integration by parts with $u = x$, $du = dx$, and

$$dv = \cos^2 x \; dx = \frac{1+\cos(2x)}{2} \; dx$$

$$v = \frac{x}{2} + \frac{1}{4}\sin(2x)$$

Then we get

$$A_2 = \frac{1}{\pi}\left(\left(\frac{x^2}{2} + \frac{1}{4}x\sin(2x)\right)\Big|_0^{2\pi} - \int_0^{2\pi}\left(\frac{x}{2} + \frac{1}{4}\sin(2x)\right) \; dx\right)$$

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$$A_2 = \frac{1}{\pi}\left(2\pi^2 + 0 - \left(\frac{x^2}{4} + \frac{1}{8}\cos(2x)\right)\Big|_0^{2\pi}\right)$$

$$A_2 = \frac{1}{\pi}(2\pi^2 - \pi^2)$$

$$A_2 = \pi$$

Then A_1 is

$$A_1 = \frac{4(1^2+4)((-1)^1-1)}{\pi(1^2-4)^2}$$

$$A_1 = -\frac{40}{9\pi}$$

Then the Fourier cosine series representation is

$$f(x) = -\frac{40}{9\pi}\cos\left(\frac{x}{2}\right) + \pi\cos x + \sum_{n=3}^{\infty}\frac{4(n^2+4)((-1)^n-1)}{\pi(n^2-4)^2}\cos\left(\frac{nx}{2}\right)$$

6. Find the Fourier cosine series representation of the function $f(x) = \sin(3x+5)$ on $0 \le x \le \pi$.

Solution:

For A_0, we get

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$$A_0 = \frac{1}{L}\int_0^L f(x)\ dx$$

$$A_0 = \frac{1}{\pi}\int_0^\pi \sin(3x+5)\ dx$$

$$A_0 = -\frac{1}{3\pi}\cos(3x+5)\Big|_0^\pi$$

$$A_0 = -\frac{1}{3\pi}(\cos(3\pi+5)-\cos 5)$$

$$A_0 = -\frac{1}{3\pi}((\cos 3\pi\cos 5-\sin 3\pi\sin 5)-\cos 5)$$

$$A_0 = -\frac{1}{3\pi}(-\cos 5-\cos 5)$$

$$A_0 = \frac{2\cos 5}{3\pi}$$

For A_n, we get

$$A_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)\ dx$$

$$A_n = \frac{2}{\pi}\int_0^\pi f(x)\cos(nx)\ dx$$

$$A_n = \frac{2}{\pi}\int_0^\pi \sin(3x+5)\cos(nx)\ dx$$

Use the trigonometric identity

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$$\sin A \cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$$

we get

$$A_n = \frac{2}{\pi}\int_0^{\pi} \frac{1}{2}[\sin((n+3)x+5) + \sin((3-n)x+5)]\ dx$$

$$A_n = -\frac{1}{\pi}\left[\frac{1}{n+3}\cos((n+3)x+5) + \frac{1}{3-n}\cos((3-n)x+5)\right]\Bigg|_0^{\pi}$$

$$A_n = -\frac{1}{\pi}\left[\frac{1}{n+3}(\cos((n+3)\pi+5) - \cos 5) + \frac{1}{3-n}(\cos((3-n)\pi+5) - \cos 5)\right]$$

Use the trigonometric identity

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

we get

$$A_n = -\frac{1}{\pi}\left[\frac{1}{n+3}(\cos(\pi(n+3))\cos 5 - \sin(\pi(n+3))\sin 5 - \cos 5)\right.$$

$$\left. + \frac{1}{3-n}(\cos(\pi(3-n))\cos 5 - \sin(\pi(3-n))\sin 5 - \cos 5)\right]$$

$$A_n = -\frac{1}{\pi}\left[\frac{1}{n+3}(\cos(\pi(n+3))\cos 5 + 0 - \cos 5)\right.$$

$$\left. + \frac{1}{3-n}(\cos(\pi(3-n))\cos 5 + 0 - \cos 5)\right]$$

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$$A_n = -\frac{1}{\pi}\left[\frac{1}{n+3}(-\cos(n\pi)\cos 5 - \cos 5) + \frac{1}{3-n}(-\cos(n\pi)\cos 5 - \cos 5)\right]$$

$$A_n = \frac{1}{\pi}\left(\cos 5((-1)^n + 1)\left(\frac{1}{n+3} + \frac{1}{3-n}\right)\right)$$

$$A_n = \frac{6\cos 5((-1)^n + 1)}{\pi(9-n^2)} \text{ with } n \neq 3$$

We need to find A_1, A_2, and A_3. We get

$$A_1 = \frac{6\cos 5((-1)^1 + 1)}{\pi(9-1^2)} = 0$$

$$A_2 = \frac{6\cos 5((-1)^2 + 1)}{\pi(9-2^2)} = \frac{12\cos 5}{5\pi}$$

For A_3, we find

$$A_3 = \frac{2}{\pi}\int_0^{\pi} \sin(3x+5)\cos(3x)\ dx$$

Use the trigonometric identity

$$\sin A \cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$$

to get

$$A_3 = \frac{1}{\pi}\int_0^{\pi} \sin(6x+5) + \sin 5\ dx$$

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$$A_3 = \frac{1}{\pi}\left(\frac{1}{6}\cos(6x+5)\Big|_0^{\pi} + (\sin 5)x\Big|_0^{\pi}\right)$$

$$A_3 = \frac{1}{\pi}\left(\frac{1}{6}\cos(6\pi+5) - \frac{1}{6}\cos 5 + \pi \sin 5\right)$$

$$A_3 = \frac{1}{\pi}\left(\frac{1}{6}\cos 5 - \frac{1}{6}\cos 5 + \pi \sin 5\right)$$

$$A_3 = \frac{1}{\pi}\sin 5 \cdot \pi = \sin 5$$

Then the Fourier cosine series representation is

$$f(x) = \frac{2\cos 5}{3\pi} + \frac{12\cos 5}{5\pi}\cos(2x) + \sin 5\cos(3x) + \sum_{n=4}^{\infty}\frac{6\cos 5((-1)^n+1)}{\pi(9-n^2)}\cos(nx)$$

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FOURIER SINE SERIES

■ 1. Find the Fourier sine series representation of the function $f(x) = 2x^3 - 3x$ on $-L \leq x \leq L$.

Solution:

Since $f(-x) = -f(x)$, the function is odd.

$$f(-x) = 2(-x)^3 - 3(-x)$$

$$f(-x) = -(2x^3 - 3x)$$

For B_n, we get

$$B_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2}{L}\int_0^L (2x^3 - 3x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{4}{L}\int_0^L x^3 \sin\left(\frac{n\pi x}{L}\right)\,dx - \frac{6}{L}\int_0^L x\sin\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = x^3$, $du = 3x^2\,dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$ in the first integral and $u = x$, $du = dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$ in the second integral.

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$$B_n = \left[\frac{4}{L}\cdot\frac{-L}{n\pi}x^3\cos\left(\frac{n\pi x}{L}\right) - \frac{4}{L}\cdot\frac{-L}{n\pi}\int 3x^2\cos\left(\frac{n\pi x}{L}\right)\,dx\right.$$

$$\left. -\frac{6}{L}\cdot\frac{-L}{n\pi}x\cos\left(\frac{n\pi x}{L}\right) + \frac{6}{L}\cdot\frac{-L}{n\pi}\int\cos\left(\frac{n\pi x}{L}\right)\,dx\right]\Bigg|_0^L$$

$$B_n = \left[\frac{-4}{n\pi}x^3\cos\left(\frac{n\pi x}{L}\right) + \frac{6}{n\pi}x\cos\left(\frac{n\pi x}{L}\right) + \frac{12}{n\pi}\int x^2\cos\left(\frac{n\pi x}{L}\right)\,dx\right.$$

$$\left. -\frac{6}{n\pi}\int\cos\left(\frac{n\pi x}{L}\right)\,dx\right]\Bigg|_0^L$$

Use integration by parts with $u = x^2$, $du = 2x\ dx$, $dv = \cos(n\pi x/L)\ dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

$$B_n = \left[\frac{-4}{n\pi}x^3\cos\left(\frac{n\pi x}{L}\right) + \frac{6}{n\pi}x\cos\left(\frac{n\pi x}{L}\right)\right.$$

$$\left. +\frac{12}{n\pi}\left(\frac{L}{n\pi}x^2\sin\left(\frac{n\pi x}{L}\right) - \frac{2L}{n\pi}\int x\sin\left(\frac{n\pi x}{L}\right)\,dx\right) - \frac{6L}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\right]\Bigg|_0^L$$

$$B_n = \left[\frac{-4}{n\pi}x^3\cos\left(\frac{n\pi x}{L}\right) + \frac{6}{n\pi}x\cos\left(\frac{n\pi x}{L}\right) + \frac{12L}{(n\pi)^2}x^2\sin\left(\frac{n\pi x}{L}\right)\right.$$

$$\left. -\frac{24L}{(n\pi)^2}\int x\sin\left(\frac{n\pi x}{L}\right)\,dx - \frac{6L}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\right]\Bigg|_0^L$$

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Use integration by parts with $u = x$, $du = dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$ in the second integral.

$$B_n = \left[\frac{-4}{n\pi}x^3\cos\left(\frac{n\pi x}{L}\right) + \frac{6}{n\pi}x\cos\left(\frac{n\pi x}{L}\right) + \frac{12L}{(n\pi)^2}x^2\sin\left(\frac{n\pi x}{L}\right)\right.$$

$$\left. -\frac{24L}{(n\pi)^2}\left(\frac{-L}{n\pi}x\cos\left(\frac{n\pi x}{L}\right) - \frac{-L}{n\pi}\int\cos\left(\frac{n\pi x}{L}\right)dx\right) - \frac{6L}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\right]\Bigg|_0^L$$

$$B_n = \left[\frac{-4}{n\pi}x^3\cos\left(\frac{n\pi x}{L}\right) + \frac{6}{n\pi}x\cos\left(\frac{n\pi x}{L}\right) + \frac{12L}{(n\pi)^2}x^2\sin\left(\frac{n\pi x}{L}\right)\right.$$

$$\left. +\frac{24L^2}{(n\pi)^3}x\cos\left(\frac{n\pi x}{L}\right) - \frac{24L^2}{(n\pi)^3}\int\cos\left(\frac{n\pi x}{L}\right)dx - \frac{6L}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\right]\Bigg|_0^L$$

$$B_n = \left[\frac{-4}{n\pi}x^3\cos\left(\frac{n\pi x}{L}\right) + \frac{6}{n\pi}x\cos\left(\frac{n\pi x}{L}\right) + \frac{12L}{(n\pi)^2}x^2\sin\left(\frac{n\pi x}{L}\right)\right.$$

$$\left. +\frac{24L^2}{(n\pi)^3}x\cos\left(\frac{n\pi x}{L}\right) - \frac{24L^3}{(n\pi)^4}\sin\left(\frac{n\pi x}{L}\right) - \frac{6L}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\right]\Bigg|_0^L$$

$$B_n = \frac{-4}{n\pi}L^3(-1)^n + \frac{6}{n\pi}L(-1)^n + \frac{24L^3}{(n\pi)^3}L(-1)^n$$

$$B_n = \frac{(-4L^3 + 6L)}{n\pi}(-1)^n + \frac{24L^3}{(n\pi)^3}(-1)^n$$

Then the Fourier sine series representation of $f(x) = 2x^3 - 3x$ on $-L \le x \le L$ is

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$$f(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \sum_{n=1}^{\infty} \left(\frac{-4L^3 + 6L}{n\pi}(-1)^n + \frac{24L^3}{(n\pi)^3}(-1)^n\right) \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{2L}{\pi^3} \sum_{n=1}^{\infty} ((-2L^2 + 3)\pi^2 n^2 + 12L^2)\frac{(-1)^n}{n^3} \sin\left(\frac{n\pi x}{L}\right)$$

■ 2. Find the Fourier sine series representation of the function $f(x) = -x^3$ on $-L \leq x \leq L$.

Solution:

Since $f(-x) = -f(x)$, the function is odd.

$$f(-x) = -(-x)^3$$

$$f(-x) = -(-x^3)$$

For B_n, we get

$$B_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2}{L}\int_0^L (-x^3)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

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$$B_n = -\frac{2}{L}\int_0^L x^3 \sin\left(\frac{n\pi x}{L}\right)\, dx$$

Use integration by parts with $u = x^3$, $du = 3x^2\, dx$, $dv = \sin(n\pi x/L)\, dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$B_n = \left[\frac{-2}{L}\cdot\frac{-L}{n\pi}x^3\cos\left(\frac{n\pi x}{L}\right) - \frac{-2}{L}\cdot\frac{-L}{n\pi}\int 3x^2\cos\left(\frac{n\pi x}{L}\right)\, dx\right]\Bigg|_0^L$$

$$B_n = \left[\frac{2}{n\pi}x^3\cos\left(\frac{n\pi x}{L}\right) - \frac{6}{n\pi}\int x^2\cos\left(\frac{n\pi x}{L}\right)\right]\Bigg|_0^L$$

Use integration by parts with $u = x^2$, $du = 2x\, dx$, $dv = \cos(n\pi x/L)\, dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

$$B_n = \left[\frac{2}{n\pi}x^3\cos\left(\frac{n\pi x}{L}\right) - \frac{6}{n\pi}\cdot\frac{L}{n\pi}x^2\sin\left(\frac{n\pi x}{L}\right) + \frac{6}{n\pi}\cdot\frac{L}{n\pi}\cdot\int 2x\sin\left(\frac{n\pi x}{L}\right)\, dx\right]\Bigg|_0^L$$

$$B_n = \left[\frac{2}{n\pi}x^3\cos\left(\frac{n\pi x}{L}\right) - \frac{6L}{(n\pi)^2}x^2\sin\left(\frac{n\pi x}{L}\right) + \frac{12L}{(n\pi)^2}\int x\sin\left(\frac{n\pi x}{L}\right)\, dx\right]\Bigg|_0^L$$

Use integration by parts with $u = x$, $du = dx$, $dv = \sin(n\pi x/L)\, dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$B_n = \left[\frac{2}{n\pi}x^3\cos\left(\frac{n\pi x}{L}\right) - \frac{6L}{(n\pi)^2}x^2\sin\left(\frac{n\pi x}{L}\right)\right.$$

$$\left. + \frac{12L}{(n\pi)^2}\cdot\frac{-L}{n\pi}x\cos\left(\frac{n\pi x}{L}\right) - \frac{12L}{(n\pi)^2}\cdot\frac{-L}{n\pi}\int\cos\left(\frac{n\pi x}{L}\right)\, dx\right]\Bigg|_0^L$$

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$$B_n = \left[\frac{2}{n\pi}x^3\cos\left(\frac{n\pi x}{L}\right) - \frac{6L}{(n\pi)^2}x^2\sin\left(\frac{n\pi x}{L}\right)\right.$$

$$\left. -\frac{12L^2}{(n\pi)^3}x\cos\left(\frac{n\pi x}{L}\right) + \frac{12L^3}{(n\pi)^4}\sin\left(\frac{n\pi x}{L}\right)\right]\Bigg|_0^L$$

Substitute $\cos(n\pi) = (-1)^n$.

$$B_n = \frac{2}{n\pi}L^3(-1)^n - \frac{12L^3}{(n\pi)^3}(-1)^n$$

Then the Fourier sine series representation of $f(x) = -x^3$ on $-L \leq x \leq L$ is

$$f(x) = \sum_{n=1}^{\infty}\left(\frac{2}{n\pi}L^3 - \frac{12L^3}{(n\pi)^3}\right)(-1)^n\sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{2L^3}{\pi^3}\sum_{n=1}^{\infty}(\pi^2n^2 - 6)\frac{(-1)^n}{n^3}\sin\left(\frac{n\pi x}{L}\right)$$

■ 3. Find the Fourier sine series representation of the function $f(x) = x^3 - 2x$ on $-L \leq x \leq L$.

Solution:

Since $f(-x) = -f(x)$, the function is odd.

$$f(-x) = (-x)^3 - 2(-x)$$

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$$f(-x) = -(x^3 - 2x)$$

For B_n, we get

$$B_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2}{L}\int_0^L (x^3 - 2x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2}{L}\int_0^L x^3 \sin\left(\frac{n\pi x}{L}\right)\,dx - \frac{4}{L}\int_0^L x\sin\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = x^3$, $du = 3x^2\,dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$ in the first integral and $u = x$, $du = dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$ in the second integral.

$$B_n = \left[\frac{2}{L}\cdot\frac{-L}{n\pi}x^3\cos\left(\frac{n\pi x}{L}\right) - \frac{2}{L}\cdot\frac{-L}{n\pi}\int 3x^2\cos\left(\frac{n\pi x}{L}\right)\,dx\right.$$

$$\left. -\frac{4}{L}\cdot\frac{-L}{n\pi}x\cos\left(\frac{n\pi x}{L}\right) + \frac{4}{L}\cdot\frac{-L}{n\pi}\int\cos\left(\frac{n\pi x}{L}\right)\,dx\right]\Bigg|_0^L$$

$$B_n = \left[\frac{-2}{n\pi}x^3\cos\left(\frac{n\pi x}{L}\right) + \frac{6}{n\pi}\int x^2\cos\left(\frac{n\pi x}{L}\right)\,dx\right.$$

$$\left. +\frac{4}{n\pi}x\cos\left(\frac{n\pi x}{L}\right) - \frac{4}{n\pi}\int\cos\left(\frac{n\pi x}{L}\right)\,dx\right]\Bigg|_0^L$$

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Use integration by parts with $u = x^2$, $du = 2x\ dx$, $dv = \cos(n\pi x/L)\ dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

$$B_n = \left[\frac{-2}{n\pi}x^3\cos\left(\frac{n\pi x}{L}\right) + \frac{6}{n\pi}\cdot\frac{L}{n\pi}x^2\sin\left(\frac{n\pi x}{L}\right) - \frac{6}{n\pi}\cdot\frac{L}{n\pi}\int 2x\sin\left(\frac{n\pi x}{L}\right)\ dx\right.$$

$$\left. + \frac{4}{n\pi}x\cos\left(\frac{n\pi x}{L}\right) - \frac{4}{n\pi}\cdot\frac{-L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\right]\Bigg|_0^L$$

$$B_n = \left[\frac{-2}{n\pi}x^3\cos\left(\frac{n\pi x}{L}\right) + \frac{6L}{(n\pi)^2}x^2\sin\left(\frac{n\pi x}{L}\right) - \frac{6L}{(n\pi)^2}\int x\sin\left(\frac{n\pi x}{L}\right)\ dx\right.$$

$$\left. + \frac{4}{n\pi}x\cos\left(\frac{n\pi x}{L}\right) + \frac{4L}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\right]\Bigg|_0^L$$

Use integration by parts with $u = x$, $du = dx$, $dv = \sin(n\pi x/L)\ dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$B_n = \left[\frac{-2}{n\pi}x^3\cos\left(\frac{n\pi x}{L}\right) + \frac{6L}{(n\pi)^2}x^2\sin\left(\frac{n\pi x}{L}\right) - \frac{12L}{(n\pi)^2}\cdot\frac{-L}{n\pi}x\cos\left(\frac{n\pi x}{L}\right)\right.$$

$$\left. + \frac{12L}{(n\pi)^2}\cdot\frac{-L}{n\pi}\int\cos\left(\frac{n\pi x}{L}\right)\ dx + \frac{4}{n\pi}x\cos\left(\frac{n\pi x}{L}\right) + \frac{4L}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\right]\Bigg|_0^L$$

$$B_n = \left[\frac{-2}{n\pi}x^3\cos\left(\frac{n\pi x}{L}\right) + \frac{6L}{(n\pi)^2}x^2\sin\left(\frac{n\pi x}{L}\right) + \frac{12L^2}{(n\pi)^3}x\cos\left(\frac{n\pi x}{L}\right)\right.$$

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$$-\frac{12L^3}{(n\pi)^4}\sin\left(\frac{n\pi x}{L}\right)+\frac{4}{n\pi}x\cos\left(\frac{n\pi x}{L}\right)+\frac{4L}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\Bigg]\Bigg|_0^L$$

For $n = 1, 2, 3, \ldots$, $\sin(n\pi) = 0$ and $\cos(\pm n\pi) = (-1)^n$, so the expression for B_n simplifies to

$$B_n = \frac{-2L^3}{n\pi}(-1)^n + \frac{12L^3}{(n\pi)^3}(-1)^n + \frac{4L}{n\pi}(-1)^n$$

$$B_n = \frac{2L(-1)^n}{\pi^3 n^3}(6L^2 - L^2\pi^2 n^2 + 2\pi^2 n^2)$$

Then the Fourier series representation of $f(x) = x^3 - 2x$ on $-L \leq x \leq L$ is

$$f(x) = \sum_{n=1}^{\infty} \frac{2L(-1)^n}{\pi^3 n^3}(6L^2 - L^2\pi^2 n^2 + 2\pi^2 n^2)\sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{2L}{\pi^3}\sum_{n=1}^{\infty}(6L^2 - L^2\pi^2 n^2 + 2\pi^2 n^2)\frac{(-1)^n}{n^3}\sin\left(\frac{n\pi x}{L}\right)$$

■ 4. Find the Fourier sine series representation of the function $f(x) = x - x^2$ on $0 \leq x \leq L$.

Solution:

The function isn't odd, so we'll find its odd extension on $-L \leq x \leq L$.

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$$g(x) = \begin{cases} f(x) & 0 \le x \le L \\ -f(-x) & -L \le x < 0 \end{cases}$$

$$g(x) = \begin{cases} x - x^2 & 0 \le x \le L \\ -((-x) - (-x)^2) & -L \le x < 0 \end{cases}$$

$$g(x) = \begin{cases} x - x^2 & 0 \le x \le L \\ x + x^2 & -L \le x < 0 \end{cases}$$

Because the odd extension is an odd function, we can find its Fourier sine series representation, starting with calculating B_n.

$$B_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2}{L}\int_0^L (x - x^2)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2}{L}\int_0^L x\sin\left(\frac{n\pi x}{L}\right)\,dx - \frac{1}{L}\int_0^L x^2\sin\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = x$, $du = dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$ in the first integral and $u = x^2$, $du = 2x\,dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$ in the second integral.

$$B_n = \left[\frac{2}{L}\cdot\frac{-L}{n\pi}x\cos\left(\frac{n\pi x}{L}\right) - \frac{2}{L}\cdot\frac{-L}{n\pi}\int\cos\left(\frac{n\pi x}{L}\right)\,dx\right.$$

$$\left. -\frac{2}{L}\cdot\frac{-L}{n\pi}x^2\cos\left(\frac{n\pi x}{L}\right) + \frac{2}{L}\cdot\frac{-L}{n\pi}\int 2x\cos\left(\frac{n\pi x}{L}\right)\,dx\right]\Bigg|_0^L$$

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$$B_n = \left[\frac{-2}{n\pi}x\cos\left(\frac{n\pi x}{L}\right) + \frac{2L}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\right.$$

$$\left. + \frac{2}{n\pi}x^2\cos\left(\frac{n\pi x}{L}\right) - \frac{4}{n\pi}\int x\cos\left(\frac{n\pi x}{L}\right)\,dx\right]\Bigg|_0^L$$

Use integration by parts with $u = x$, $du = dx$, $dv = \cos(n\pi x/L)\,dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

$$B_n = \left[\frac{-2}{n\pi}x\cos\left(\frac{n\pi x}{L}\right) + \frac{2L}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right) + \frac{2}{n\pi}x^2\cos\left(\frac{n\pi x}{L}\right)\right.$$

$$\left. - \frac{4L}{(n\pi)^2}x\sin\left(\frac{n\pi x}{L}\right) + \frac{4L}{(n\pi)^2}\int \sin\left(\frac{n\pi x}{L}\right)\,dx\right]\Bigg|_0^L$$

$$B_n = \left[\frac{-2}{n\pi}x\cos\left(\frac{n\pi x}{L}\right) + \frac{2L}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right) + \frac{2}{n\pi}x^2\cos\left(\frac{n\pi x}{L}\right)\right.$$

$$\left. - \frac{4L}{(n\pi)^2}x\sin\left(\frac{n\pi x}{L}\right) - \frac{4L^2}{(n\pi)^3}\cos\left(\frac{n\pi x}{L}\right)\right]\Bigg|_0^L$$

Evaluate the interval using $\sin(n\pi) = 0$ and $\cos(n\pi) = (-1)^n$.

$$B_n = \frac{-2L}{n\pi}(-1)^n + \frac{2L^2}{n\pi}(-1)^n - \frac{4L^2}{(n\pi)^3}(-1)^n + \frac{4L^2}{(n\pi)^3}$$

$$B_n = \frac{2L}{\pi^3 n^3}((L\pi^2 n^2 - \pi^2 n^2 - 2L)(-1)^n + 2L)$$

Then the Fourier sine series representation of $f(x) = x - x^2$ on $0 \le x \le L$ is

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$$f(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{2L}{\pi^3} \sum_{n=1}^{\infty} ((L\pi^2 n^2 - \pi^2 n^2 - 2L)(-1)^n + 2L)\frac{1}{n^3} \sin\left(\frac{n\pi x}{L}\right)$$

■ 5. Find the Fourier sine series representation of $f(x) = x^2 - Lx$ on $0 \leq x \leq L$.

Solution:

The function isn't odd, so we'll find its odd extension on $-L \leq x \leq L$.

$$g(x) = \begin{cases} f(x) & 0 \leq x \leq L \\ -f(-x) & -L \leq x < 0 \end{cases}$$

$$g(x) = \begin{cases} x^2 - Lx & 0 \leq x \leq L \\ -((-x)^2 - (-x)L) & -L \leq x < 0 \end{cases}$$

$$g(x) = \begin{cases} x^2 - Lx & 0 \leq x \leq L \\ -x^2 - Lx & -L \leq x < 0 \end{cases}$$

Because the odd extension is an odd function, we can find its Fourier sine series representation, starting with calculating B_n.

$$B_n = \frac{1}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

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$$B_n = \frac{2}{L}\int_0^L (x^2 - Lx)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2}{L}\left[\int_0^L x^2 \sin\left(\frac{n\pi x}{L}\right)\,dx - \int_0^L Lx \sin\left(\frac{n\pi x}{L}\right)\,dx\right]$$

Use integration by parts with $u = x$, $du = dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$ in the second integral and $u = x^2$, $du = 2x\,dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$ in the first integral.

$$B_n = \frac{2}{L}\left[\frac{-L}{n\pi}x^2\cos\left(\frac{n\pi x}{L}\right) - \frac{-L}{n\pi}\int 2x\cos\left(\frac{n\pi x}{L}\right)\,dx\right.$$

$$\left. -\frac{-L}{n\pi}x\cos\left(\frac{n\pi x}{L}\right) + \frac{-L}{n\pi}\int\cos\left(\frac{n\pi x}{L}\right)\,dx\right]\Bigg|_0^L$$

$$B_n = \left[-\frac{2}{n\pi}x^2\cos\left(\frac{n\pi x}{L}\right) + \frac{4}{n\pi}\int x\cos\left(\frac{n\pi x}{L}\right)\,dx\right.$$

$$\left. +\frac{2L}{n\pi}x\cos\left(\frac{n\pi x}{L}\right) - \frac{2L^2}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\right]\Bigg|_0^L$$

Use integration by parts with $u = x$, $du = dx$, $dv = \cos(n\pi x/L)\,dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

$$B_n = \left[-\frac{2}{n\pi}x^2\cos\left(\frac{n\pi x}{L}\right) + \frac{4L}{(n\pi)^2}x\sin\left(\frac{n\pi x}{L}\right) - \frac{2L}{(n\pi)^2}\int\sin\left(\frac{n\pi x}{L}\right)\,dx\right.$$

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$$+\frac{2L}{n\pi}x\cos\left(\frac{n\pi x}{L}\right)-\frac{2L^2}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\Bigg]\Bigg|_0^L$$

$$B_n=\left[-\frac{2}{n\pi}x^2\cos\left(\frac{n\pi x}{L}\right)+\frac{4L}{(n\pi)^2}x\sin\left(\frac{n\pi x}{L}\right)+\frac{2L^2}{(n\pi)^3}\cos\left(\frac{n\pi x}{L}\right)\right.$$

$$+\frac{2L}{n\pi}x\cos\left(\frac{n\pi x}{L}\right)-\frac{2L^2}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\Bigg]\Bigg|_0^L$$

Evaluate the interval using $\sin(n\pi)=0$ and $\cos(n\pi)=(-1)^n$.

$$B_n=\frac{-2L^2}{n\pi}(-1)^n+\frac{4L^2}{(n\pi)^3}(-1)^n+\frac{2L^2}{n\pi}(-1)^n-\frac{4L^2}{(n\pi)^3}$$

$$B_n=\frac{4L^2}{(n\pi)^3}((-1)^n-1)$$

Then the Fourier sine series representation of $f(x)=x^2-Lx$ on $0\le x\le L$ is

$$f(x)=\sum_{n=1}^{\infty}B_n\sin\left(\frac{n\pi x}{L}\right)$$

$$f(x)=\frac{4L^2}{\pi^3}\sum_{n=1}^{\infty}\frac{(-1)^n-1}{n^3}\sin\left(\frac{n\pi x}{L}\right)$$

■ 6. Find the Fourier sine series representation of $f(x)=e^{L-x}+5$ on $0\le x\le L$.

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Solution:

The function isn't odd, so we'll find its odd extension on $-L \leq x \leq L$.

$$g(x) = \begin{cases} f(x) & 0 \leq x \leq L \\ -f(-x) & -L \leq x < 0 \end{cases}$$

$$g(x) = \begin{cases} e^{L-x} + 5 & 0 \leq x \leq L \\ -(e^{L-(-x)} + 5) & -L \leq x < 0 \end{cases}$$

$$g(x) = \begin{cases} e^{L-x} + 5 & 0 \leq x \leq L \\ -e^{L+x} - 5 & -L \leq x < 0 \end{cases}$$

Because the odd extension is an odd function, we can find its Fourier sine series representation, starting with calculating B_n.

$$B_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2}{L}\int_0^L (e^{L-x} + 5)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2}{L}\int_0^L e^{L-x}\sin\left(\frac{n\pi x}{L}\right)\,dx + \frac{1}{L}\int_0^L 5\sin\left(\frac{n\pi x}{L}\right)\,dx$$

First, Let's evaluate the second integral:

$$\frac{2}{L}\int_0^L 5\sin\left(\frac{n\pi x}{L}\right)\,dx = \frac{2}{L}\cdot\frac{-5L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_0^L$$

$$= -\frac{10}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_0^L$$

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Evaluate the interval using $\sin(n\pi) = 0$ and $\cos(n\pi) = (-1)^n$.

$$= -\frac{10}{n\pi}((-1)^n - (-1))$$

$$= \frac{10(-(-1)^n - 1)}{n\pi}$$

To evaluate the first integral, use integration by parts with $u = e^{L-x}$, $du = -e^{L-x}\,dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$\frac{2}{L}\int_0^L e^{L-x}\sin\left(\frac{n\pi x}{L}\right)\,dx = \left[\frac{2}{L}\cdot\frac{-L}{n\pi}e^{L-x}\cos\left(\frac{n\pi x}{L}\right) - \frac{2}{L}\cdot\frac{-L}{n\pi}\int(-e^{L-x})\cos\left(\frac{n\pi x}{L}\right)\,dx\right]\Bigg|_0^L$$

$$\frac{2}{L}\int_0^L e^{L-x}\sin\left(\frac{n\pi x}{L}\right)\,dx = \left[-\frac{2}{n\pi}e^{L-x}\cos\left(\frac{n\pi x}{L}\right) - \frac{2}{n\pi}\int e^{L-x}\cos\left(\frac{n\pi x}{L}\right)\,dx\right]\Bigg|_0^L$$

Use integration by parts with $u = e^{L-x}$, $du = -e^{L-x}\,dx$, $dv = \cos(n\pi x/L)\,dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

$$\frac{2}{L}\int_0^L e^{L-x}\sin\left(\frac{n\pi x}{L}\right)\,dx = \left[-\frac{2}{n\pi}e^{L-x}\cos\left(\frac{n\pi x}{L}\right) - \frac{2}{n\pi}\cdot\frac{L}{n\pi}e^{L-x}\sin\left(\frac{n\pi x}{L}\right)\right.$$

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$$+\frac{2}{n\pi}\cdot\frac{L}{n\pi}\int(-e^{L-x})\sin\left(\frac{n\pi x}{L}\right)\,dx\Bigg]\Bigg|_0^L$$

$$\frac{2}{L}\int_0^L e^{L-x}\sin\left(\frac{n\pi x}{L}\right)\,dx=\left[-\frac{2}{n\pi}e^{L-x}\cos\left(\frac{n\pi x}{L}\right)-\frac{2L}{(n\pi)^2}e^{L-x}\sin\left(\frac{n\pi x}{L}\right)\right.$$

$$\left.-\frac{2L}{(n\pi)^2}\int e^{L-x}\sin\left(\frac{n\pi x}{L}\right)\,dx\right]\Bigg|_0^L$$

Now we can move the integral on the right to the other side of the equation and solve.

$$\left(\frac{2}{L}+\frac{2L}{(n\pi)^2}\right)\int_0^L e^{L-x}\sin\left(\frac{n\pi x}{L}\right)\,dx=\left[-\frac{2}{n\pi}e^{L-x}\cos\left(\frac{n\pi x}{L}\right)\right.$$

$$\left.-\frac{2L}{(n\pi)^2}e^{L-x}\sin\left(\frac{n\pi x}{L}\right)\right]\Bigg|_0^L$$

$$\frac{2}{L}\left(\frac{(n\pi)^2+L^2}{(n\pi)^2}\right)\int_0^L e^{L-x}\sin\left(\frac{n\pi x}{L}\right)\,dx=\left[-\frac{2}{n\pi}e^{L-x}\cos\left(\frac{n\pi x}{L}\right)\right.$$

$$\left.-\frac{2L}{(n\pi)^2}e^{L-x}\sin\left(\frac{n\pi x}{L}\right)\right]\Bigg|_0^L$$

$$\frac{2}{L}\int_0^L e^{L-x}\sin\left(\frac{n\pi x}{L}\right)\,dx=\left(\frac{(n\pi)^2}{(n\pi)^2+L^2}\right)\left[-\frac{2}{n\pi}e^{L-x}\cos\left(\frac{n\pi x}{L}\right)\right.$$

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$$-\frac{2L}{(n\pi)^2}e^{L-x}\sin\left(\frac{n\pi x}{L}\right)\Bigg]\Bigg|_0^L$$

Evaluate the interval using $\sin(n\pi) = 0$ and $\cos(n\pi) = (-1)^n$.

$$\frac{2}{L}\int_0^L e^{L-x}\sin\left(\frac{n\pi x}{L}\right)\,dx = \left(\frac{(n\pi)^2}{(n\pi)^2+L^2}\right)\left(-\frac{2}{n\pi}(-1)^n+\frac{2e^L}{n\pi}\right)$$

$$= \left(\frac{(n\pi)^2}{(n\pi)^2+L^2}\right)\left(\frac{2e^L-2(-1)^n}{n\pi}\right)$$

$$= \frac{2(n\pi e^L-n\pi(-1)^n)}{(n\pi)^2+L^2}$$

Then

$$B_n = \frac{2(n\pi e^L-n\pi(-1)^n)}{(n\pi)^2+L^2}+\frac{10(-(-1)^n-1)}{n\pi}$$

Then the Fourier sine series representation of $f(x) = e^{L-x}+5$ on $0 \le x \le L$ is

$$f(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \sum_{n=1}^{\infty}\left(\frac{2(n\pi e^L-n\pi(-1)^n)}{(n\pi)^2+L^2}+\frac{10(-(-1)^n-1)}{n\pi}\right)\sin\left(\frac{n\pi x}{L}\right)$$

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COSINE AND SINE SERIES OF PIECEWISE FUNCTIONS

■ 1. Find the Fourier cosine series representation of the piecewise function on $0 \leq x \leq L$.

$$f(x) = \begin{cases} 0 & 0 \leq x \leq \frac{L}{2} \\ x^2 & \frac{L}{2} < x \leq L \end{cases}$$

Solution:

The function is even so $B_n = 0$. For A_0, we get

$$A_0 = \frac{1}{L}\int_0^L f(x)\ dx$$

$$A_0 = \frac{1}{L}\left(\int_0^{\frac{L}{2}} 0\ dx + \int_{\frac{L}{2}}^L x^2\ dx\right)$$

$$A_0 = \frac{1}{L} \cdot \frac{x^3}{3}\bigg|_{\frac{L}{2}}^L$$

$$A_0 = \frac{1}{L}\left(\frac{L^3}{3} - \frac{L^3}{24}\right)$$

$$A_0 = \frac{7L^2}{24}$$

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For A_n, we get

$$A_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = \frac{2}{L}\left(\int_0^{\frac{L}{2}} 0\cos\left(\frac{n\pi x}{L}\right)\,dx + \int_{\frac{L}{2}}^L x^2\cos\left(\frac{n\pi x}{L}\right)\,dx\right)$$

Use integration by parts with $u = x^2$, $du = 2x\,dx$, $dv = \cos(n\pi x/L)\,dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

$$A_n = \frac{2}{L}\left(x^2\frac{L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_{\frac{L}{2}}^L - \frac{2L}{n\pi}\int_{\frac{L}{2}}^L x\sin\left(\frac{n\pi x}{L}\right)\,dx\right)$$

Use integration by parts with $u = x$, $du = dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$A_n = \frac{2}{L}\left(0 - \frac{L^3}{4n\pi}\sin\left(\frac{n\pi}{2}\right)\right) - \frac{4}{n\pi}\left(-x\frac{L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_{\frac{L}{2}}^L + \frac{L}{n\pi}\int_{\frac{L}{2}}^L \cos\left(\frac{n\pi x}{L}\right)\,dx\right)$$

$$A_n = -\frac{L^2}{2n\pi}\sin\left(\frac{n\pi}{2}\right) + \frac{4L^2}{n^2\pi^2}\left((-1)^n - \frac{1}{2}\cos\left(\frac{n\pi}{2}\right)\right) - \frac{4L^2}{n^3\pi^3}\sin\left(\frac{n\pi x}{L}\right)\Big|_{\frac{L}{2}}^L$$

$$A_n = -\frac{L^2}{2n\pi}\sin\left(\frac{n\pi}{2}\right) + \frac{4L^2}{n^2\pi^2}\left((-1)^n - \frac{1}{2}\cos\left(\frac{n\pi}{2}\right)\right) + \frac{4L^2}{n^3\pi^3}\sin\left(\frac{n\pi}{2}\right)$$

$$A_n = \left(\frac{4L^2}{n^3\pi^3} - \frac{L^2}{2n\pi}\right)\sin\left(\frac{n\pi}{2}\right) + \frac{4L^2}{n^2\pi^2}\left((-1)^n - \frac{1}{2}\cos\left(\frac{n\pi}{2}\right)\right)$$

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Then the Fourier cosine series representation of the piecewise function on $0 \leq x \leq L$ is

$$f(x) = \frac{7L^2}{24} + \sum_{n=1}^{\infty} \left(\left(\frac{4L^2}{n^3\pi^3} - \frac{L^2}{2n\pi} \right) \sin\left(\frac{n\pi}{2}\right) + \frac{4L^2}{n^2\pi^2} \left((-1)^n - \frac{1}{2}\cos\left(\frac{n\pi}{2}\right) \right) \right) \cos\left(\frac{n\pi x}{L}\right)$$

■ 2. Find the Fourier sine series representation of the piecewise function on $0 \leq x \leq 2\pi$.

$$f(x) = \begin{cases} 2x & 0 \leq x \leq \pi \\ 4\pi - 2x & \pi < x \leq 2\pi \end{cases}$$

Solution:

The odd extension $g(x)$ of the piecewise function $f(x)$ will be

$$g(x) = \begin{cases} f(x) & 0 \leq x \leq L \\ -f(-x) & -L \leq x < 0 \end{cases}$$

$$g(x) = \begin{cases} 2x & 0 \leq x \leq \pi \\ 4\pi - 2x & \pi < x \leq 2\pi \\ 2x & -\pi \leq x < 0 \\ -4\pi - 2x & -2\pi < x < -\pi \end{cases}$$

To find the Fourier sine series, we'll calculate B_n.

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$$B_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{1}{\pi}\left(\int_0^{\pi} 2x\sin\left(\frac{nx}{2}\right)\,dx + \int_{\pi}^{2\pi}(4\pi - 2x)\sin\left(\frac{nx}{2}\right)\,dx\right)$$

Use integration by parts with $u = 2x$, $du = 2\,dx$, $dv = \sin(nx/2)\,dx$, and $v = -(2/n)\cos(nx/2)$ in the first integral and with $u = 4\pi - 2x$, $du = -2\,dx$, $dv = \sin(nx/2)\,dx$, and $v = -(2/n)\cos(nx/2)$ in the second integral.

$$B_n = \frac{1}{\pi}\left(-\frac{4x}{n}\cos\left(\frac{nx}{2}\right)\Big|_0^{\pi} + \frac{4}{n}\int_0^{\pi}\cos\left(\frac{nx}{2}\right)\,dx\right.$$

$$\left.+(4\pi - 2x)\left(-\frac{2}{n}\cos\left(\frac{nx}{2}\right)\right)\Big|_{\pi}^{2\pi} - \frac{4}{n}\int_{\pi}^{2\pi}\cos\left(\frac{nx}{2}\right)\,dx\right)$$

$$B_n = \frac{1}{\pi}\left(-\frac{4\pi}{n}\cos\left(\frac{n\pi}{2}\right) + 0 + \frac{8}{n^2}\sin\left(\frac{nx}{2}\right)\Big|_0^{\pi}\right.$$

$$\left.+0 + \frac{4\pi}{n}\cos\left(\frac{n\pi}{2}\right) - \frac{8}{n^2}\sin\left(\frac{nx}{2}\right)\Big|_{\pi}^{2\pi}\right)$$

$$B_n = \frac{1}{\pi}\left(\frac{8}{n^2}\sin\left(\frac{n\pi}{2}\right) + \frac{8}{n^2}\sin\left(\frac{n\pi}{2}\right)\right)$$

$$B_n = \frac{16}{n^2\pi}\sin\left(\frac{n\pi}{2}\right)$$

Then the Fourier cosine series representation of the piecewise function on $0 \le x \le 2\pi$ is

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$$f(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \sum_{n=1}^{\infty} \frac{16}{n^2\pi} \sin\left(\frac{n\pi}{2}\right) \sin\left(\frac{nx}{2}\right)$$

■ 3. Find the Fourier cosine series representation of the piecewise function on $0 \leq x \leq \pi$.

$$f(x) = \begin{cases} x^2 & 0 \leq x \leq \frac{\pi}{2} \\ x & \frac{\pi}{2} < x \leq \pi \end{cases}$$

Solution:

The even extension $g(x)$ of the piecewise function $f(x)$ will be

$$g(x) = \begin{cases} f(x) & 0 \leq x \leq L \\ f(-x) & -L \leq x < 0 \end{cases}$$

$$g(x) = \begin{cases} x & \frac{\pi}{2} < x \leq \pi \\ x^2 & 0 \leq x \leq \frac{\pi}{2} \\ x^2 & -\frac{\pi}{2} \leq x < 0 \\ -x & -\pi \leq x < -\frac{\pi}{2} \end{cases}$$

To find the Fourier cosine series, we'll calculate A_0 and A_n. For A_0, we get

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$$A_0 = \frac{1}{\pi}\int_0^{\pi} f(x)\ dx$$

$$A_0 = \frac{1}{\pi}\left(\int_0^{\frac{\pi}{2}} x^2\ dx + \int_{\frac{\pi}{2}}^{\pi} x\ dx\right)$$

$$A_0 = \frac{1}{\pi}\left(\frac{x^3}{3}\bigg|_0^{\frac{\pi}{2}} + \frac{x^2}{2}\bigg|_{\frac{\pi}{2}}^{\pi}\right)$$

$$A_0 = \frac{1}{\pi}\left(\frac{\pi^3}{24} + \frac{\pi^2}{2} - \frac{\pi^2}{8}\right)$$

$$A_0 = \frac{\pi^2}{24} + \frac{3\pi}{8}$$

For A_n, we get

$$A_n = \frac{2}{L}\int_0^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)\ dx$$

$$A_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos(nx)\ dx$$

$$A_n = \frac{2}{\pi}\left(\int_0^{\frac{\pi}{2}} x^2\cos(nx)\ dx + \int_{\frac{\pi}{2}}^{\pi} x\cos(nx)\ dx\right)$$

Use integration by parts with $u = x^2$, $du = 2x\ dx$, $dv = \cos(nx)\ dx$, and $v = (1/n)\sin(nx)$ in the first integral and with $u = x$, $du = dx$, $dv = \cos(nx)\ dx$, and $v = (1/n)\sin(nx)$ in the second integral.

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$$A_n = \frac{2}{\pi}\left(\frac{x^2}{n}\sin(nx)\Big|_0^{\frac{\pi}{2}} - \frac{2}{n}\int_0^{\frac{\pi}{2}} x\sin(nx)\,dx + \frac{x}{n}\sin(nx)\Big|_{\frac{\pi}{2}}^{\pi} - \frac{1}{n}\int_{\frac{\pi}{2}}^{\pi}\sin(nx)\,dx\right)$$

Use integration by parts with $u = x$, $du = dx$, $dv = \sin(nx)\,dx$, and $v = -(1/n)\cos(nx)$.

$$A_n = \frac{2}{\pi}\left(\frac{\pi^2}{4n}\sin\left(\frac{n\pi}{2}\right) - 0 - \frac{2}{n}\left(-\frac{x}{n}\cos(nx) + \frac{1}{n^2}\sin(nx)\right)\Big|_0^{\frac{\pi}{2}}\right.$$

$$\left. +0 - \frac{\pi}{2n}\sin\left(\frac{n\pi}{2}\right) + \frac{1}{n^2}\cos(nx)\Big|_{\frac{\pi}{2}}^{\pi}\right)$$

$$A_n = \frac{2}{\pi}\left[\frac{\pi^2}{4n}\sin\left(\frac{n\pi}{2}\right) + \frac{\pi}{n^2}\cos\left(\frac{n\pi}{2}\right) - \frac{2}{n^3}\sin\left(\frac{n\pi}{2}\right)\right.$$

$$\left. -\frac{\pi}{2n}\sin\left(\frac{n\pi}{2}\right) + \frac{1}{n^2}(-1)^n - \frac{1}{n^2}\cos\left(\frac{n\pi}{2}\right)\right]$$

$$A_n = \frac{2}{\pi}\left[\left(\frac{\pi^2}{4n} - \frac{2}{n^3} - \frac{\pi}{2n}\right)\sin\left(\frac{n\pi}{2}\right) + \frac{\pi - 1}{n^2}\cos\left(\frac{n\pi}{2}\right) + \frac{1}{n^2}(-1)^n\right]$$

$$A_n = \left(\frac{\pi}{2n} - \frac{4}{n^3\pi} - \frac{1}{n}\right)\sin\left(\frac{n\pi}{2}\right) + \frac{2(\pi - 1)}{n^2\pi}\cos\left(\frac{n\pi}{2}\right) + \frac{2}{n^2\pi}(-1)^n$$

Then the Fourier cosine series representation of the piecewise function on $0 \le x \le \pi$ is

$$f(x) = \frac{\pi^2}{24} + \frac{3\pi}{8}$$

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$$+\sum_{n=1}^{\infty}\left(\left(\frac{\pi}{2n}-\frac{4}{n^3\pi}-\frac{1}{n}\right)\sin\left(\frac{n\pi}{2}\right)+\frac{2(\pi-1)}{n^2\pi}\cos\left(\frac{n\pi}{2}\right)+\frac{2}{n^2\pi}(-1)^n\right)\cos(nx)$$

■ 4. Find the Fourier sine series representation of the piecewise function on $0 \leq x \leq \pi$.

$$f(x)=\begin{cases}\sin x & 0 \leq x \leq \frac{\pi}{2}\\ \sin 2x & \frac{\pi}{2} < x \leq \pi\end{cases}$$

Solution:

The function is odd so $A_0 = A_n = 0$. For B_n, we get

$$B_n=\frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n=\frac{2}{\pi}\int_0^{\pi} f(x)\sin(nx)\,dx$$

$$B_n=\frac{2}{\pi}\left(\int_0^{\frac{\pi}{2}}\sin x\sin(nx)\,dx+\int_{\frac{\pi}{2}}^{\pi}\sin(2x)\sin(nx)\,dx\right)$$

Use the trigonometric identity

$$\sin A\sin B=\frac{1}{2}(\cos(A-B)-\cos(A+B))$$

to get

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$$B_n = \frac{1}{\pi}\left(\int_0^{\frac{\pi}{2}} \cos(n-1)x - \cos(n+1)x \; dx \right.$$

$$\left. + \int_{\frac{\pi}{2}}^{\pi} \cos(n-2)x - \cos(n+2)x \; dx \right)$$

$$B_n = \frac{1}{\pi}\left(\left(\frac{1}{n-1}\sin(n-1)x - \frac{1}{n+1}\sin(n+1)x \right) \Big|_0^{\frac{\pi}{2}} \right.$$

$$\left. + \left(\frac{1}{n-2}\sin(n-2)x - \frac{1}{n+2}\sin(n+2)x \right) \Big|_{\frac{\pi}{2}}^{\pi} \right)$$

$$B_n = \frac{1}{\pi}\left(\frac{1}{n-1}\sin\left(\frac{n\pi}{2} - \frac{\pi}{2}\right) - \frac{1}{n+1}\sin\left(\frac{n\pi}{2} + \frac{\pi}{2}\right) \right.$$

$$\left. + 0 - \frac{1}{n-2}\sin\left(\frac{n\pi}{2} - \pi\right) + \frac{1}{n+2}\sin\left(\frac{n\pi}{2} + \pi\right) \right)$$

Use the trigonometric identities

$$\sin(A - B) = \sin A \cos B - \cos A \sin B$$

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

to get

$$B_n = \frac{1}{\pi}\left(-\frac{1}{n-1}\cos\left(\frac{n\pi}{2}\right) - \frac{1}{n+1}\cos\left(\frac{n\pi}{2}\right) \right.$$

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$$+\frac{1}{n-2}\sin\left(\frac{n\pi}{2}\right)-\frac{1}{n+2}\sin\left(\frac{n\pi}{2}\right)\Bigg)$$

$$B_n=\frac{1}{\pi}\left(-\frac{2n}{n^2-1}\cos\left(\frac{n\pi}{2}\right)+\frac{4}{n^2-4}\sin\left(\frac{n\pi}{2}\right)\right) \text{ with } n\neq 1,2$$

$$B_1=\frac{2}{\pi}\left(\int_0^{\frac{\pi}{2}}\sin x\sin x\ dx+\int_{\frac{\pi}{2}}^{\pi}\sin(2x)\sin x\ dx\right)$$

Use the trigonometric identities

$$\sin^2 x=\frac{1-\cos(2x)}{2}$$

$$\sin A\sin B=\frac{1}{2}(\cos(A-B)-\cos(A+B))$$

to get

$$B_1=\frac{2}{\pi}\left(\int_0^{\frac{\pi}{2}}\frac{1-\cos(2x)}{2}\ dx+\frac{1}{2}\int_{\frac{\pi}{2}}^{\pi}(\cos x-\cos(3x))\ dx\right)$$

$$B_1=\frac{2}{\pi}\left(\left(\frac{x}{2}-\frac{\sin(2x)}{4}\right)\Bigg|_0^{\frac{\pi}{2}}+\frac{1}{2}\left(\sin x-\frac{1}{3}\sin(3x)\right)\Bigg|_{\frac{\pi}{2}}^{\pi}\right)$$

$$B_1=\frac{2}{\pi}\left(\frac{\pi}{4}-0+\frac{1}{2}\left(0-1-0-\frac{1}{3}\right)\right)$$

$$B_1=\frac{2}{\pi}\left(\frac{\pi}{4}-\frac{2}{3}\right)$$

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$$B_1 = \frac{1}{2} - \frac{4}{3\pi}$$

and

$$B_2 = \frac{2}{\pi}\left(\int_0^{\frac{\pi}{2}} \sin x \sin(2x)\ dx + \int_{\frac{\pi}{2}}^{\pi} \sin^2(2x)\ dx \right)$$

Use the trigonometric identities

$$\sin^2 x = \frac{1 - \cos(2x)}{2}$$

$$\sin A \sin B = \frac{1}{2}(\cos(A - B) - \cos(A + B))$$

to get

$$B_2 = \frac{2}{\pi}\left(\frac{1}{2}\int_0^{\frac{\pi}{2}} \cos x - \cos(3x)\ dx + \int_{\frac{\pi}{2}}^{\pi} \frac{1 - \cos(4x)}{2}\ dx \right)$$

$$B_2 = \frac{2}{\pi}\left(\frac{1}{2}\left(\sin x - \frac{1}{3}\sin(3x) \right)\Big|_0^{\frac{\pi}{2}} + \left(\frac{x}{2} - \frac{\sin(4x)}{8} \right)\Big|_{\frac{\pi}{2}}^{\pi} \right)$$

$$B_2 = \frac{2}{\pi}\left(\frac{1}{2}\left(1 - 0 + \frac{1}{3} - 0 \right) + \frac{\pi}{2} - 0 - \frac{\pi}{4} + 0 \right)$$

$$B_2 = \frac{2}{\pi}\left(\frac{2}{3} + \frac{\pi}{4} \right)$$

$$B_2 = \frac{4}{3\pi} + \frac{1}{2}$$

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Then the Fourier sine series representation of the piecewise function on $0 \leq x \leq \pi$ is

$$f(x) = \left(\frac{1}{2} - \frac{4}{3\pi}\right)\sin x + \left(\frac{1}{2} + \frac{4}{3\pi}\right)\sin(2x)$$

$$+ \sum_{n=3}^{\infty} \frac{1}{\pi}\left(-\frac{2n}{n^2-1}\cos\left(\frac{n\pi}{2}\right) + \frac{4}{n^2-4}\sin\left(\frac{n\pi}{2}\right)\right)\sin(nx)$$

■ 5. Find the Fourier cosine series representation of the piecewise function on $0 \leq x \leq L$.

$$f(x) = \begin{cases} 3x & 0 \leq x \leq \frac{L}{3} \\ 3x - L & \frac{L}{3} < x \leq \frac{2L}{3} \\ 3x - 2L & \frac{2L}{3} < x < L \end{cases}$$

Solution:

The even extension $g(x)$ of the piecewise function $f(x)$ will be

$$g(x) = \begin{cases} f(x) & 0 \leq x \leq L \\ f(-x) & -L \leq x < 0 \end{cases}$$

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$$g(x) = \begin{cases} 3x - 2L & \frac{2L}{3} < x < L \\ 3x - L & \frac{L}{3} < x \leq \frac{2L}{3} \\ 3x & 0 \leq x \leq \frac{L}{3} \\ -3x & -\frac{L}{3} \leq x < 0 \\ -3x - L & -\frac{2L}{3} \leq x < -\frac{L}{3} \\ -3x - 2L & -L < x < -\frac{2L}{3} \end{cases}$$

To find the Fourier cosine series, we'll calculate A_0 and A_n. For A_0, we get

$$A_0 = \frac{1}{L}\int_0^L f(x)\ dx$$

$$A_0 = \frac{1}{L}\left(\int_0^{\frac{L}{3}} 3x\ dx + \int_{\frac{L}{3}}^{\frac{2L}{3}} (3x - L)\ dx + \int_{\frac{2L}{3}}^{L} (3x - 2L)\ dx \right)$$

$$A_0 = \frac{1}{L}\left(\frac{3x^2}{2}\bigg|_0^{\frac{L}{3}} + \left(\frac{3x^2}{2} - Lx \right)\bigg|_{\frac{L}{3}}^{\frac{2L}{3}} + \left(\frac{3x^2}{2} - 2Lx \right)\bigg|_{\frac{2L}{3}}^{L} \right)$$

$$A_0 = \frac{1}{L}\left(\frac{L^2}{6} + \frac{2L^2}{3} - \frac{2L^2}{3} - \frac{L^2}{6} + \frac{L^2}{3} + \frac{3L^2}{2} - 2L^2 - \frac{2L^2}{3} + \frac{4L^2}{3} \right)$$

$$A_0 = \frac{1}{L}\left(\frac{L^2}{2} \right)$$

$$A_0 = \frac{L}{2}$$

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For A_n, we get

$$A_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = \frac{2}{L}\left(\int_0^{\frac{L}{3}} 3x\cos\left(\frac{n\pi x}{L}\right)\,dx + \int_{\frac{L}{3}}^{\frac{2L}{3}} (3x-L)\cos\left(\frac{n\pi x}{L}\right)\,dx\right.$$

$$\left. + \int_{\frac{2L}{3}}^{L} (3x-2L)\cos\left(\frac{n\pi x}{L}\right)\,dx\right)$$

Use integration by parts with $u = 3x$, $du = 3\,dx$, $dv = \cos(n\pi x/L)\,dx$, and $v = (L/n\pi)\sin(n\pi x/L)$ in the first integral, with $u = 3x - L$, $du = 3\,dx$, $dv = \cos(n\pi x/L)\,dx$, and $v = (L/n\pi)\sin(n\pi x/L)$ in the second integral, and with $u = 3x - 2L$, $du = 3\,dx$, $dv = \cos(n\pi x/L)\,dx$, and $v = (L/n\pi)\sin(n\pi x/L)$ in the third integral.

$$A_n = \frac{2}{L}\left(3x\frac{L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_0^{\frac{L}{3}} - \int_0^{\frac{L}{3}} \frac{3L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\,dx\right.$$

$$+(3x-L)\frac{L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_{\frac{L}{3}}^{\frac{2L}{3}} - \int_{\frac{L}{3}}^{\frac{2L}{3}} \frac{3L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$\left.+(3x-2L)\frac{L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_{\frac{2L}{3}}^{L} - \int_{\frac{2L}{3}}^{L} \frac{3L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\,dx\right)$$

$$A_n = \frac{2}{L}\left(3x\frac{L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_0^{\frac{L}{3}} + (3x-L)\frac{L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_{\frac{L}{3}}^{\frac{2L}{3}}\right.$$

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$$+(3x-2L)\frac{L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_{\frac{2L}{3}}^{L}-\int_0^L\frac{3L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\,dx\Bigg)$$

$$A_n=\frac{2}{L}\left(\frac{L^2}{n\pi}\sin\left(\frac{n\pi}{3}\right)-0+\frac{L^2}{n\pi}\sin\left(\frac{2n\pi}{3}\right)-0+\frac{L^2}{n\pi}\sin(n\pi)\right.$$

$$\left.-0+\frac{3L^2}{n^2\pi^2}\cos\left(\frac{n\pi x}{L}\right)\Big|_0^L\right)$$

$$A_n=\frac{2L}{n\pi}\left(\sin\left(\frac{n\pi}{3}\right)+\sin\left(\frac{2n\pi}{3}\right)\right)+\frac{6L}{n^2\pi^2}((-1)^n-1)$$

Then the Fourier cosine series representation of the piecewise function on $0\le x\le L$ is

$$f(x)=\frac{L}{2}$$

$$+\sum_{n=1}^{\infty}\left(\frac{2L}{n\pi}\left(\sin\left(\frac{n\pi}{3}\right)+\sin\left(\frac{2n\pi}{3}\right)\right)+\frac{6L}{n^2\pi^2}((-1)^n-1)\right)\cos\left(\frac{n\pi x}{L}\right)$$

■ 6. Find the Fourier sine series representation of the piecewise function on $0\le x\le 2$.

$$f(x)=\begin{cases}x^2 & 0\le x\le 1\\ x^2-1 & 1<x\le 2\end{cases}$$

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Solution:

The odd extension $g(x)$ of the piecewise function $f(x)$ will be

$$g(x) = \begin{cases} f(x) & 0 \le x \le L \\ -f(-x) & -L \le x < 0 \end{cases}$$

$$g(x) = \begin{cases} x^2 - 1 & 1 < x \le 2 \\ x^2 & 0 \le x \le 1 \\ -x^2 & -1 \le x < 0 \\ -x^2 + 1 & -2 \le x < -1 \end{cases}$$

To find the Fourier sine series, we'll calculate B_n.

$$B_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2}{2}\int_0^2 f(x)\sin\left(\frac{n\pi x}{2}\right)\,dx$$

$$B_n = \int_0^1 x^2 \sin\left(\frac{n\pi x}{2}\right)\,dx + \int_1^2 (x^2 - 1)\sin\left(\frac{n\pi x}{2}\right)\,dx$$

$$B_n = \int_0^1 x^2 \sin\left(\frac{n\pi x}{2}\right)\,dx + \int_1^2 x^2 \sin\left(\frac{n\pi x}{2}\right)\,dx - \int_1^2 \sin\left(\frac{n\pi x}{2}\right)\,dx$$

$$B_n = \int_0^2 x^2 \sin\left(\frac{n\pi x}{2}\right)\,dx - \int_1^2 \sin\left(\frac{n\pi x}{2}\right)\,dx$$

Use integration by parts with $u = x^2$, $du = 2x\ dx$, $dv = \sin(n\pi x/L)\ dx$, and $v = -(2/n\pi)\cos(n\pi x/2)$.

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$$B_n = -\frac{2x^2}{n\pi}\cos\left(\frac{n\pi x}{2}\right)\Big|_0^2 + \frac{4}{n\pi}\int_0^2 x\cos\left(\frac{n\pi x}{2}\right)\,dx + \frac{2}{n\pi}\cos\left(\frac{n\pi x}{2}\right)\Big|_1^2$$

Use integration by parts with $u = x$, $du = dx$, $dv = \cos(n\pi x/2)\,dx$, and $v = (2/n\pi)\sin(n\pi x/2)$.

$$B_n = -\frac{8}{n\pi}(-1)^n + 0 + \frac{4}{n\pi}\left(\frac{2x}{n\pi}\sin\left(\frac{n\pi x}{2}\right)\Big|_0^2 - \frac{2}{n\pi}\int_0^2 \sin\left(\frac{n\pi x}{2}\right)\,dx\right)$$

$$+\frac{2}{n\pi}(-1)^n - \frac{2}{n\pi}\cos\left(\frac{n\pi}{2}\right)$$

$$B_n = -\frac{6}{n\pi}(-1)^n - \frac{2}{n\pi}\cos\left(\frac{n\pi}{2}\right) + \frac{4}{n\pi}\left(0 + \frac{4}{(n\pi)^2}\cos\left(\frac{n\pi x}{2}\right)\Big|_0^2\right)$$

$$B_n = -\frac{2}{n\pi}\left(\cos\left(\frac{n\pi}{2}\right) + 3(-1)^n\right) + \frac{16}{n^3\pi^3}\cos\left(\frac{n\pi x}{2}\right)\Big|_0^2$$

$$B_n = -\frac{2}{n\pi}\left(\cos\left(\frac{n\pi}{2}\right) + 3(-1)^n\right) + \frac{16}{n^3\pi^3}((-1)^n - 1)$$

Then the Fourier sine series representation of the piecewise function on $0 \le x \le 2$ is

$$f(x) = \sum_{n=1}^{\infty}\left(-\frac{2}{n\pi}\left(\cos\left(\frac{n\pi}{2}\right) + 3(-1)^n\right) + \frac{16}{n^3\pi^3}((-1)^n - 1)\right)\sin\left(\frac{n\pi x}{2}\right)$$

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